Intel's Core-based client processors

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- 1. Introduction and overview of Intel's Core-based client processors
- 2. Naming of Intel's Core-based client processors
- 3. Key enhancements in the microarchitecture of Intel's Core-based client processors over time
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- 5. References

Intel’s Core-based client processors

- This Chapter focuses on client processors that power desktops and laptops.
- We begin with an introduction and a generational overview of Intel’s Core-based client processors.
- Next, we discuss Intel naming conventions for these processors.
- The main Section explores the key architectural features over time, including their CPU architecture, core count, core width, and memory subsystem.
- Finally, we present two case studies of Intel’s latest processor, including its Lunar Lake desktop and Arrow Lake laptop processors, highlighting their microarchitecture and relevance for the company.

A remark on the terminology

In this Lecture Note, we use the terms “processor family”, “processor line”, or “processor series” as synonyms.

1. Introduction and overview of Intel’s Core-based client processors

1. Introduction and overview of Intel's Core-based client processors

What are client processors?

- Processors powering desktops (DTs) and laptops are named client processors.

- In contrast, Intel and AMD refer to their laptop processors as mobile processors; nevertheless, lecture, we will use the original term "laptop processors".
- Using this terminology, client and mobile processors can be contrasted as follows:

  First, client processors, which include both DT and laptop processors, are typically built on the x86 ISA, while mobile processors, such as those found in tablets and smartphones, primarily use the ARM ISA.

  Secondly, client devices - namely DTs and laptops - generally do not provide touchscreen displays, unlike mobile devices, which include tablets and smartphones, that typically do.

Intel and AMD likely chose the term "mobile" for its marketing appeal.

Remark

The above statement about the use of the x86 and ARM ISA has recently become blurred" as an increasing number of ARM ISA-based clients are emerging in the market offering lower power consumption, as discussed in a separate Chapter.

- In the following Figure, we provide an overview of how Intel's processors can be classified according to their market aim.

1. Introduction and overview of Intel's Core-based client processors (2)

The client processor category in Intel's Core family

Main processor categories in Intel's Core family

Server processors
Xeon E7/E5/E3
Platinum/Gold etc.

Workstation processors
Xeon W7/W5/W3

HEDT processors (High-End Desktops)
Core i7/i9
(Extreme Edition or X models)

Client processors (for desktops/laptops)
Core i9/i7/i5/i3
(Basic architectures)

Mobile processors (for tablets, smartphones)
Atom lines
(+ LP Core models)

Clients

Mobiles

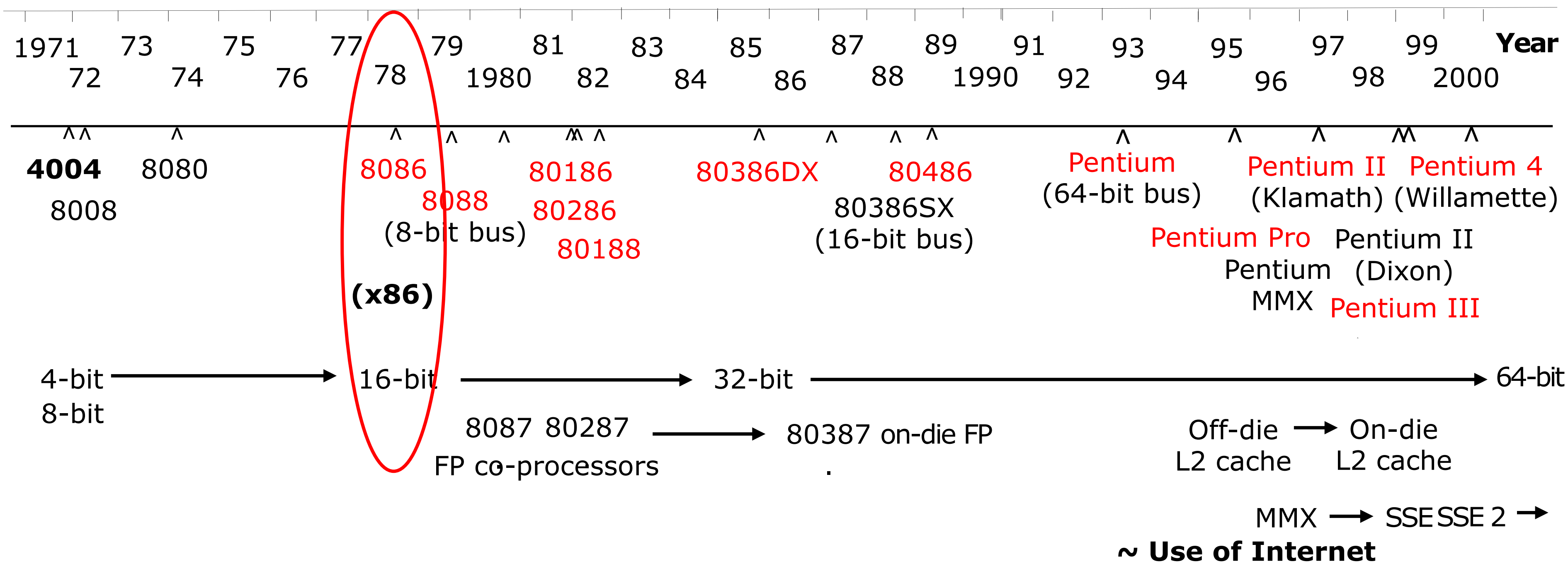
Recent config.

Servers	Workstations	High-End Desktops	Clients	Mobiles
Up to 144 C	Up to 60 C	Up to 32 C	Up to (8 + 16) C + G	Up to 10 C+ G
(See related Chapter)	(Not discussed)	(See Annex)	(See related Chapter)	(See Annex)

Intel’s early mainstream processors and DRAM memories -1

Intel’s journey to the Core family -1

The Figure below illustrates Intel’s processor lines that preceded the Core family.



See the next slide for a brief discussion of the above processors.

Intel's journey to the Core family -2

- Intel introduced the x86 ISA with its 16-bit 8086 processor in 1978.
- This was soon followed by the release of the 8088 along with two other 16-bit processors - the 80186 and 80286.
- IBM selected the 8088 for its original IBM PC, launched in 1981.

  The 8088 was essentially an 8086 with an 8-bit wide IO bus, making the IBM PC more cost-effective.

  Later, IBM released the PC AT, an upgraded version of the original IBM PC, featuring a hard drive and adopting the more advanced 8086.
- As the IBM PC gained dominance in the global market, Intel's x86 processor family became the most successful and widely adopted architecture in personal computing.
- Intel subsequently introduced a long line of 32-bit processors, including the 80386, 80486, Pentium, Pentium II, Pentium III, and Pentium 4.
- These increasingly advanced processors allowed Intel to maintain a dominant position in the market, consistently capturing around 90 % market share.
- In 2004, Intel entered the 64-bit era with its Prescott F Pentium 4 model, marking a significant evolution in its processor architecture.
- Then, in 2006, Intel launched its first 64-bit dual-core processor line, the Core family, specifically designed to deliver maximum performance per Watt.
- Since that milestone, Intel has introduced new generations of Core-based client processor lines almost every year, continually improving performance and power efficiency, as will be discussed subsequently.

Overview of Intel's Core-based client processor lines and their key features -1

Gen.	Processor	Techno-logy	Date of intro.	Seg-ment	Max core count	Connecting memory	Mem. speed (up to)	Mem. channels	Socket	MCH/ PCH
1.	Core 2	65 nm	6/2006	L D	2C	Via the NB	DDR2-800		LGA 775	945-975
	Core 2 Quad	65 nm	6/2007	D	2x2C		DDR3-1067		LGA 775	3 Series
	Penryn	45 nm	6/2008	D	2x2C		DDR3-1067		LGA 775	4 Series
	2. G. Nehalem (Lynnfield)	45 nm	9/2009	D	4C		DDR3-1333		LGA 1156	5 Series
	Westmere (L: Arrandale) D: Clarkdale/	32 nm	1/2010	L D	2C+G				L:BGA 1288 D:LGA 1156	5 Series
2.	Sandy Bridge	32 nm	1/2011	L D	2C/4C+G		DDR3-1333		L: PGA 988 D: LGA 1155	6 Series
3.	Ivy Bridge	22 nm	4/2012	L D	4C+G		DDR3-1600			7 Series
4.	Haswell	22 nm	6/2013	L D	4C+G		DDR3-1600		L: BGA 1168 D: LGA 1150	D: 8 Ser.
4.	Haswell Refr.	22 nm	5/2014	D	4C+G	Via the processor	DDR3-1666			D: 9 Ser.
5.	Broadwell	14 nm	9/2014	L D	4C+G		DDR3-1866	2		D: 9 Ser.
6.	Skylake	14 nm	10/2015	L D	4C+G		DDR4-2133		L: BGA 1356 D: LGA 1151	D: 100 Series
7.	Kaby Lake	14 nm	8/2016	L D	4C+G		DDR4-2400			D: 200 Series
8.	Coffee Lake	14 nm	10/2017	L D	6C+G		DDR4-2666		L: BGA 1440 D: LGA 1151	D:300 Series
8.	Amber Lake	14 nm	8/2018	L	2C+G		LPDDR3-2133		BGA 1515	--
8.	Whiskey Lake	14 nm	8/2018	L	4C+G		DDR4-2400		BGA 1528	--
9.	Coffee Lake R.	14 nm	10/2018	D	8C+G		DDR4-2666		LGA 1151	300 Series
10.	Comet Lake	14 nm	8/2019	L D	10C+G		DDR4-2666		L: BGA 1528 D: LGA 1200	D: 400 Series
11.	Rocket Lake	14 nm	3/2021	D	8C+G		DDR4-3200		LGA 1200	

Overview of Intel's Core-based client processor lines and their key features -2

	Processor	Techno-logy	Date of intro.	Market Segment	Max. core count	Mem. speed (up to)	Mem. channels	Socket	PCH
8.	Cannon Lake	10 nm	5/2018	L:U	2C	DDR4-2400	2x64- bit	BGA 1440	--
10.	Ice Lake	10 nm	7/2019	L: G	4C+G	DDR4-3200 LPDDR4- 3733	2x64- bit	BGA 1526	--
11.	Tiger Lake	10 nm	09/2020 01/2021 05/2021	L: G L: H L: H	4C+G 4C+G 8C+G	LPDDR4x-4267 DDR4-3200	2x64- bit	BGA 1449 BGA 1449 BGA 1787	--
11.	Tiger Lake	10 nm	01/2021	D	8C+G	DDR4-3200	2x64- bit	LGA 1700	500 Series
12.	Alder Lake	Intel 7	11/2021	D	8+8C+G	DDR5-4800	2x64- bit	D: LGA 1700	600 Series
12.	Alder Lake	Intel 7	02/2022	L:H,P L: U	6+8C+G 2+8+G	DDR5-4800 LPDDR5-5200 LPDDR4x-4267	2x64- bit	BGA 1744	--
12.	Alder Lake	Intel 7	05/2022	L: HX	8+8+G	DDR5-4800	2x64- bit	BGA: 1781	600 Series
13	Raptor Lake	Intel 7	10/2022	D	8+16C+G	DDR5-5600	2x64- bit	LGA 1700	700 Series
13.	Raptor Lake	Intel 7	01/2023	L:HX	8+16C+G	DDR5-5600	2x64- bit	BGA 1964	700 Series
13.	Raptor Lake	Intel 7	01/2023	L:H,P L:U	6+8C+G 2+8C8C	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267	2x64- bit	BGA 1744	--
14	Raptor Lake Refresh	Intel 7	10/2023	D	8+16C+G	DDR5-5600	2x64- bit	LGA 1700	700 Series
14	Raptor Lake Refresh	Intel 7	01/2024	L:HX	8+16C+G	DDR5-5600 DDR4-3200	2x64- bit	BGA 1964	700 Series
14.	Raptor Lake Refresh	Intel 7	01/2024	L:U	2+8C+G	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267	2x64- bit	BGA-1744	--

L: Laptop D: Desktop

Overview of Intel's Core Ulta client processor lines and their key features

	Processor	Techno-logy	Date of intro.	Seg-ment	Max. core count	Memory (up to)	TDP	Socket	PCH
14/ CUS 1	Meteor Lake (Series 1)	CPU: Intel 4 GPU: N5 SOC: N6	12/2023	L:H L:U	6+8+2C+G 2+8+2C+G+NPU	DDR5-5600 LPDDR5X-7467	H: 45/28 W U: 15 W	BGA 2049	-integrated
14/ CUS 1	Meteor Lake (Series 1)	CPU: Intel 4 GPU: N5 SOC: N6	12/2023	L:U-9W	2+8+2C+G+NPU	LPDDR5X-6400	9 W	BGA 2551	integrated
CUS 2	Lunar Lake (Series 2) (2 dies)	Compute: N3B Platform.: N6	09/2024	L:V	4+4C+G+ NPU (no hyperthreading)	LPDDR5x-8533 (in-package integrated)	17/30W	BGA 2833	Integrated
CUS 2	Arrow Lake (Series 2)	CPU: N3B GPU: N5 SOC/IO: N6	10/2024 01/2025	D:S	8+16C+G+ NPU (no hyperthreading)	DDR5-5600	125 W	LGA 1851	800 Series
CUS 2	Arrow Lake (Series 2)	CPU: N3B GPU: N5 SOC/IO: N6	01/2025	L:HX	8+16C+G+ NPU (no hyperthreading)	DDR5-6400	55 W	BGA 2114	800 Series
CUS 2	Arrow Lake (Series 2)	CPU: N3B GPU: N5 SOC/IO: N6	01/2025	L:H	4+8+2C+G+ NPU (no hyperthreading)	DDR5-6400 LPDDR5x-8400	28 W	BGA 2049	Integrated
CUS 2	Arrow Lake (Series 2)	CPU: Intel 3 GPU: N5 SOC/IO: N6	07/2025	L:UA	2+8+2C+G+ NPU (P: hyperthreading)	DDR5-6400 LPDDR5x-8400	15 W	BGA 2049	Integrated
CUS 3	Panther Lake (Series 3)	CPU: Intel 1.8A GPU: N3E SOC/IO: N6	01/2026	L:H	4+8+4+G	DDR5-7200 LPDDR5x-9600		BGA 2540	Integrated

The tiles with the specified IC technologies N3B, N3E N5, and N6 are manufactured by TSMC.
CUS: Core Ultra Series

2. Naming of Intel’s Core-based client processors

2. Naming of Intel's Core-base client processors

- Beginning with its 2nd gen. Sandy Bridge family (in 2011), Intel introduced a standard branding for its client, HEDT, and server processors, as follows:

  - Client processors are branded as Core i3/i5/i7/i9,
  - HEDT processors are branded as Core i7/i9 and
  - Server processors are branded as Xeon E3/E5/E7, where E refers to Enterprise.

- From these categories, we will now focus solely on Intel's naming conventions for client processors.

  However, Intel has adopted two distinct approaches to naming its processors:

  - The first naming convention was used through the 13th generation of Intel's Core processors.
  - Beginning with the 14th generation Meteor Lake client lineup, Intel introduced a new naming convention for its Core Ultra processor lines.

- Both naming schemes will be explained in the following sections.

Branding: Márkanév

a) Intel's original naming scheme for Core-based client processors (1-14 generations)

- From the Sandy Bridge line (2011) through to the 14th gen. Raptor Lake Refresh line (2023), Intel named its client processors by three name segments.
  - The first segments identifies the brand name, which is the "Intel Core" designation.
  - the next part is the brand modifier, typically i3/i5/i7/i9, followed by a hyphen.
  - The final segment consists of three parts, which are:
    - The generation identifier, it may consist of 1 or 2 digits (2-14) while
    - the model number (SKU (Stock Keeping Unit)) it is 3 digits long and
    - the product line suffix, which consists of one or two letters, and will be discussed in detail later.

- The Figure below presents a naming example; it refers to the 7th gen. i7-7500U (Kaby Lake DT) processor with the suffix U.

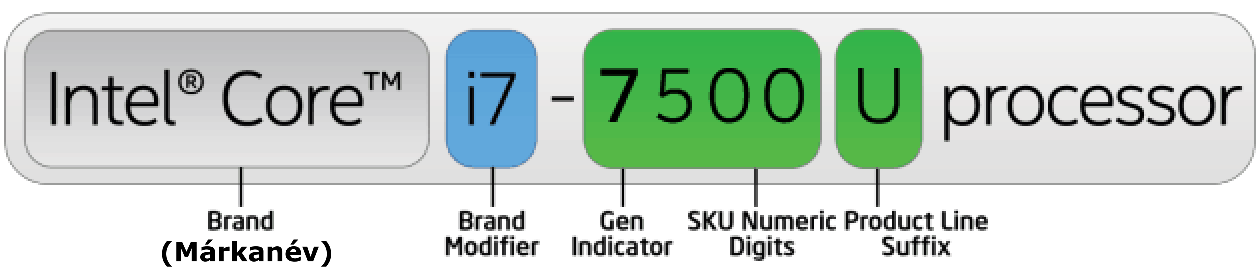


Figure: Intel's naming scheme used from the 2. up the 14. generations of the Core family (2011-2023) [43]

- The following Figures illustrate the brand modifiers, generation identifiers and model numbers of Intel's client processors across the 2nd to 14th generation Core processors.

2. Naming of Intel's Core-base client processors (3)

Brand modifiers, generation identifiers, and model numbers of Intel's Core-based client processors up to the 14 nm lines

1. gen.				2. gen.	3. gen.	4. gen.	5. gen.
Core 2	**Penryn**	**Nehalem**	**West-mere**	**Sandy Bridge**	**Ivy Bridge**	**Haswell**	**Broad-well**
New microarch.	**New process**	**New microarch.**	**New process**	**New microarch.**	**New process**	**New microarchi.**	**New process**
65 nm	**45 nm**	**45 nm**	**32 nm**	**32 nm**	**22 nm**	**22 nm**	**14 nm**
	TICK	***TOCK***	***TICK***	***TOCK***	***TICK***	***TOCK***	***TICK***
(2006)	*(2007)*	*(2008)*	*(2010)*	*(2011)*	*(2012)*	*(2013)*	*(2014)*
4/5/6xxx	6/7/8/9xxx	**i5/i7**-xxx	**i3/i5/i7**-xxx	**i3-i7**-2xxx	**i3-i7**-3xxx	**i3-i7**-4xxx	**i3-i7**-5xxx

6. gen.	7. gen.	8. gen.[1]	9. gen.	10. gen.
Skylake	**Kaby Lake**	**Kaby Lake R/G Coffee Lake Amber Lake-Y Whiskey Lake-U Cannon Lake (10 nm)**	**Coffee Lake R**	**Comet Lake**
New microarch.	**New microarch.**		**New microarch.**	**New microarch.**
14 nm	**14 nm**	**14 nm**	**14 nm**	**14 nm**
TOCK	***TOCK***	***TOCK***	***TOCK***	***TOCK***
(2015)	*(2016)*	*(2017/18)*	*(2018)*	*(2019)*
i3/i5/i7--6xxx	-7xxx	-8xxx (-10xxx for AL-Y R)	**i3-i9**-9xxx	**i3-i9**-10xxx

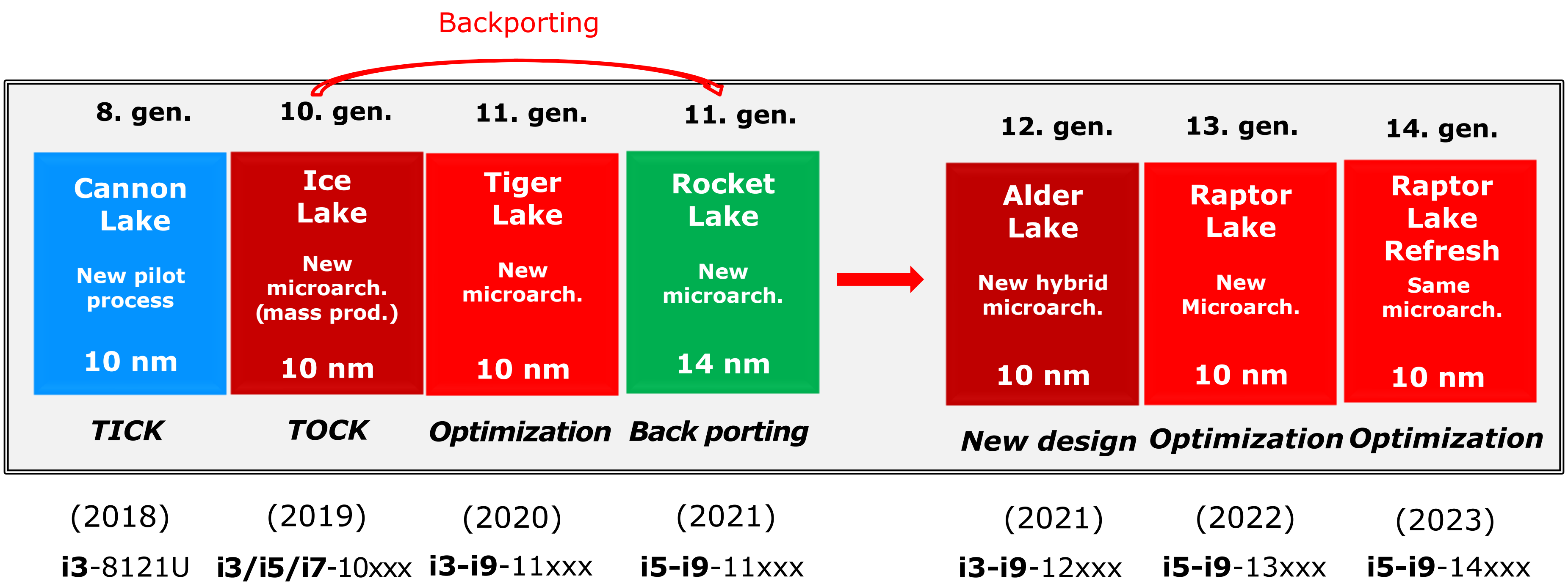
[1]The 8th generation encompasses the following processor lines:

- Kaby Lake Refresh
- Kaby Lake G with AMD Vega graphics
- Coffee Lake
- Amber Lake Y
- Whiskey Lake U
- (all 14 nm) and
- Cannon Lake (10 nm)

lines [29].

R: Refresh

Brand modifiers and model numbers of Intel’s Core-based 10-nm client processors



Remarks on discrepancies in the naming of Intel’s processor generations

- Note that Intel named both its 14 nm Comet Lake (2019) and the 10 nm Ice Lake (2019) processors as part of the 10th generation.
- Similarly, the company assigned the 11th generation label to both its 10 nm Tiger Lake (2020) and the 14 nm Rocket Lake (2021) lines.
- In addition, Intel has confusingly labeled both the Raptor Lake Refresh and Meteor Lake lines as 14th-generation (the latter not indicated in the Figure above).

2. Naming of Intel's Core-base client processors (5)

The product line suffix of Core-based client processors up to the 10 nm lines

- It is the final part of the third segment, it indicates both the target market sector and the associated TDP (Thermal Design Power) of the processor, as outlined below.

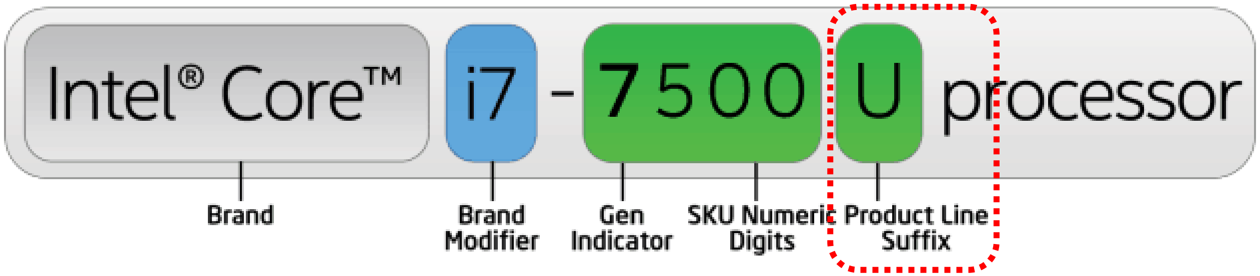


Figure: Naming scheme of client processors in Intel's Core family [43] (used from the 2nd gen. Sandy Bridge line until the 14th Core generation (2011-2023)

- Most frequently used client suffixes and related TDP values:

Mobile suffixes (SoC: System on Chip)

Y: ultra-low-power (11.5/13 W), when regular naming (i3/i5/i7-nxxxY) is used or extreme-low-power (4.5/5 W, when irregular naming m3/m5/m7-nYxx) is used
U: low power (typically 15/28 W)
H: high-performance laptop (typically 35/45 W)

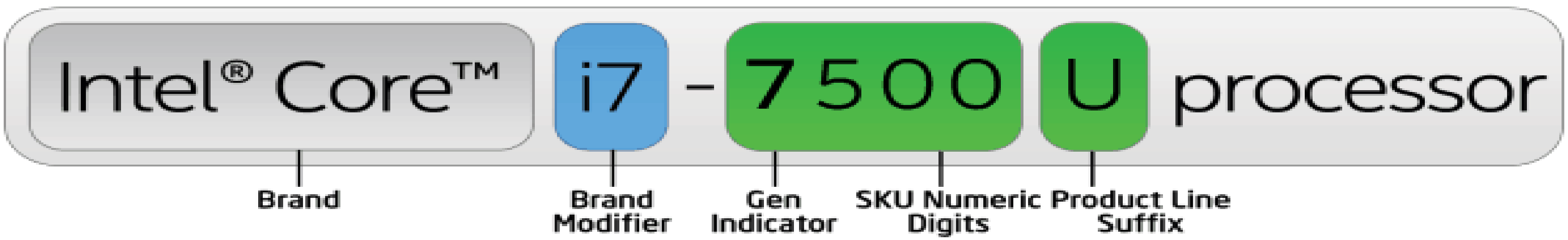
Desktop suffixes

T: power-optimized models (ultra-low power with 35/45 W TDP)
S: performance-optimized models (low power with 65 W TDP)
K: unlocked (adjustable clock multiplier (60-95 W)

b) Intel's new naming scheme for the Core Ultra series (introduced in 2023) [44]

- With the introduction of the 14th gen. Meteor Lake Refresh line, Intel revised its client processor naming scheme, replacing the format it had used for over 10 years.

For a comparison, we show both naming schemes, first the original format.



The new naming scheme is, as shown below:

- It consists further on of three parts:
  1) the brand name
  2) the brand level (called brand modifier previously) and
  3) the processor number

however, they have modified interpretations, which will be described in the following.

2. Naming of Intel’s Core-base client processors (7)

b1) Interpretation of the new brand names

There are two differences compared to the original naming scheme:

- Intel interprets now the original “Intel Core” brand as the mainstream-oriented brand,
- in addition, the company introduces a new – premium brand, and designates it as the “Intel Core Ultra” brand.

2. Naming of Intel’s Core-base client processors (8)

b2) Interpretation of the brand levels

- The brand levels (called previously brand modifier) refer further on to the performance level of the processor concerned.

  However, in the new naming system they no longer include the letter "i“- i.e. instead of i3, i5 etc. Intel uses now only 3, 5. etc.

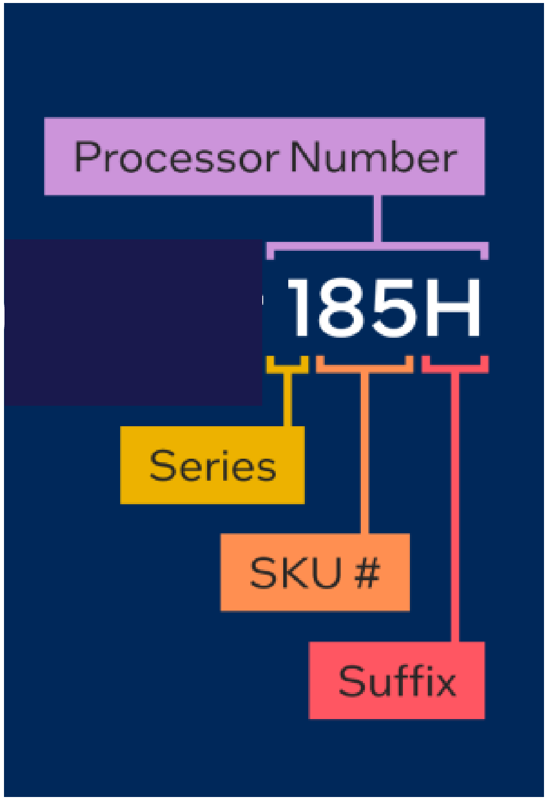
- Mainstream-oriented “Core processors” may have the brand levels 3/5/7, whereas
- Premium “Core Ultra processors” may have the brand levels 5/7/9.

2. Naming of Intel’s Core-base client processors (9)

b3) Interpretation of the processor numbers [45]

The 3rd naming segments, called Processor numbers, continue to have three parts, as shown to the right.

These are: the Series identifier, the model number (SKU: Stock Keeping Unit) and the Suffix, as detailed next.



Series identifier: It disrupts the initially introduced generation numbering, instead, it introduces a new Series sequencing, beginning with Meteor Lake as Series 1.

SKU#: Model number; it only includes two digits.

Suffix: It relates to the market segment and the base TDP value, as before.

Most frequently used client suffixes and related TDP values:

Mobile suffixes

- Y: ultra-low-power (9 W
- U: low power (typically 15 W)
- V: low-power (typically 17/30 W)
- H: high-performance laptop (typically 28/45 W)
- HX: extreme performance (typically 65 W)

Desktop suffixes

- T: power-optimized models (35 W)
- K: unlocked (adjustable clock multiplier (125 W)
- KF: unlocked without a GPU (adjustable clock multiplier (125 W)

2. Naming of Intel's Core-base client processors (10)

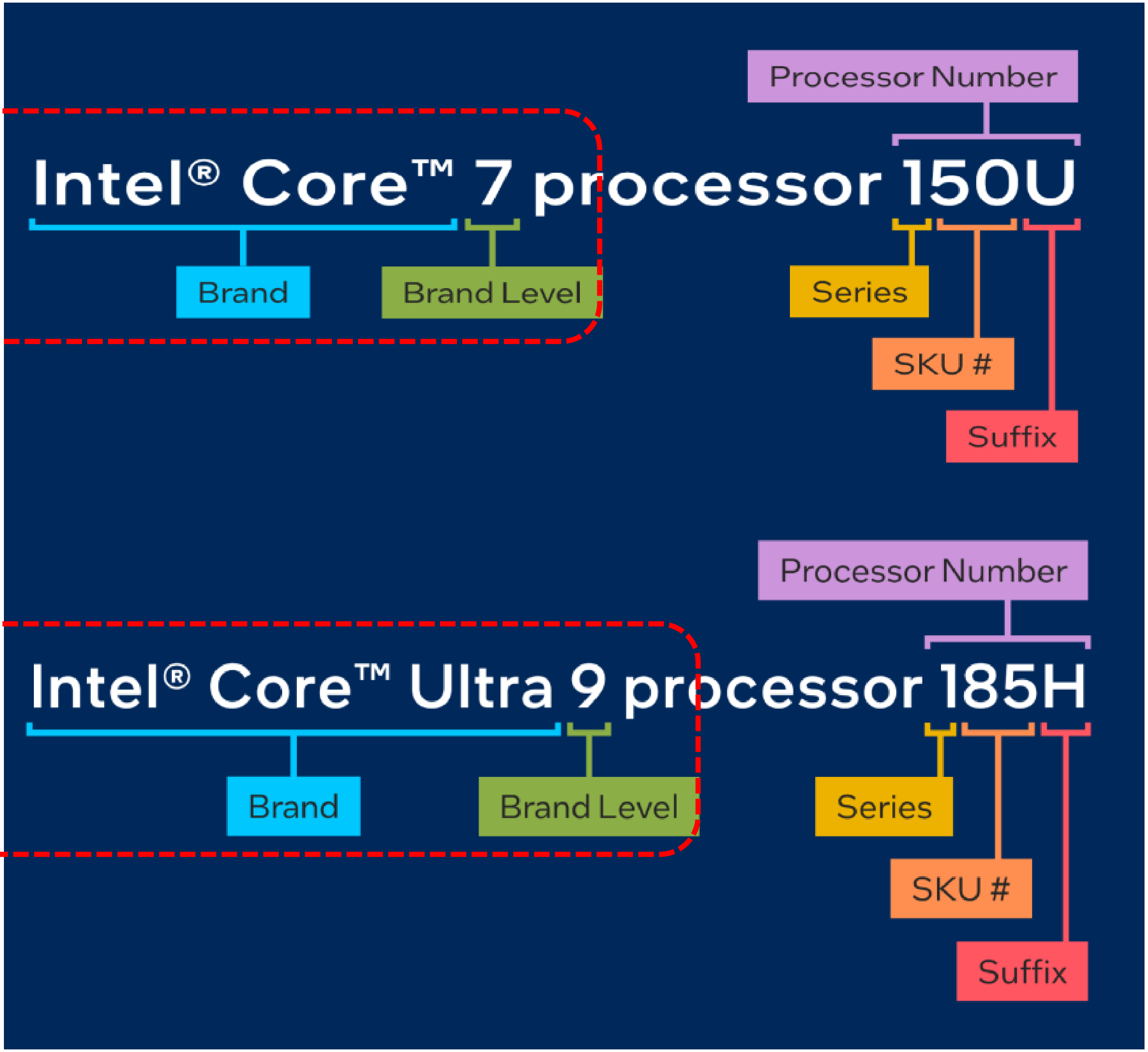
Remark

It is worth noting that the new TDP values are generally similar to those of previous ones, however, the HX and K/KF suffixes correspond to higher TDP ratings than before, reaching up to 15/28 W and 125 W respectively.

2. Naming of Intel's Core-base client processors (11)

Mainstream processors

Premium processors



Raptor Lake Refresh
U series 01/2124

2. Naming of Intel's Core-base client processors (12)

Using the new naming scheme of client processors beginning with the 14th Core generation (called Meteor Lake)

Multi-die heterogeneous big. little designs

(14th gen.) Core Ultra Series 1	Core Ultra Series 2	Core Ultra Series 2	Core Series 2	Core Ultra Series 3
Meteor Lake	**Lunar Lake**	**Arrow Lake**	**Meteor Lake R.**	**Panther Lake**
New process and μarch.	**New process and μarch.**	**New μarch.**	**New μarch.**	**New process new μarch.**
Intel 4 TSMC N5/N6	**TSMC N3/N6**	**TSMC N3/N5/N6**	**Intel 4 TSMC N5/N6**	**Intel 1.8A/3 TSMC N3/N6**
4 tiles	2 tiles	4 tiles	4 tiles	***3 tiles***
(12/2023) 100U/H	(09/2024) 200V	(10/2024) 200K (01/2025) 200H/HX/T	(01/2025) 200U	(01/2026) 300H
(Laptops)	(Low-Power Laptops)	(Laptops/ DTs)	(Laptops)	(Laptops)

Remark

There is an inconsistency in Intel's naming scheme, as follows:

In 01/2014, Intel introduced the Raptor Lake Refresh U-series DT processors and retroactively applied its new naming scheme to it, labelling it as Core Ultra Series 1 .

Previous and recent badges of Intel's Core processors [45], [44]

Intel®	Intel® Core™ Ultra	Intel® Core™ Processors

The previous badge

Renewed badges (since the 14th gen.)

Badge: jelkép

3. Key enhancements in the microarchitecture of Intel's Core-based client processors over time

- 3.1 Overview of the key enhancements of the microarchitecture
- 3.2 Evolution of the CPU core configuration and introduction of multi-die designs
- 3.3 How are the cores of big.little architecutures scheduled?
- 3.4 How are Intel's recent multicore processors packaged?
- 3.5 How has the width of the CPU cores increased?
- 3.6 How has the core count in Intel's client CPUs increased?
- 3.7 How has the memory subsystem been enhanced?
- 3.8 Basics of 3D graphics processing

3.1 Overview of the key enhancements in the microarchitecture (1)

3. Key enhancements in the microarchitecture of Intel's Core-based client processors

3.1 Overview of the key microarchitectural enhancements

Processor microarchitectures can be enhanced in two different aspects, as shown below:

Key microarchitectural enhancements in Intel's Core-based client processors

- **Key enhancements made to increase processor performance**
- **Key enhancements made to improve power management** (Not discussed)

- For processor performance, CPU cores play a crucial role, particularly regarding single-core or single-thread performance.
- For this reason, it is appropriate to discuss improvements in CPU core designs separately from those made in other processor components, as indicated below.

Key enhancements made to increase processor performance

- **Key microarchitectural enhancements made to the CPU cores**
- **Further microarchitectural enhancements made to the processor**

- The following Sections outline these improvements made to bosting processor performance.

3.1 Overview of the key enhancements in the microarchitecture (2)

What key microarchitectural enhancements have been made to CPU cores?

- *Increasing the CPU core width (from 4 to 9)*
- *Increasing the back-end width of CPU cores (from 5 to 17)*
- *Shared L3 cache with private per-core L2 caches*
- *Segmented L3 cache*
- *Hyperthreading*
- *Widening the SIMD execution width (initially to 128-bit, then to 256-bit and finally to 512-bit).*

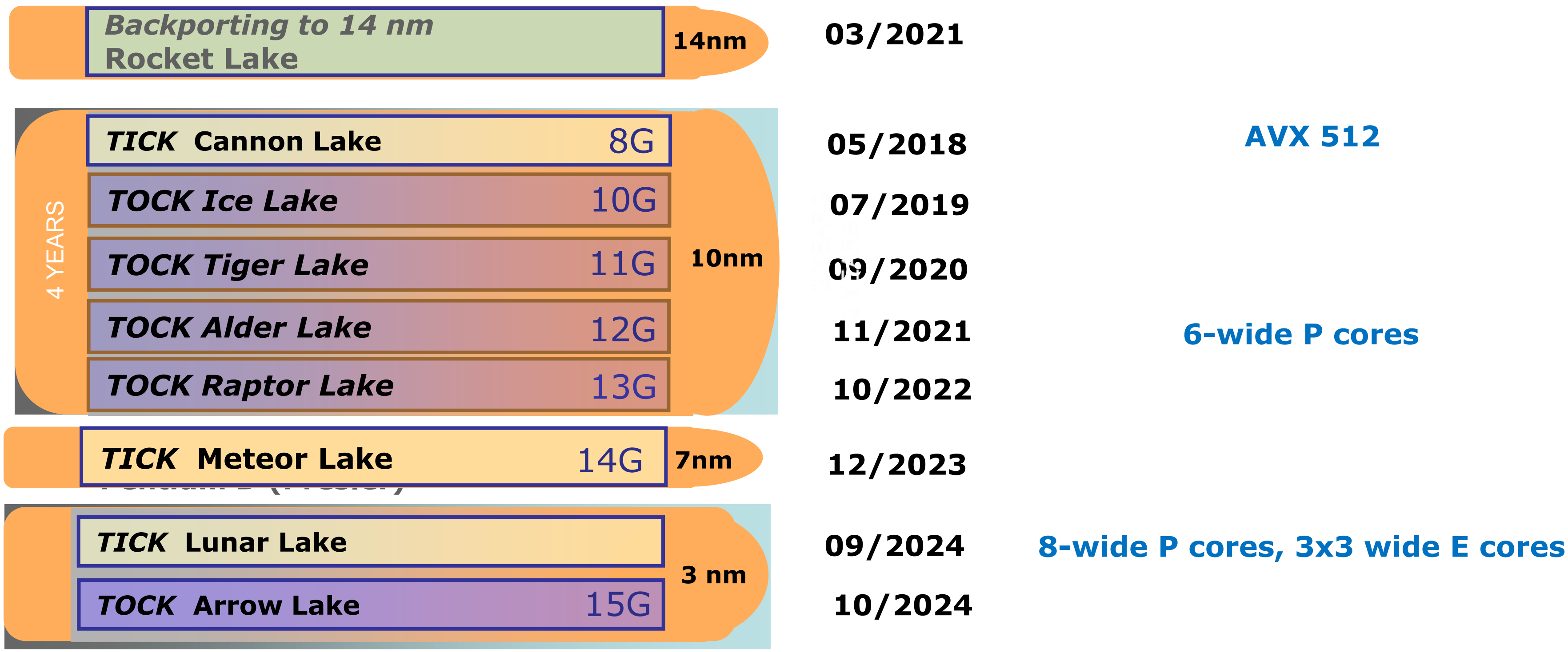
The next two Figures show when Intel introduced the above-mentioned enhancements.

3.1 Overview of the key enhancements in the microarchitecture (3)

Key microarchitectural enhancements of the CPU cores -1

Period		Core	Gen.	Process	Date	Key enhancements
2 YEARS	*TICK*	Pentium 4 (Cedar Mill) Pentium D (Presler)		65nm	01/2006	
	TOCK	Core 2	1G	65nm	07/2006	4-wide core, 128-bit SIMD FX/FP EUs, shared L2,
2 YEARS	*TICK*	Penryn Family		45nm	11/2007	
	TOCK	Nehalem		45nm	11/2008	private L2, monolithic shared L3, HT
2 YEARS	*TICK*	Westmere		32nm	01/2010	
	TOCK	Sandy Bridge	2G	32nm	01/2011	256-bit (FP) AVX, segmented L3, integrated GPU
2 YEARS	*TICK*	Ivy Bridge	3G	22nm	04/2012	
	TOCK	Haswell	4G	22nm	06/2013	256-bit (FX/FP) AVX2
6 YEARS	*TICK*	Broadwell	5G	14nm	09/2014	
	TOCK	*Skylake*	6G	14nm	10/2015	
	TOCK	*Kaby Lake*	7G	14nm	08/2016	
	TOCK	*Kaby Lake Refresh*	8G	14nm	08/2017	
	TOCK	*Coffee Lake*	8G	14nm	10/2017	
	TOCK	*Amber/Whismain Lake*	8G	14nm	08/2018	
	TOCK	*Coffee Lake Refresh*	9G	14nm	10/2018	
	TOCK	*Comet Lake*	10G	14nm	08/2019	

Key microarchitectural enhancements of the CPU cores -2



These enhancements focus on boosting performance.

3.1 Overview of the key enhancements in the microarchitecture (5)

What further microarchitectural enhancements have been made to the processors?

- *Evolution of the CPU core configuration*
- *Increasing the core count (from 2 up to 24 in laptops/DTs*
- *Introducing the multi-die (heterogeneous) approach*
- *Ring bus interconnect*
- *Integrated memory controller (NUMA architecture)*
- *Integrated PCI controller*
- *Integrated ISP (Signal Processor)*
- *Integrated GPU*
- *Integrated FIVR (Fully Integrated Voltage Regulator)*
- *Integrated NPU) Neural Processing Units)*
- *Enhancing the memory subsystem*
- *Enhancing the PCIe subsystem*

The next two Figures show when Intel introduced the above-mentioned enhancements.

3.1 Overview of the key enhancements in the microarchitecture (6)

Introducing further key microarchitectural enhancements to the processors -1

Period	Step	Processor	Gen.	Process	Date	Key microarchitectural enhancements
2 YEARS	*TICK*	Pentium 4 (Cedar Mill) Pentium D (Presler)		65nm	01/2006	2 cores, symmetrical multicores
	TOCK	Core 2	1G		07/2006	
2 YEARS	*TICK*	Penryn Family		45nm	11/2007	
	TOCK	Nehalem			11/2008	4 cores, integr. MC, integr. PCI,
2 YEARS	*TICK*	Westmere		32nm	01/2010	In package integrated GPU
	TOCK	Sandy Bridge	2G		01/2011	Ring bus, integrated GPU
2 YEARS	*TICK*	Ivy Bridge	3G	22nm	04/2012	
	TOCK	Haswell	4G		06/2013	FIVR
6 YEARS	*TICK*	Broadwell	5G	14nm	09/2014	
	TOCK	*Skylake*	6G		10/2015	ISP, no FIVR
	TOCK	*Kaby Lake*	7G		08/2016	In KBL G series: in package integr. CPU, GPU, HBM2
	TOCK	*Kaby Lake Refresh*	8G		08/2017	
	TOCK	*Coffee Lake*	8G		10/2017	6C
	TOCK	*Amber/Whismain Lake*	8G		08/2018	
	TOCK	*Coffee Lake Refresh*	9G		10/2018	8C
	TOCK	*Comet Lake*	10G		08/2019	10C

3.1 Overview of the key enhancements in the microarchitecture (7)

Introducing further key microarchitectural enhancements to the processors -2

Backporting to 14 nm Rocket Lake		14nm	03/2021	
TICK Cannon Lake	8G	10nm (4 YEARS)	05/2018	
TOCK *Ice Lake*	10G		07/2019	FIVR in the PCH
TOCK *Tiger Lake*	11G		09/2020	
TOCK *Alder Lake*	12G		11/2021	(big.little architecture)* 8+8 cores
TOCK *Raptor Lake*	13G		10/2022	8+16 cores
TICK Meteor Lake	14G	7nm	12/2023	(Heterogeneous multi-die big-little architecture)* NPU
TICK Lunar Lake		3 nm	09/2024	On-chip LPDDR5X
TOCK Arrow Lake	15G		10/2024	

*These enhancements focus on other aspects than boosting performance.

Particularly, big.little architectures aim to reduce power consumption, while multi-die (heterogeneous) implementations seek to lower manufacturing costs.

Evolution of selected microarchitectural features in Intel's Core-based client processors

- It would be tempting to trace the dynamic evolution of the microarchitecture in the Core family across all the enhancements mentioned, but space constraints make that impractical.
- Instead, we'll focus on a few key enhancements, which highlight the major trends in CPU core design.

The selected design aspects discussed in the following Sections are:

3.2 Evolution of the CPU core configuration and introduction of multi-die designs
3.3 How are the cores of big.little architecutures scheduled?
3.4 How are Intel's recent multicore processors packaged?
3.5 How has the CPU core width increased?
3.6 How has the core count in Intel's client CPUs increased?
3.7 How has the memory subsystem been enhanced? and
3.8 Basics of 3D graphics processing.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs

3.2 Evolution of the CPU core configuration and introduction of multi-die designs -1

The CPU core configurations and the multi-die approaches have evolved in Intel's Core family through three consecutive steps, as shown below.

Evolution of CPU core configurations in Intel's Core-based client processors

Symmetrical multicore designs	Single-die big.little designs	Heterogeneous multi-die big.little designs
All cores participate equally and concurrently in executing workloads, this results in parallel processing.	There is a mix of high-performance but power-hungry big cores, and lower-performance more energy-efficient little cores available. The OS assigns less demanding tasks to the little cores, conserving power, while assigning the big cores for more-intensive workloads.	Such processors are divided into multiple parts, mplemented on multiple dies, called tiles. This results in higher manufacturing yield. Each tile can be fabricated using a suitable IC technology, which may be either more advanced but costly or less advanced and less costly.
It aims to increase performance.	reduce power consumption.	increase yield and provide a more efficient design with lower fabrication costs.
First implemented in: *Pentium Extreme Edition 840 (2005)*	*12th gen. Alder Lake (2021)*	*14th gen. Meteor Lake (2023)*

Evolution of the CPU core configuration and introduction of multi-die designs -2

The above-mentioned evolution steps will be discussed next.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (3)

a) The 1st evolution step: symmetrical multicore designs (2006-2021)

- From the very beginning up to the 11th generation Core processors, which were released in 2021, Intel's CPUs were designed as symmetrical multicores.

  In these designs, all cores participate equally and concurrently in executing workloads.
- Obviously, the primary goal of symmetrical multicores is to increase performance through parallel processing.

Intel's first symmetrical multicore design within the Core family: the Core 2 (released 2006)

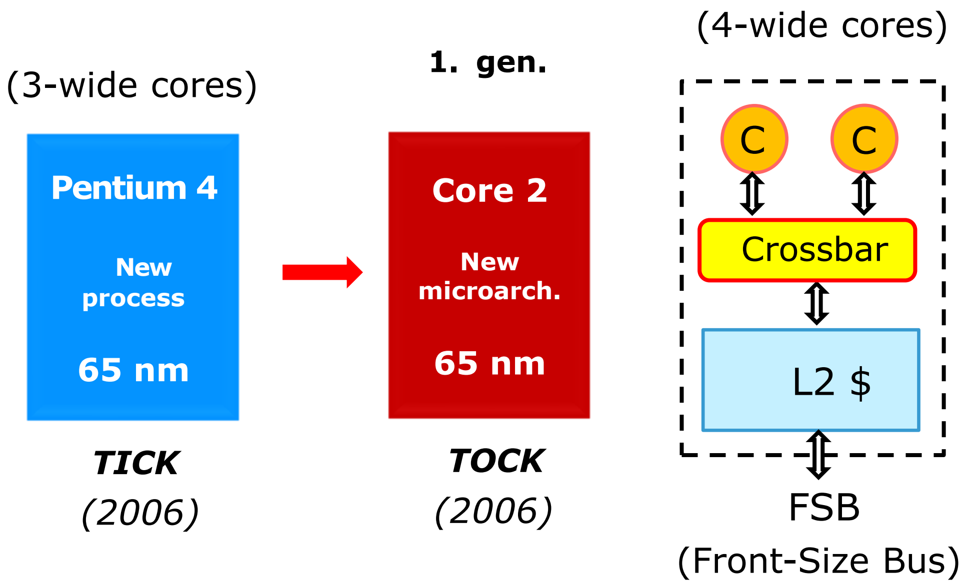
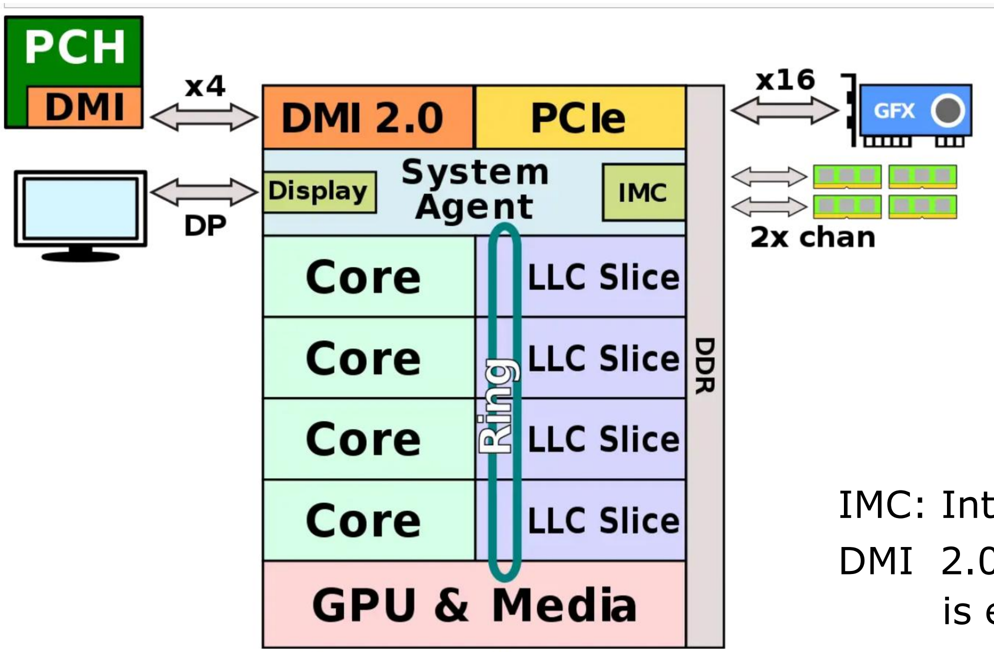


Figure: simplified block diagram of Intel's Core 2 processor

- The Core 2 was a 4-wide superscalar design, including two cores which were connected via a crossbar switch to the 1 MB shared L2 cache.
- The L2 cache was linked to the system bus, known as the Front-Side Bus (FSB) through a bus unit (which is not shown in the Figure above).
- The FSB provided the connection to the chipset, which comprised the North Bridge (NB) and the South Bridge (SB) (although these components are not indicated).

Example of Intel's advanced symmetrical multicore designs – the microarchitecture of the 2nd gen. Sandy Bridge processor [46]



IMC: Integrated Memory Controller

DMI 2.0 (Direct Media Interface) is essentially a PCIe 2.0 link.

- 2. gen. Core processors became ring bus-based and included a segmented L3 cache.
- These designs introduced two key components: a GPU + MM and a System Agent (SA).
- The SA is responsible for providing memory accesses and handling PCIe-based connections to
  - the external graphics (GFX) through x16 PCIe 3.0 links,
  - the Platform Control Hub (PCH) via an x4 DMI link and
  - the display using a Display Port (DP) link.

The last symmetrical multicore designs – Intel's 10 nm Tiger Lake processors

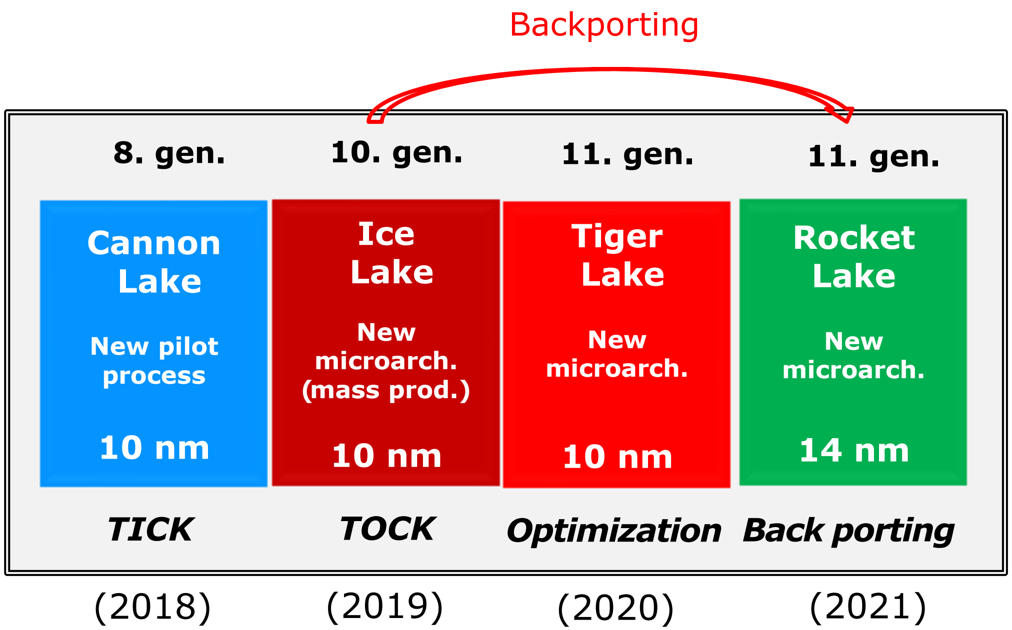
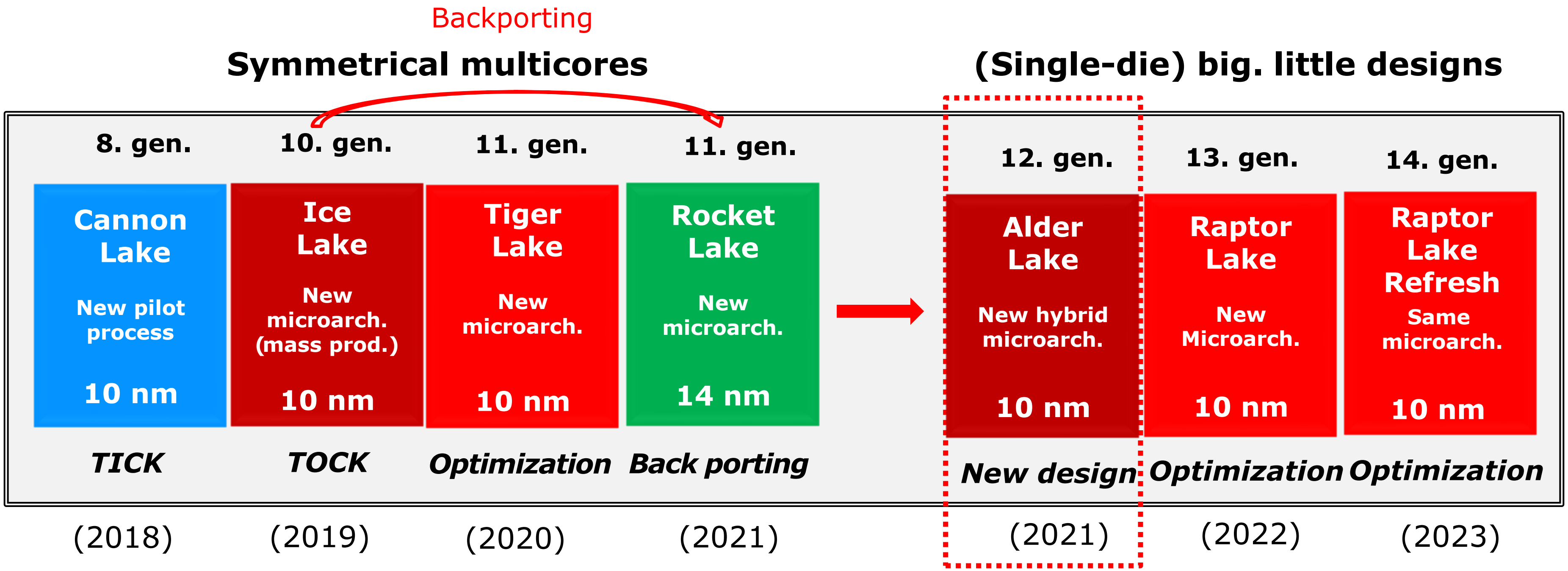


Figure: Intel's last symmetrical multicore designs implemented using the 10 nm process

- Following Ice Lake, Intel launched the performance-optimized 11th gen. Tiger Lake line.
- Next, Intel released the Rocket Lake design, which is a backported implementation of the 10th gen. 10 nm Ice Lake line and also got the 11. gen. label.
- Backporting refers to the implementation of a given microarchitecture on an older, less advanced technology, typically to reduce manufacturing costs.
- Note that confusingly, Intel labeled both the 10 nm Ice Lake and the 14 nm Rocket Lake processors are 11th-generation products.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (7)

b) The 2nd evolution step: introducing big.little processor designs -1



- With the introduction of the 12th gen. Alder Lake lineup, Intel transitioned to big.little (also known as hybrid) architectures in 2021 to reduce power consumption.
- Intel's big.little designs incorporate high-performance (P) cores and high-efficiency (E) cores (as detailed next).

Principle of big.little architectures at the time of their introduction by ARM in 2012 (assuming the use of L2 caches and crossbar interconnects)

Use of homogeneous cores or big and little cores in multicore CPUs

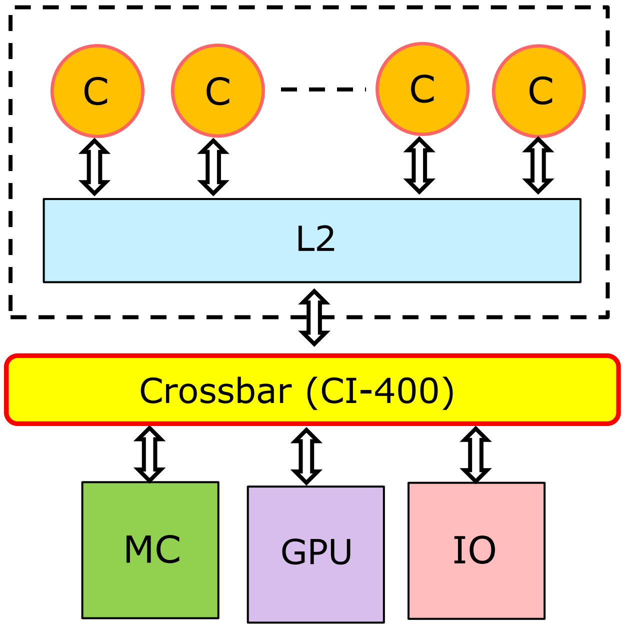
Symmetric multicores

They include homogeneous cores..

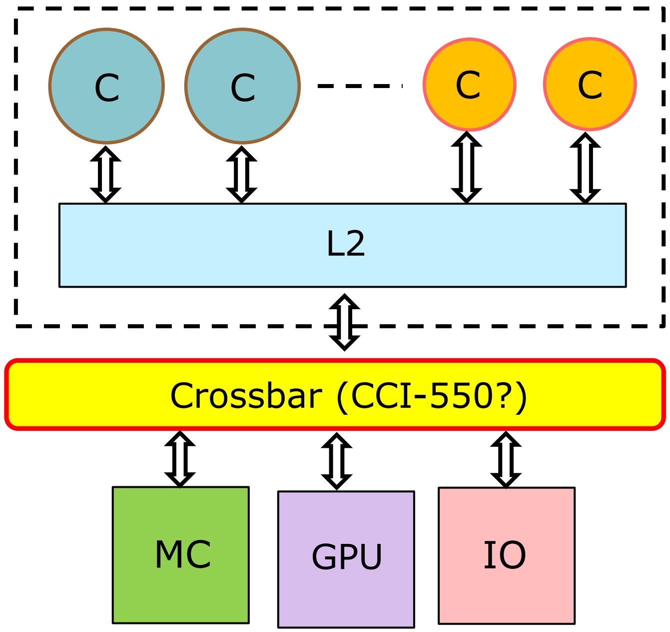
big.little multicores

They include hybrid cores:
higher performance/higher power consuming big cores,
and lower performance/lower power consuming little cores.

The OS allocates less demanding tasks to less power-hungry little cores, thereby reducing the overall power consumption.



Symmetrical multicore



big.little multicore

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (9)

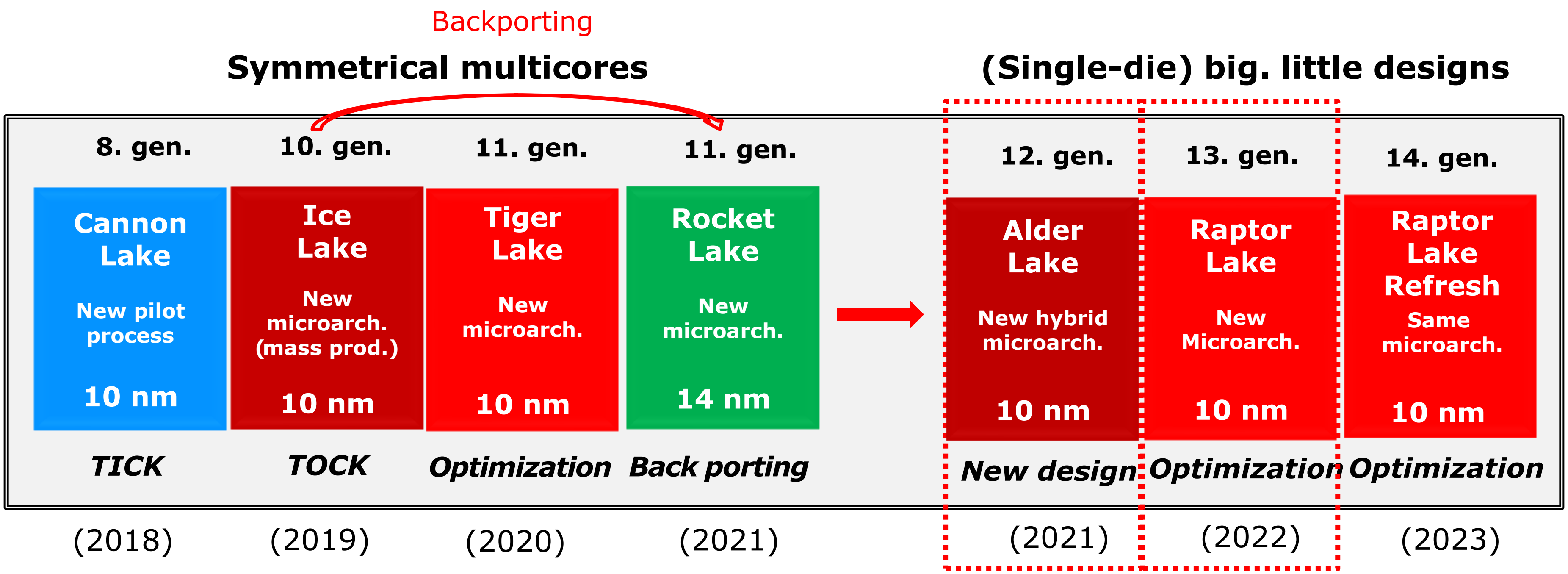
Comment on the terms "hybrid" and "heterogeneous"

- In the previous Figure, big.little architectures were classified as hybrid designs. However, they are sometimes referred to as heterogeneous designs as well.
- Both terms are closely related but used in slightly different contexts.

- The term "hybrid" is more commonly applied to CPU designs, emphasizing that the CPU consists of different types of cores, which have different characteristics, e.g., there exist both high-performance and high-efficiency cores in the CPU.
- Relating processor designs, the term "heterogeneous" typically denotes a microarchitecture that incorporates different types of execution units, such as CPUs, GPUs, NPUs, etc.
- In multi-die designs, the term "heterogeneous" refers to implementations using different types of dies rather than identical ones.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (10)

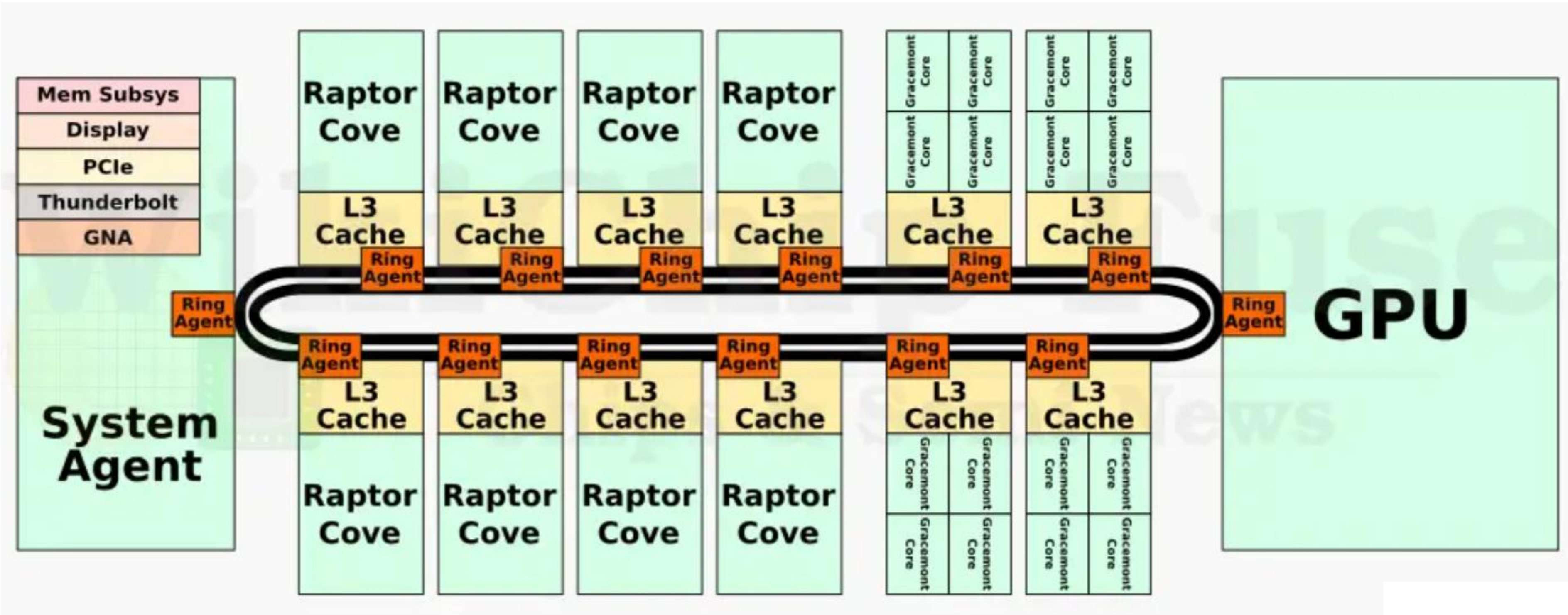
The 2nd evolution step: introducing big.little processor designs -2



- 12th gen. Alder Lake processors, in particular, include up to 8 P (Performance) cores and up to 8 E (Efficiency) cores.
- In contrast, the upcoming 13. gen. Raptor Lake line enhances this design by providing up to 8 P cores and up to 16 E cores.

We note that Intel's server processors did not transition from symmetrical multicores to big.little designs, since server workloads often demand consistent performance across the cores, e.g., database servers.

Example: Layout of the 13th gen. Raptor Lake big.little processor featuring 8 big and 16 little cores (2022) [48]



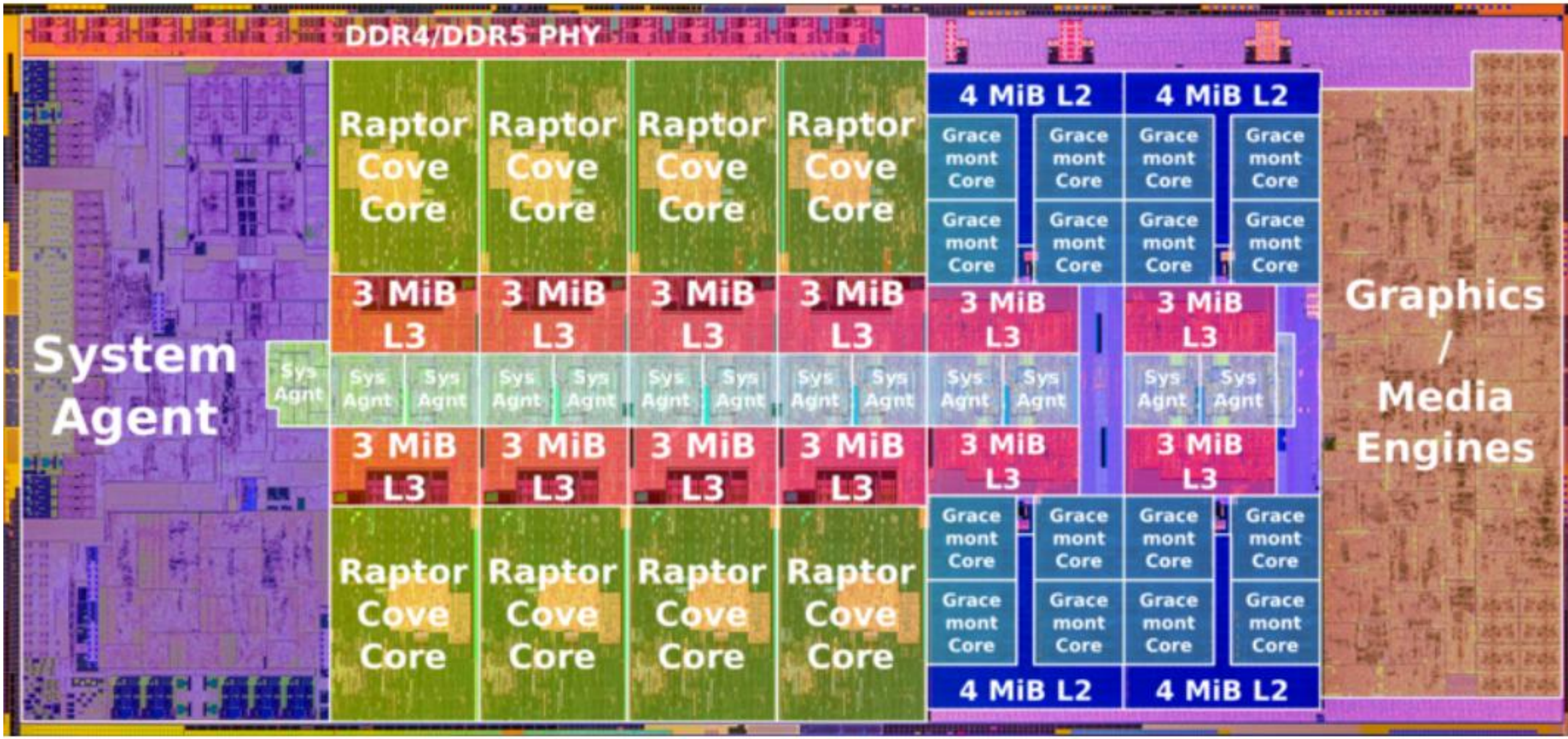
Note that the little cores are grouped into core blocks.

- There are four core blocks in total, each containing 4-cores.
- Each core block includes a shared L2 cache (not shown here, but illustrated in the next Figure), and an L3 cache segment.

GNA: Gaussian and Neural Accelerator

Thunderbolt: High-speed connectivity technology

Floor plan of the 13th gen. Raptor Lake big.little processor faturing 8 big and 16 little cores (2022) [49]



The die size is ~ 215 mm^2

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (13)

Remarks on Intel's first big.little processor implementation

- It's important to clarify that Intel's Alder Lake line is not its first hybrid (big.little) design.

  Prior to Alder Lake, Intel introduced the Lakefield processor in 2020, featuring one big and four little cores.

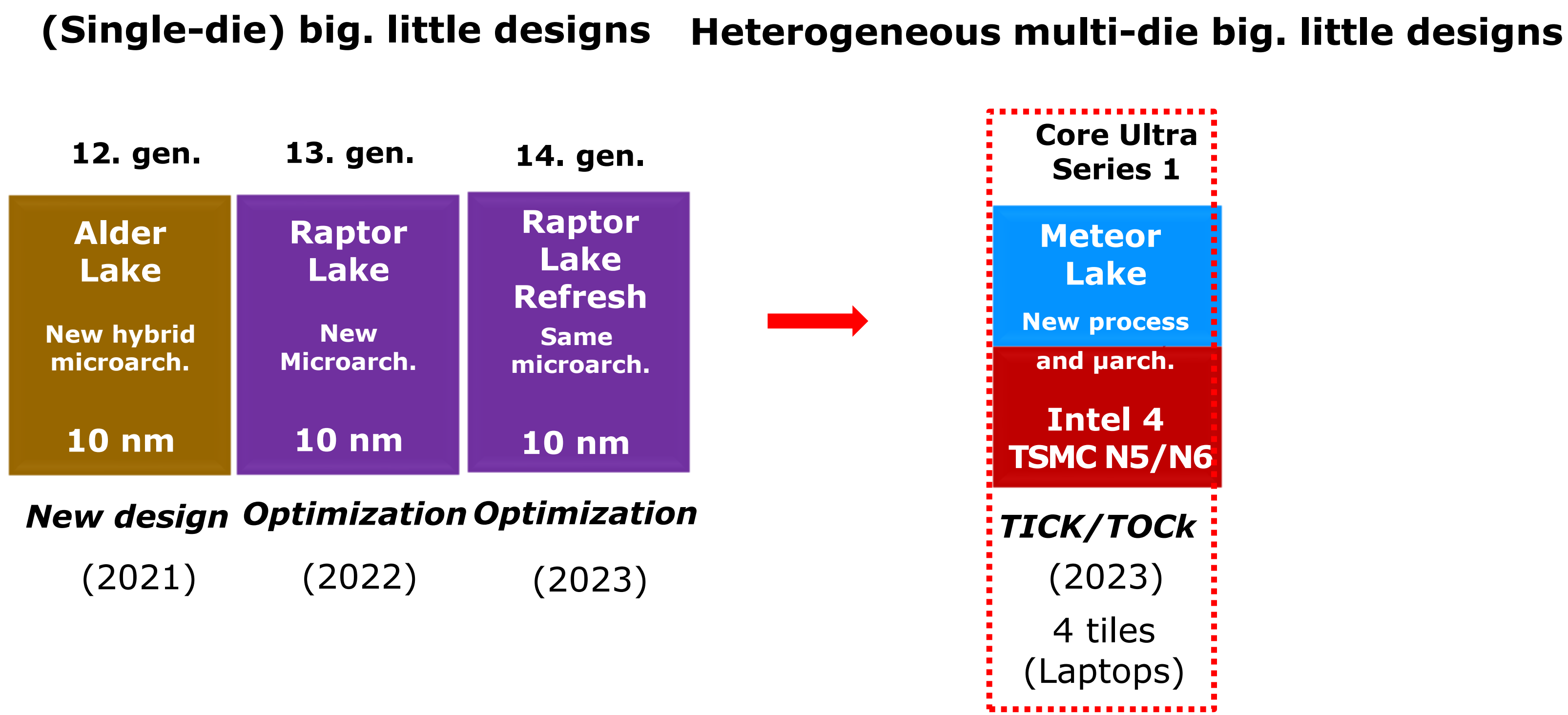
  Additionally, the Lakefield processor employed a dual-die implementation, rather than a single-die design (22 nm Base die/10 nm Compute die).

- However, this processor was not widely adopted, and did not receive a formal identifier.

  As such, Lakefield is best regarded as a pilot project – a preliminary step toward Intel's hybrid (big.little) designs.

c) The 3rd evolution step: introducing heterogeneous multi-die big.little designs -1

- In 2023, Intel began to implement big.little client processors on multiple dies, known as heterogeneous multi-die designs alongside their 14th gen. Meteor Lake lineup, as indicated below.



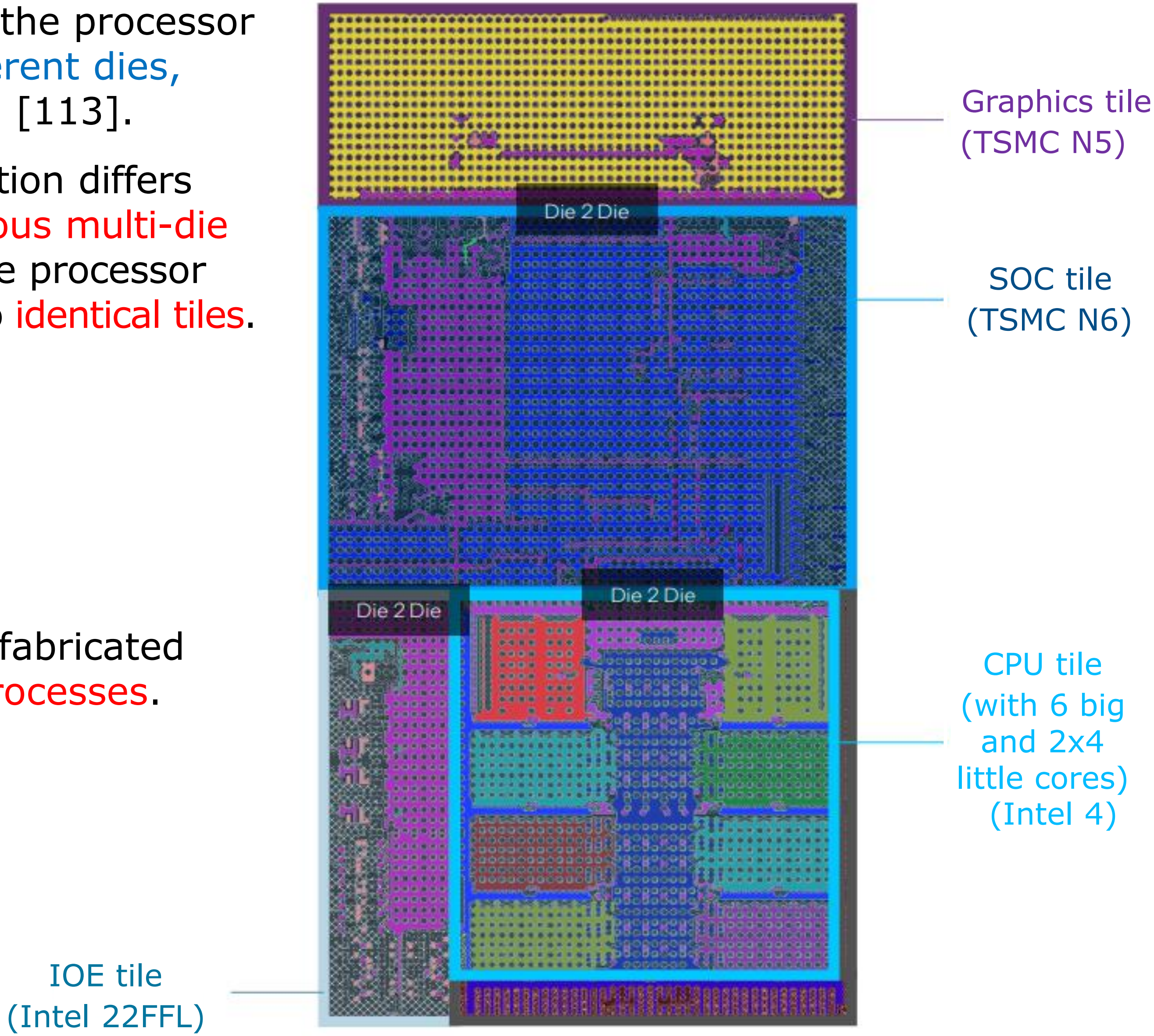
- In heterogeneous multi-die designs, the processor is divided into multiple non-identical components, known as tiles.

  These tiles are then interconnected using high-speed interfaces.

- Refer to the Meteor Lake design in the following Figure.

The tiles make up the 14th. gen. heterogeneous multi-die big.little Meteor Lake design

- As the Figure shows, the processor consists of four different dies, referred to as "tiles" [113].
- This type of segmentation differs from the homogeneous multi-die approach, where the processor die is segmented into identical tiles.
- The four tiles are:
  - CPU tile
  - Graphics tile
  - SOC tile
  - IOE tile
- Note that each tile is fabricated using different IO processes.



Benefits of heterogeneous multi-die designs compared to single-die (monolithic) designs

- Heterogeneous multi-die designs offer two key benefits.

  a) the advantage of using multi-die designs over single-die designs,
     as well as
  b) the additional benefit of using heterogeneous (multi-die) designs over single-die designs.

- Subsequently, we will discuss both advantages of heterogeneous multi-die designs separately.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (17)

a) The benefit of multi-die designs over single-die designs -1

- Enlarging dies increases the likelihood of defects, which in turn reduces the yield.

  Typically, doubling the die size will cut the yield in half.

  A lower yield, on the other hand, will lead to higher selling prices.

- The multi-die approach, which involves dividing a die into multiple smaller dies, can help improve the overall yield, as illustrated in the following Figure.

Example of increasing the yield by dividing dies into multiple smaller ones [143]

- In the Figure below, each large die is divided into 4 smaller dies, and there are 7 error spots.

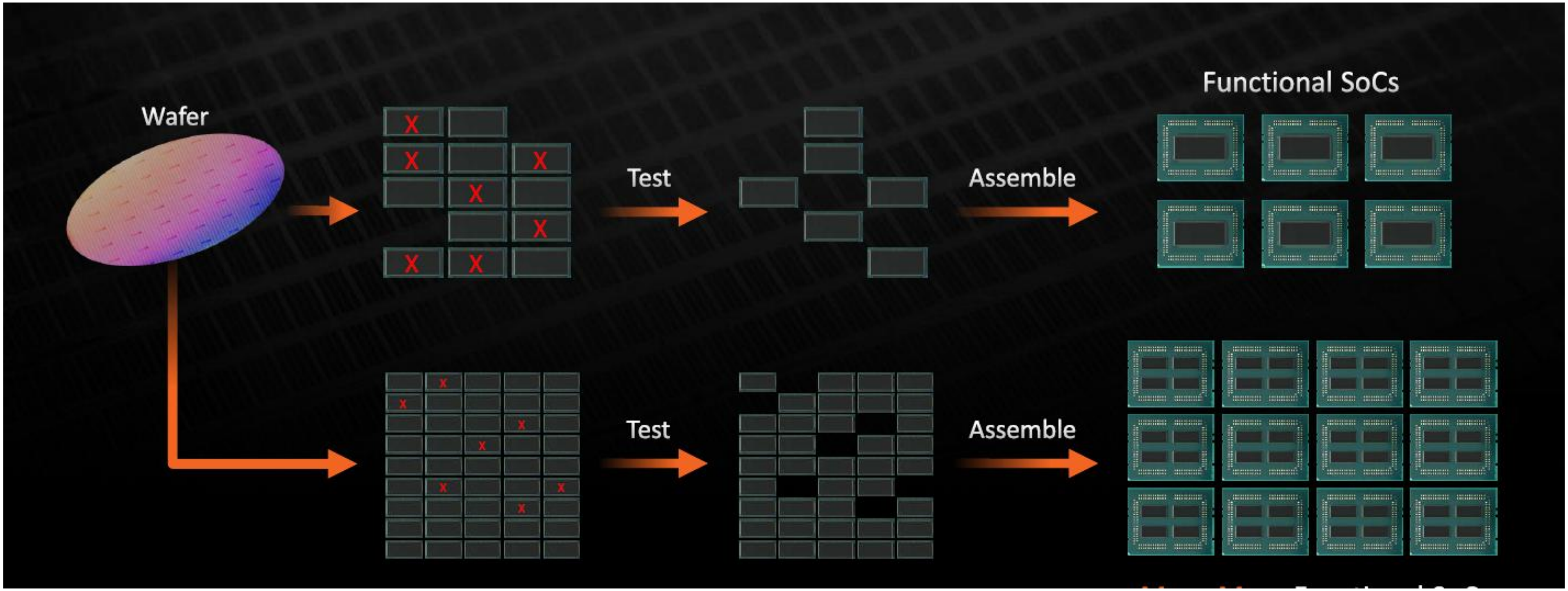


Figure: increasing the yield by dividing dies into multiple smaller ones [143]

- Without segmentation, 7 large dies become defective.
- However, after segmentation, only 7 smaller dies became faulty.

  Thus, twice as many large dies remain error-free, effectively doubling the yield.
- This benefit applies to both homogeneous and heterogeneous multi-die segmentations.

Drawbacks of the multi-die approach

When using the multi-die approach, two key requirements arise:

a) The dies need to be linked together by a proper cache coherent interconnect fabric.

The necessary interconnections increase transfer latencies, which degrade performance and incur additional power consumption due to the required communication.

b) Multiple dies must be packed together using a suitable packaging technology.

This point is discussed in Section 3.5 (How are Intel's recent multicore processors packaged?)

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (20)

b) The additional benefit of using heterogeneous multi-die designs over single-die designs

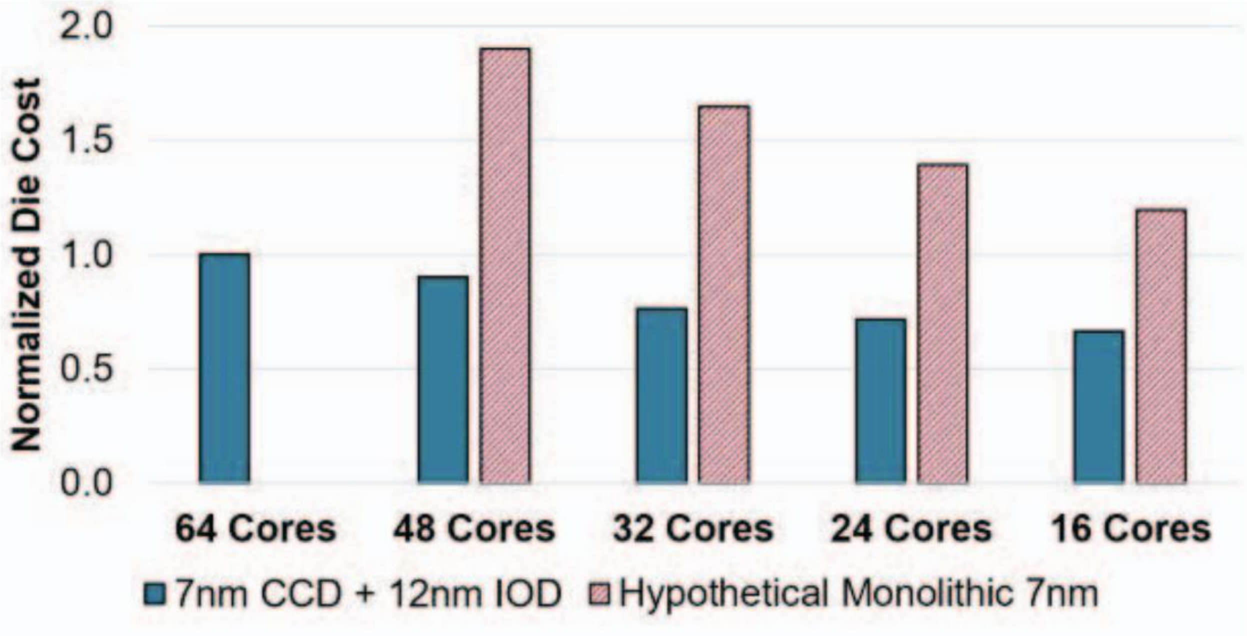
- In a heterogeneous multi-die design, a processor is built up of multiple dies, each with specific capabilities, such as computing or graphics.
- Then, different dies can be fabricated using different IC processes.
- This is crucial because not every die needs the most advanced and costly technology.

  For instance, CPU tiles require a cutting-edge technology, since they determine the processor's performance.

  In contrast, GPU dies, and first of all, IO dies can be manufactured using less advanced, more cost-effective processes without notably affecting overall system performance.
- Accordingly, by manufacturing specific die types using appropriate IC technologies — such as fabricating CPU cores by cutting-edge processes, while fabricating I/O dies with less advanced technologies — the overall manufacturing cost can be reduced compared to single die processors.

- Note that using less advanced but less costly processes for less demanding dies to reduce manufacturing costs follows the same principle as using less performant but less power-hungry cores for less-demanding workloads to reduce power consumption.

Example showing how much Zen 2 server die costs can be lowered by using heterogeneous design compared to a hypothetical monolithic layout [222]



- In the example provided, server dies are composed of CCD compute dies and an IO die, which are manufactured using 7 nm and 12 nm processes, respectively.
- As shown in the Figure, the heterogeneous design significantly reduces the die cost compared to a hypothetical monolithic design.

Remark

We note that using heterogeneous multi-die segmentation for processors is clearly more cost-effective than employing homogeneous multi-die processors, which consist of multiple dies with the same functionality, as seen in certain high-core-count Intel processors.

This is because in homogeneous multi-die processors, all dies must be manufactured using the same demanding and costly technology.

Block diagram of the 14th gen. heterogeneous multi-die big.little Meteor Lake processor

This architecture resembles that of the 13th gen. Raptor Lake processor, discussed previously, but it has been implemented across four tiles, as shown below.

M: Media Engine
IOC: IO Control Hub
PMC: Central Power Management Controller
NOC: Network on Chip
C: Additional cores to be able to switch off the CPU tile for very low CPU demand

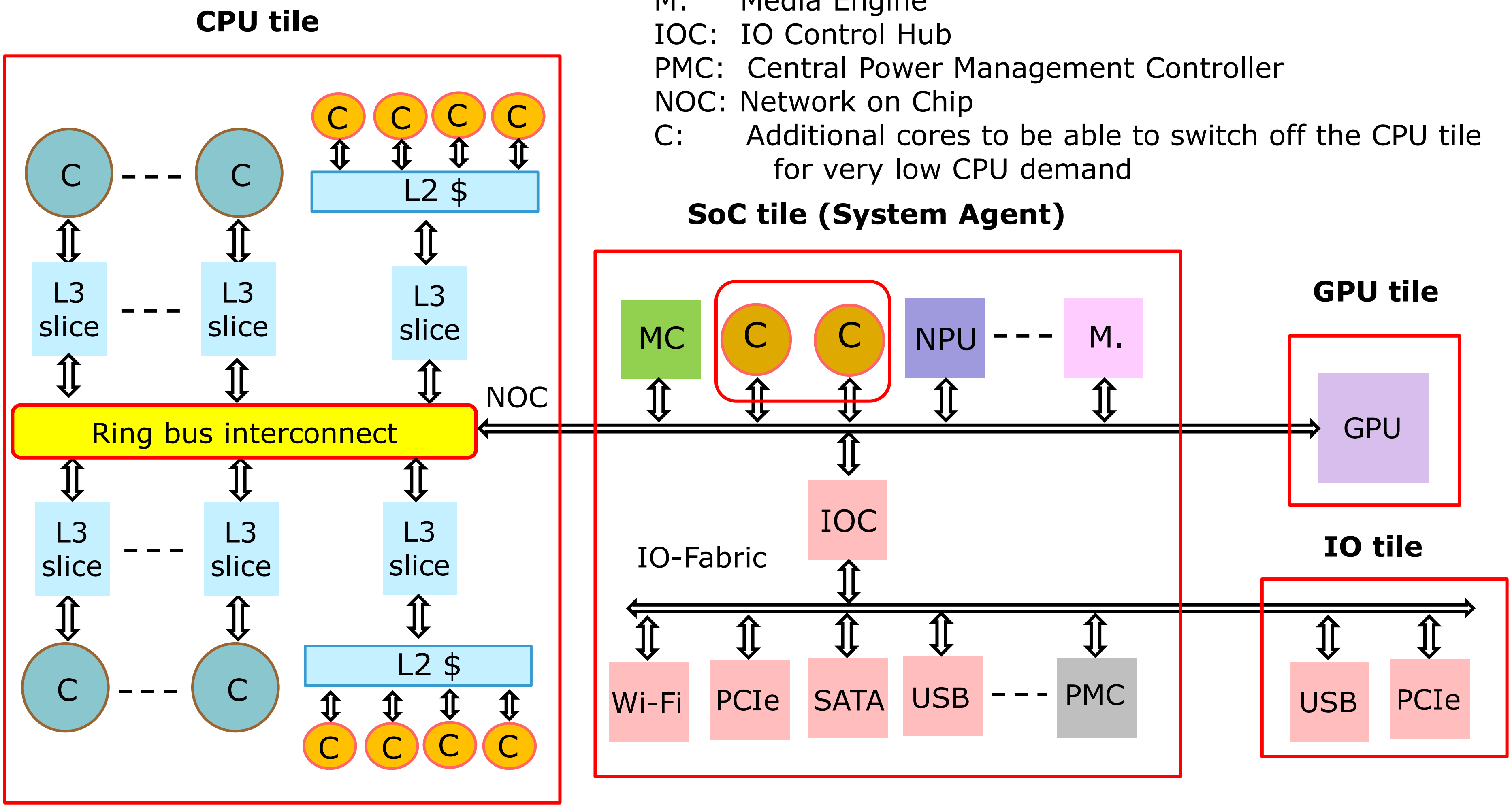
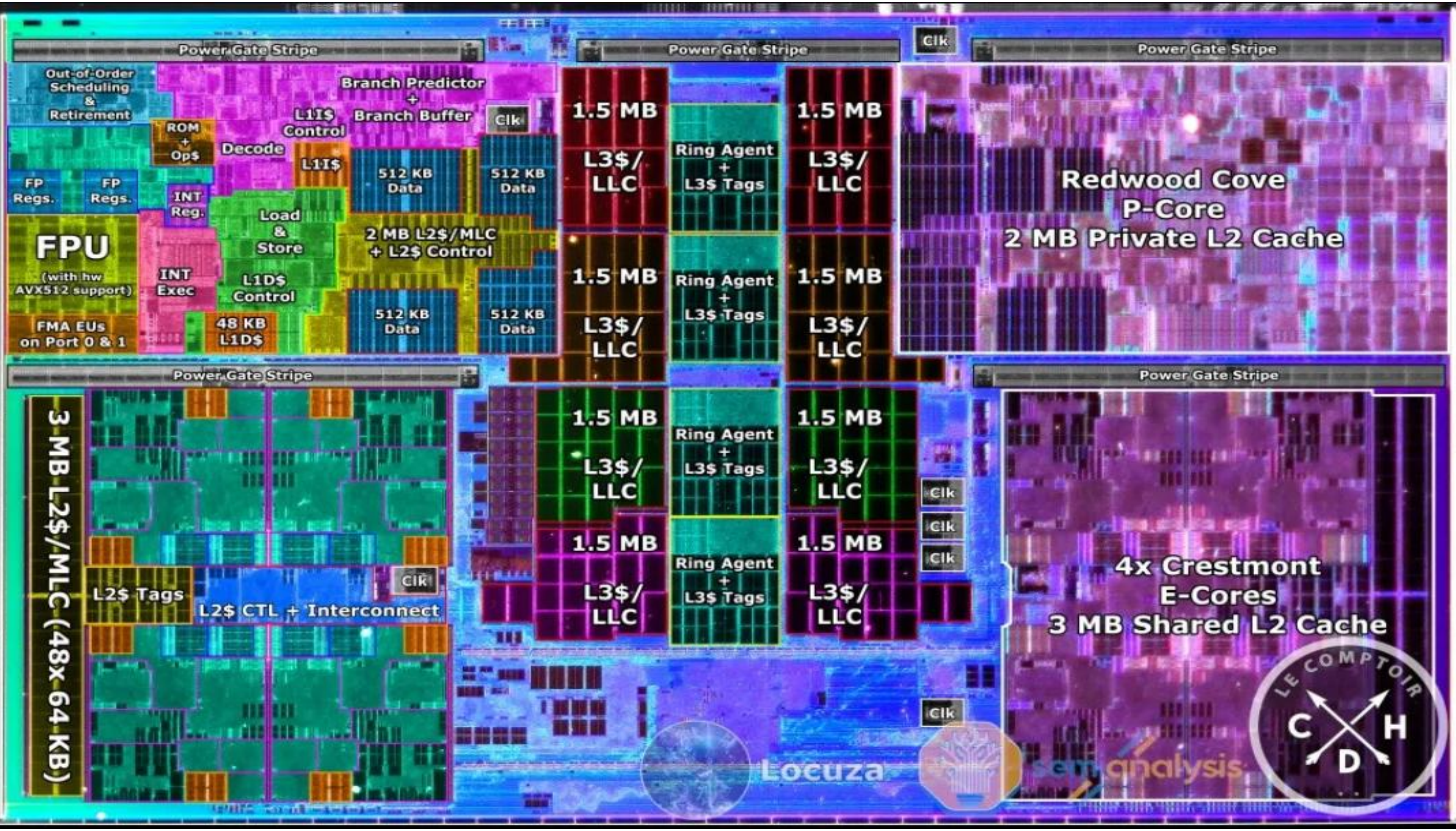


Figure: Block diagram of the 14th gen. heterogeneous multi-die big.little Meteot Lake processor

Meteor Lake Overview: In-depth with Intel Architects and Engineers - Talking Tech, Intel Technology [188]

Example: Floor plan of the CPU tile of the heterogeneous multi-die 14th gen. Meteor Lake processor with 2 P (Performance) and 8 E (Efficiency) cores [50]



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (26)

The evolution of Intel’s heterogeneous multi-die big.little designs, referred to as Intel’s Core Ultra processor series

Intel’s heterogeneous multi-die big. little designs

Core Ultra Series 1	Core Ultra Series 2	Core Ultra Series 2	Core Series 2	Core Ultra Series 3
Meteor Lake	**Lunar Lake**	**Arrow Lake**	**Meteor Lake R.**	**Panther Lake**
New process and µarch.	**New process and µarch.**	**New µarch.**	**New µarch.**	**New process new µarch.**
Intel 4 TSMC N5/N6	**TSMC N3/N6**	**TSMC N3/N5/N6**	**Intel 4 TSMC N5/N6**	**Intel 1.8A/3 TSMC N3E/N6**
TICK/Tock	***TICK/Tock***	***~ Optimization***	***Optimization***	***TICK/Tock***
(2023) 4 tiles (Laptops)	(2024) 2 tiles (LP Laptops)	(2024) 4 tiles (Laptops/DTs)	(2025) 4 tiles (Laptops)	(2026) 3 tiles (Laptops)

Remarks on the manufacturing processes used in Intel's heterogeneous multi-die big.little designs, known as the Lunar Lake and Arrow Lake processors

- Originally, Intel intended to fabricate its Lunar Lake's and Arrow Lake's processors using its Intel 20A and 18A nodes.

  However, the company encountered unexpected challenges in developing these advanced processes.

  To address its manufacturing setbacks, Intel cancelled developing the 20A process and redirected its effort toward the 18A node.

- At the same time, Intel asked TSMC to produce its Lunar Lake and Arrow Lake processors in order to avoid delays in market entries.

  Specifically, the compute dies of both lines were fabricated using TSMCs most advanced available process, the N3 process.

- A significant point to note is that in both transitions (Meteor Lake to Lunar Lake, and Meteor Lake to Arrow Lake) Intel unexpectedly changed both the IC process and the microarchitecture at the same time.

  By doing so, Intel adopted a single-phase development model in these cases, to accelerate the market entry of these processors.

  This approach marks a deviation from its original two-phase model that was introduced with the Core family.

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (28)

Remarks on the manufacturing processes used in Intel's heterogeneous multi-die big.little designs, known as the Panther Lake processor

- Again, Intel made simultaneous changes to both the IC process and the microarchitecture.

  This means that the company adopted again a single-phase development model, aimed at accelerating the market entry of its Panther Lake processors.

- This decision was likely made because Intel was not yet ready with suitable processes for these components.

- Note also that Intel let manufacture the GPU and SOC tiles by TSMC's (using its N3E and N6 processes, respectively).

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (29)

Remark to Intel's design policy: running two competing design centers -1

1. gen.				2. gen.	3. gen.	4. gen.	5. gen.
Core 2	**Penryn**	**Nehalem**	**West-mere**	**Sandy Bridge**	**Ivy Bridge**	**Haswell**	**Broad-well**
New microarch.	New process	New microarch.	New process	New microarch.	New process	New microarchi.	New process
65 nm	45 nm	45 nm	32 nm	32 nm	22 nm	22 nm	14 nm
TOCK	*TICK*	*TOCK*	*TICK*	*TOCK*	*TICK*	*TOCK*	*TICK*
(2006) Haifa	*(2007) Oregon*	*(2008) Oregon*	*(2010) Oregon*	*(2011) Haifa*	*(2012) Haifa*	*(2013) Oregon*	*(2014) Oregon*

6. gen.	7. gen.	8. gen.[1]	9. gen.	10. gen.
Skylake	**Kaby Lake**	**Kaby Lake R/G** **Coffee Lake** **Amber Lake-Y** **Whiskey Lake-U** **(Cannon Lake 10 nm)**	**Coffee Lake R**	**Comet Lake**
New microarch.	New microarch.		New microarch.	New microarch.
14 nm	14 nm	14 nm	14 nm	14 nm
TOCK	*TOCK*	*TOCK*	*TOCK*	*TOCK*
(2015) Haifa	*(2016) Haifa*	*(2017/18) Haifa*	*(2018) Haifa*	*(2019) Haifa*

[1]The 8th generation encompasses the following processor lines:

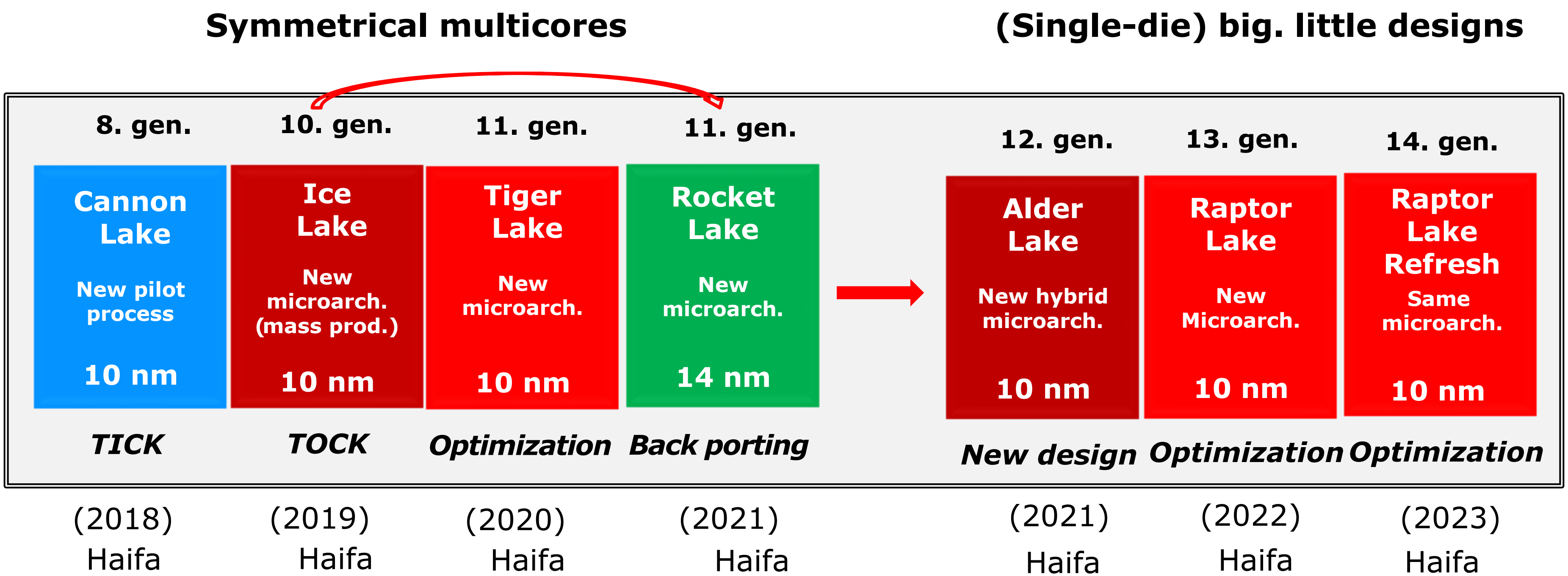
- Kaby Lake Refresh
- Kaby Lake G with AMD Vega graphics
- Coffee Lake
- Amber Lake Y
- Whiskey Lake U
- (all 14 nm) and
- Cannon Lake (10 nm)

lines [29].

R: Refresh

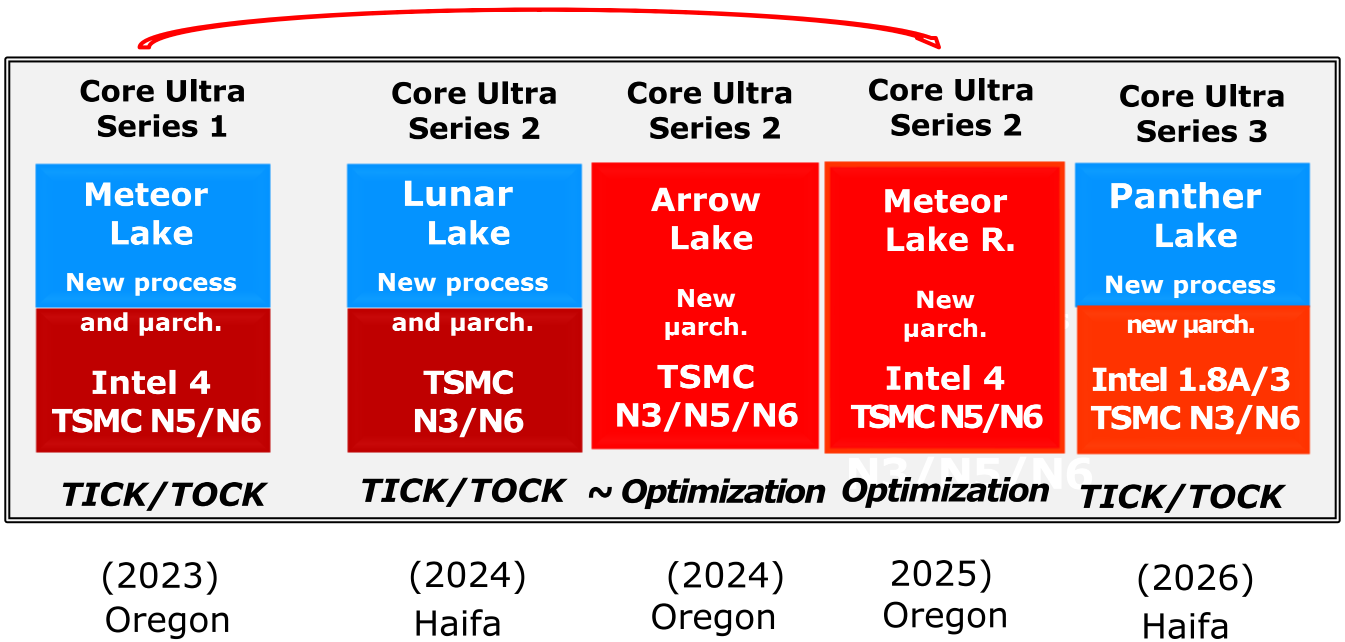
3.2 Evolution of the CPU core configuration and introduction of multi-die designs (30)

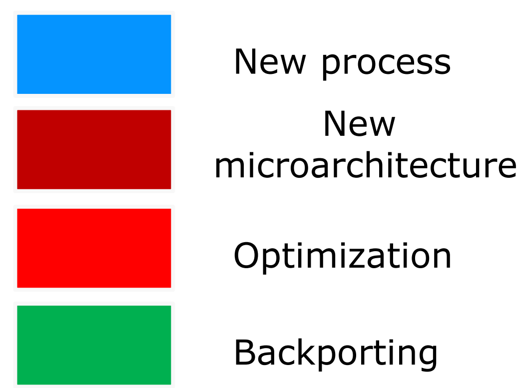
Remark to Intel's design policy: running two competing design centers -2



Remark to Intel’s design policy: running two competing design centers -3

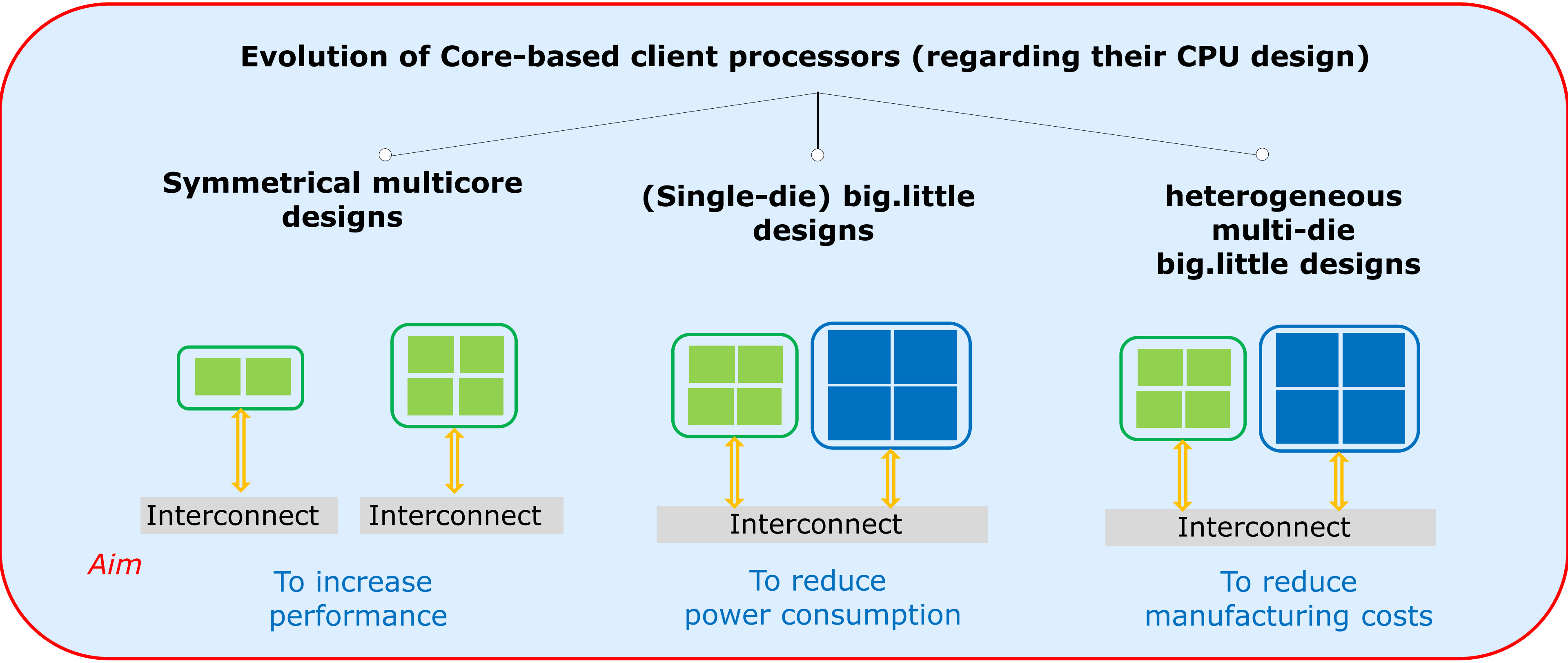
Multi-die heterogeneous big. little designs





Summing up the evolution of Intel's Core-based client processors

Considering their core configuration, Intel's Core-based client processors have undergone three main evolution steps, as illustrated in the Figure below.



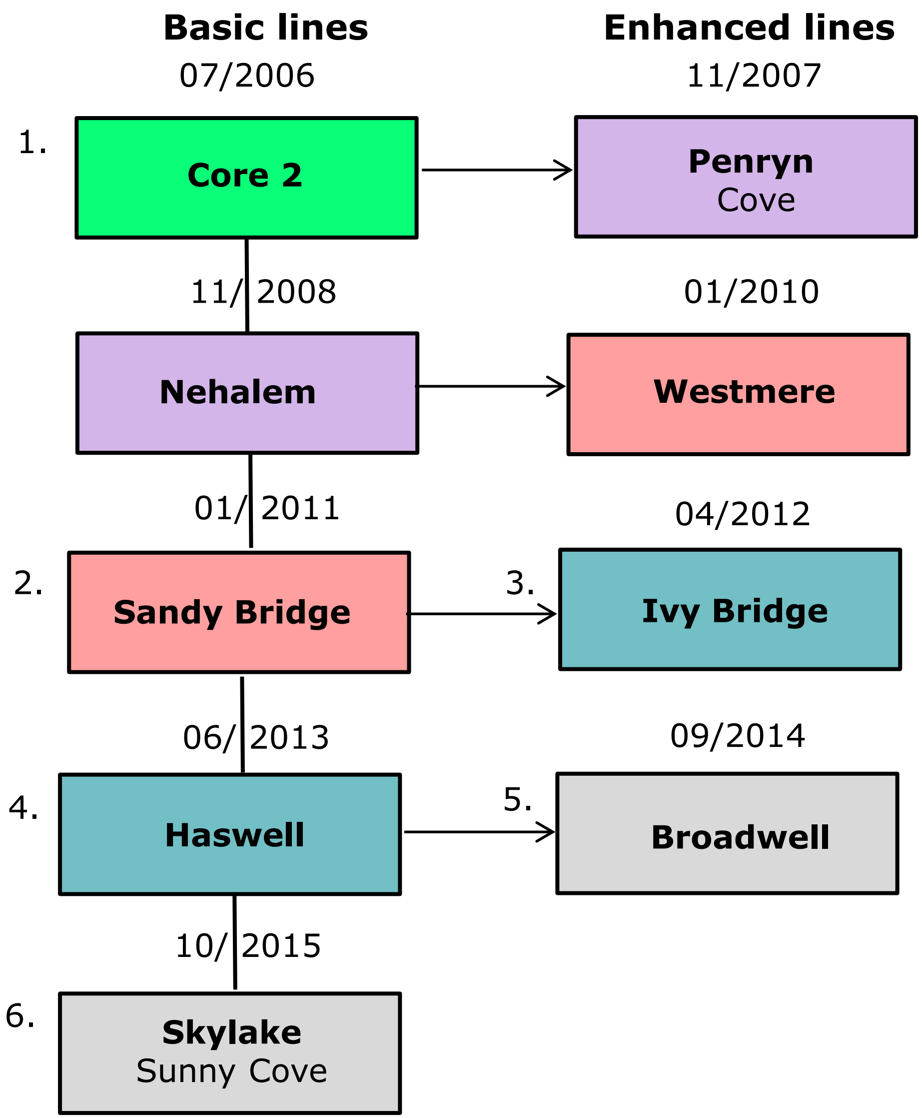
From the 1st gen. Core 2 line (2006) up to the 11th gen Raptor Lake (2021)

12th gen. Alder Lake (2021)
13th gen. Raptor Lake (2022)

From the 14th gen. Meteor Lake (2023) on

Note that the primary objective of each core configuration is highlighted in the Figure.

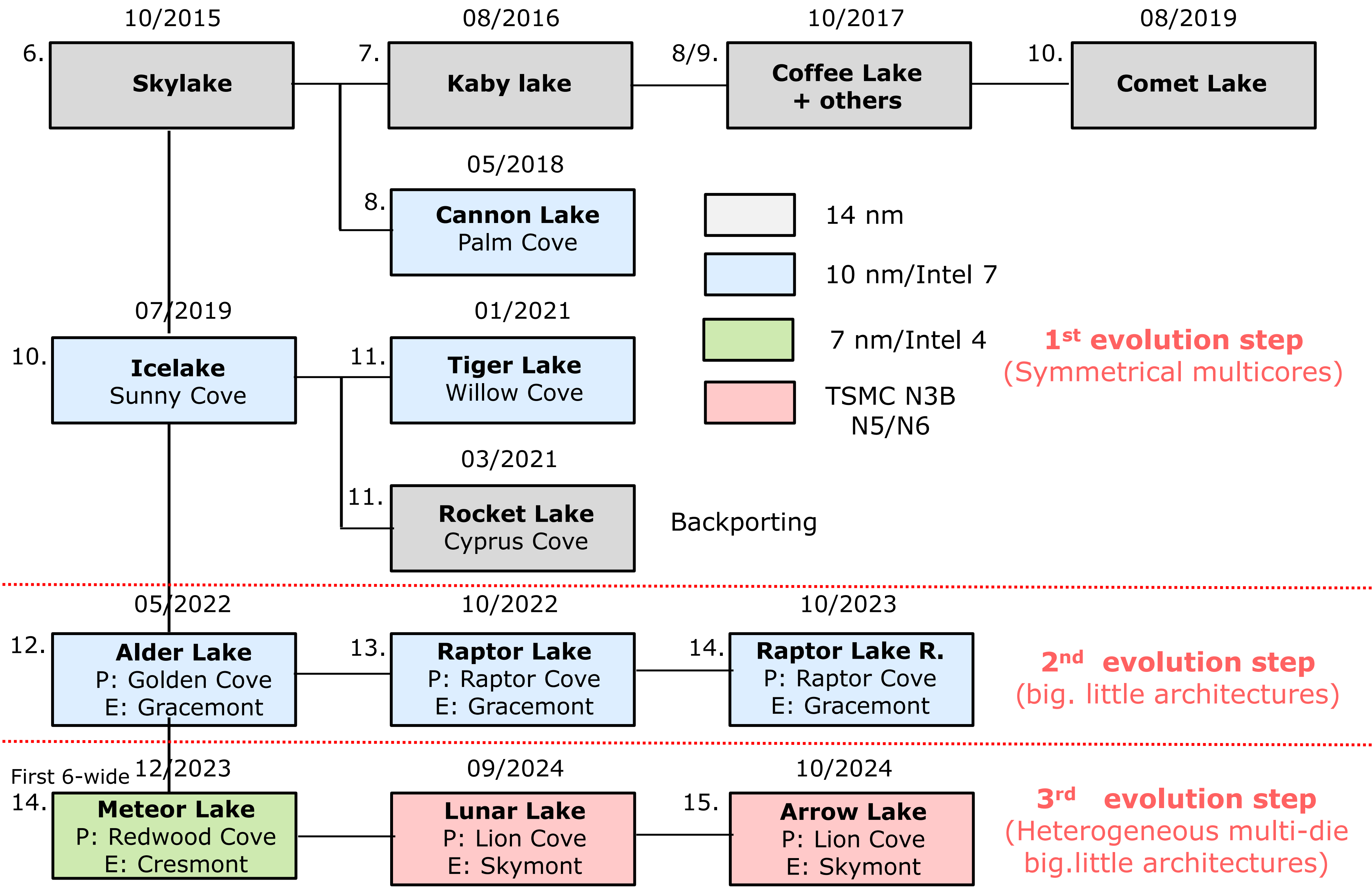
Distinguishing Intel's basic and subsequent client lines in the Core family -1



All lines shown belong to the
1st evolution step
(Symmetrical multicores)

3.2 Evolution of the CPU core configuration and introduction of multi-die designs (34)

Distinguishing Intel's basic and subsequent client lines in the Core family -2



3.2 Evolution of the CPU core configuration and introduction of multi-die designs (35)

Basic microarchitectures, their shrinks and optimizations

Basic µarchitectures	Generation	Basic µarchitectures, their shrinks and optimizations		
Core 2	**1.**	2006	65 nm	**Core 2**
		2007	45 nm	Penryn
Nehalem		2008	45 nm	**Nehalem**
		2010	32 nm	Westmere
Sandy Bridge	**2.**	2011	32 nm	**Sandy Bridge**
	3.	2012	22 nm	Ivy Bridge
Haswell	**4.**	2013	22 nm	**Haswell**
	5.	2014	14 nm	Broadwell
Skylake	**6.**	2015	14 nm	**Skylake**
	7.	2016	14 nm	Kaby Lake
	8.	2017	14 nm	Kaby Lake Refresh + more
	8.	2018	10 nm	Cannon Lake
	9.	2019	14 nm	Coffee Lake Refresh
	10.	2020	14 nm	Comet Lake
Ice Lake	**10.**	2019	10 nm	**Ice Lake**
	11.	2020	10 nm	Tiger Lake
Alder Lake	**12.**	2021	10 nm	**Alder Lake**
	13.	2022	10 nm	Raptor Lake
	14.	2023	10 nm	Raptor Lake Refresh
Meteor Lake	**14./Core Ultra Series 1**	2023	7 nm	**Meteor Lake** (CPU die)
	Core Ultra Series 2	2024	3 nm	Lunar Lake (CPU die)
	Core Ultra Series 2	2024	3 nm	Arrow Lake (CPU die)
	Core Ultra Series 3	2026	1.8A	Panther Lake

Comparing Intel's and AMD's client processor design paths, considering their CPU configurations and multi-die design approaches -1

Intel

Symmetrical multicore designs	→	Single-die big.little design	→	Heterogeneous multi-die big.little design
Pentium Extreme Edition 840 (2005)		*12th gen Alder Lake (2021)*		*14th gen Meteor Lake (2023)*

AMD

Symmetrical multicore designs	→	Modular homogeneous multi-die design	→	Modular heterogeneous multi-die designs	→	Modular heterogeneous multi-die big.little designs
Athlon 64 X2 (2005)		*Zen/Zen+ (2016/2018)*		*Zen 2 (2019)*		*Zen 5/Zen 5c (2024)*

Comparing Intel's and AMD's client processor design paths, considering their CPU configurations and multi-die design approaches -2

- The preceding Figure shows that starting from symmetrical multicore designs in 2005, both Intel and AMD have gradually transitioned to heterogeneous multi-die big.little designs, although they have followed different paths of the evolution.
- AMD first adopted homogeneous multi-die designs with its Zen family in 2017, primarily to reduce fabrication costs.
- The company also adopted a modular design approach, to utilize the same set of basic dies across various market segments, including laptops, desktops, HEDTs, or servers.

  This approach helps for reducing both development and manufacturing costs.
- Next, in 2019, AMD transitioned from homogeneous, multi-die designs to heterogeneous multi-die designs to further decrease manufacturing costs by employing less advanced processes for less demanding dies, such as the IO die.
- Finally, in 2024, AMD adopted the big.little design paradigm, to significantly lower power consumption.
- In contrast, Intel first introduced big.little designs with its 12th gen. Alder Lake line in 2021, primarily to reduce power consumption.
- This was followed by a heterogeneous multi-die design approach in its Meteor Lake line in 2023, to lower manufacturing costs.

3.3 How are the cores of big.little architectures scheduled?

3.3 How are the cores of big.little architecutures scheduled? -1

- Concerning this point, please recall that Intel introduced earlier hardware support for the OS using DVFS (Dynamic Voltage and Frequency Scheduling) to reduce dissipation in the Core family of processors, in 2006.

  With DVFS, Intel determined the actual utilization of threads (also known as logical processors) by utilizing two Model Specific Registers (MSRs) per thread, which are essentially 64-bit counters.

- While Intel introduced big.little architectures in its 12th gen. Alder Lake processor family, in 2021, it continued to provide hardware support for the OS, this time for thread scheduling.

  The introduced hardware guidance for scheduling hybrid architectures is known as Tread Director.

How are the cores of big.little architecutures scheduled? -2

- The necessity for a new approach to thread scheduling arose from the fact that big.little architectures utilize different core types, which require a distinct method of thread scheduling compared to symmetrical multicores.
- The TD supports simultaneous execution across different core types while optimizing various aspects of thread scheduling, such as performance and power consumption.
- The TD works in cooperation with the Windows OS, initially, with Windows 11 release 21H2, which was unveiled in 10/2021.
- Unfortunately, the TD is only "sparsely" documented.
- Therefore, below we can only provide a strongly simplified description of it, while referring to its implementation in the Panther Lake line, launched in 01/2026, known as Thread Director v2.

The Thread Director (TD) in the Panther Lake processor -1 [223], [224]

- The key component of the TD is the Thread Director Table.

  It keeps the actual performance and power efficiency capabilities of the logical processors.

- A logical processor is a CPU core that the OS can use to run a thread.
  - A CPU core without Hyper-Threading equals one logical processor.
  - A CPU core with Hyper-Threading provides two logical processors.

- Performance capabilities are 8-bit values (0 ... 255), specifying the relative performance level of a logical processor.

  Higher values indicate higher performance; the lowest performance level of 0 indicates a recommendation to the OS to not schedule any software threads on it for performance reasons.

- Energy Efficiency capabilities are 8-bit values (0 ... 255), specifying the relative energy efficiency level of a logical processor.

  Higher values indicate higher energy efficiency; the lowest energy efficiency capability of "0" indicates a recommendation to the OS to not schedule any software threads on it for efficiency reasons.

  An Energy Efficiency capability of 255 indicates which logical processors have the highest relative energy efficiency capability.

  In addition, the value "255" is an explicit recommendation for the OS to consolidate work on those logical processors for energy-efficiency reasons.

The Thread Director (TD) in the Panther Lake processor -2 [224]

The performance and power efficiency capabilities are kept in distinct fields of the Thread Director Table, which is structured as shown below.

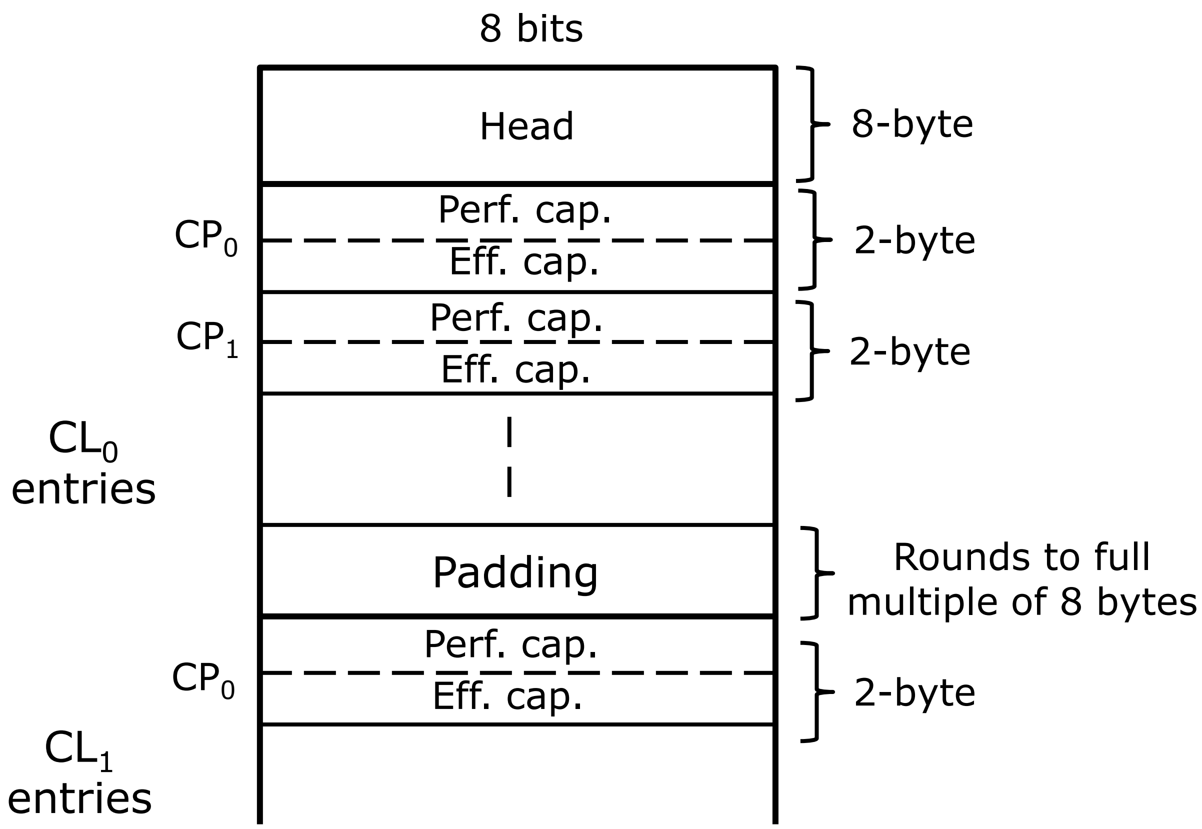
Assumed structure of the Thread Director Table [224] -1

The logical processors are ordered in classes (CL).
The Panther Lake processor has three classes, as follows:

CL_0 = LP-E cores
CL_1 = E cores
CL_2 = P cores.

The Thread Director Table lists the actual capability values ordered according to the logical processor classes Cl_i as seen in the following Figure.

Assumed structure of the Thread Director Table [224] -1



In the Table the capability values are ordered according to the logical processor classes CL_i, where the classes may be interpreted as

CL_0 = LP-E cores
CL_1 = E cores
CL_2 = P cores.

Assumed structure of the Thread Director Table [224] -3

- Within each logical core class (CL)$_{i,}$ the capabilities of the logical processors are listed as $CP_{0,}$ CP_1 etc.

How are the actual capability values determined?

- Hardware scans internal telemetry data, such as performance counters and on-die sensors in loops, and determines the capability values of the logical processors on a μs scale.
- Capabilities may change every tens of milliseconds. (223)

For instance, the operating conditions of a thread may be affected by several factors, such as a change in the supply voltage due to frequency/voltage scaling, a reduction in clock frequency and core voltage due to increased temperature or a higher load on the thread shared on the same core etc.

In such cases, the capabilities will be reduced, and the corresponding values must be updated in the TD Table.

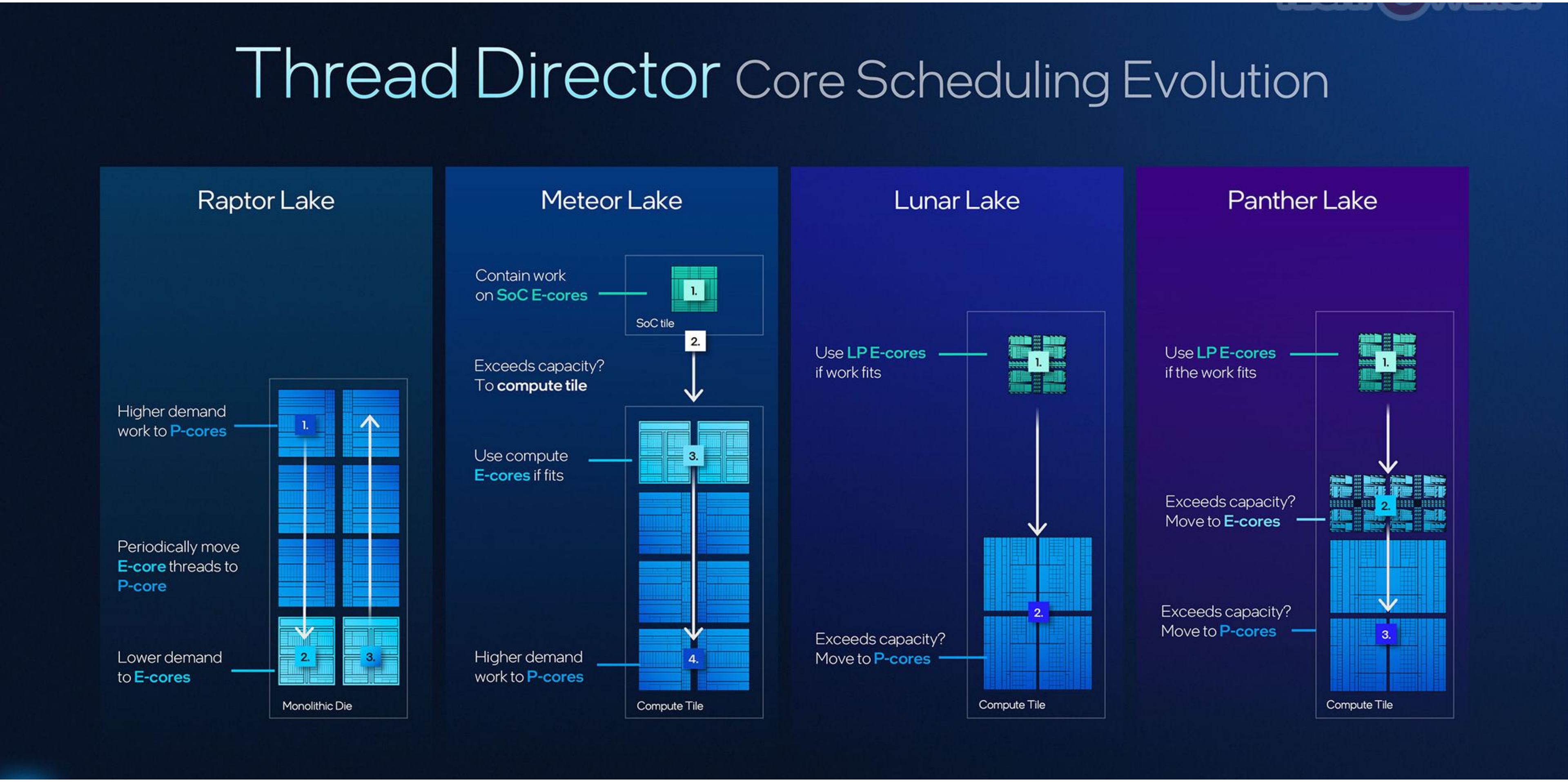
Changes of the capability values generate interrupts.

3.3 How are the cores of big.little architecutures scheduled? (8)

The Thread Director (TD) in the Panther Lake processor -3 [224]

- The OS processes these interrupts and adjusts thread scheduling accordingly while using the latest capability data in the Thread Director Table, and further data not published.
- Currently, there is no publicly available documentation revealing how the TD updates the capability data based on the telemetry data read.
- Also, there is a lack of reliable documentation on how the OS reschedules threads, making it impossible to discuss this point.

Evolution of scheduling workloads to heterogeneous cores in Intel's processors from the Raptor Lake line to the Panther line [225]



3.4 How are Intel's recent multicore processors packaged?

3.4 How are Intel's recent multicore processors packaged?

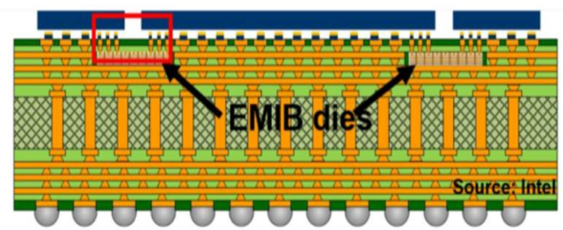
- Intel introduced its EMIB and Foveros technologies in 2017 and 2018, respectively.
- However, around 2023-2025 Intel partially reinterpreted and renamed these packaging technologies, as outlined below.

3.4 How are Intel's recent multicore processors packaged? (2)

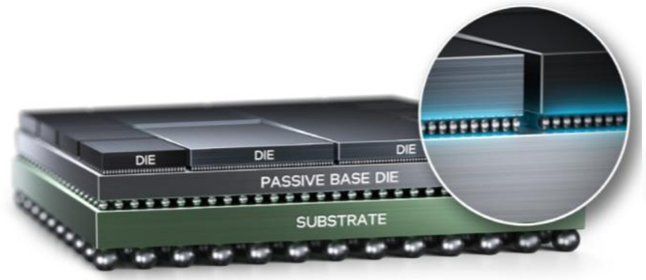
Intel's recent packaging technologies and their use

Intel's recent packaging technologies and their use

EMIB 2.5D



Foveros 2.5D



Foveros Direct 3D

Used in

EMIB 2.5D:

Stratix 10 FPGA (2017)
Server processors:
4th gen. Sapphire Rapids (2023) (4-tile CPU)
5th gen. Emerald Rapids (2024)
6th gen. Sierra Forest (2024)
6th gen. Granite Rapids (2024)

Foveros 2.5D:

Lakefield (2019)
Meteor Lake (2023)
Arrow Lake (2024)
Lunar Lake (2024)
Panther Lake (2026)

Foveros Direct 3D:

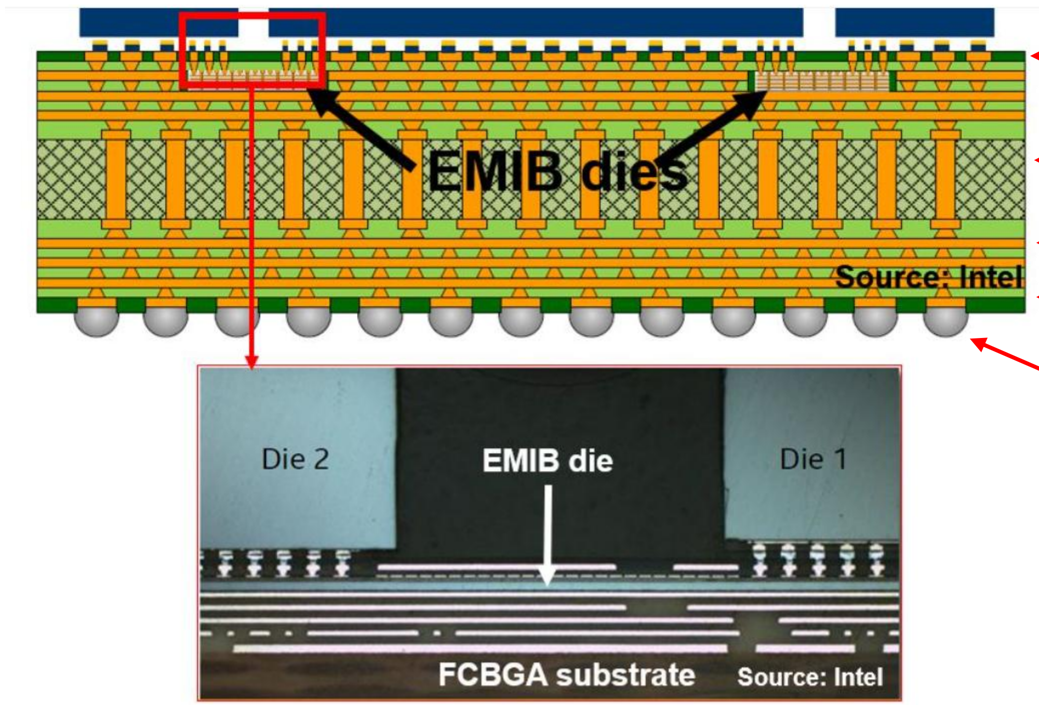
. Server processor:
6+gen. Clearwater Forest (2026)

The technologies mentioned above will be discussed subsequently.

Source of the Figures: Intel

3.4 How are Intel's recent multicore processors packaged? (3)

The EMIB 2.5D (Embedded Multi-die Interconnect Bridge) packaging technology []



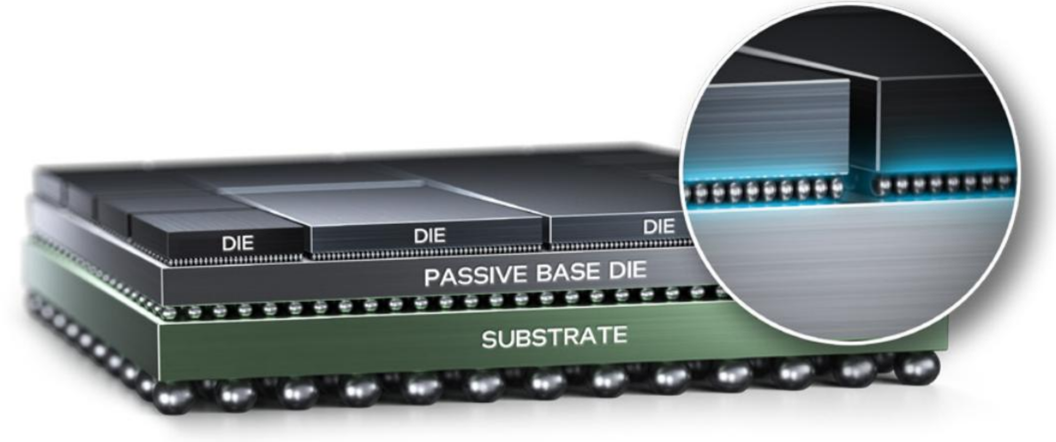
Its main components are:

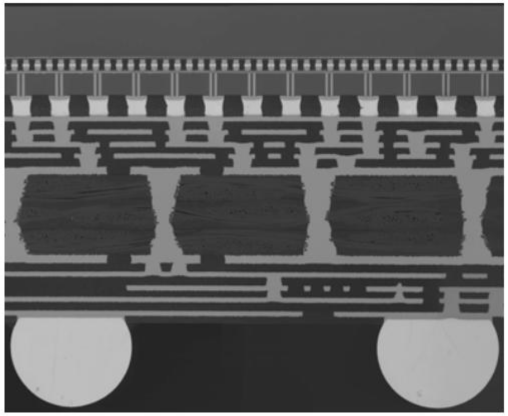
- Soldered microbumps with a pitch of 45-55 µm.
- Organic package substrate
- Cu routing layers
- Dielectric layers isolating embedded Si bridges
- BGA ball grid

It was first implemented in the FFGA chip (Stratix 10) in 2017, referred to as Intel's EMIB technology.

Soldered microbumps: forraszgöbök Pitch: osztás-távolság
Organic package substrate: Polimer-alapú tokozó anyag
Cu routing layers: réz vezetősávok BGA ball grid: BGA forraszgolyó-rács

The Foveros 2.5D packaging technology [199]



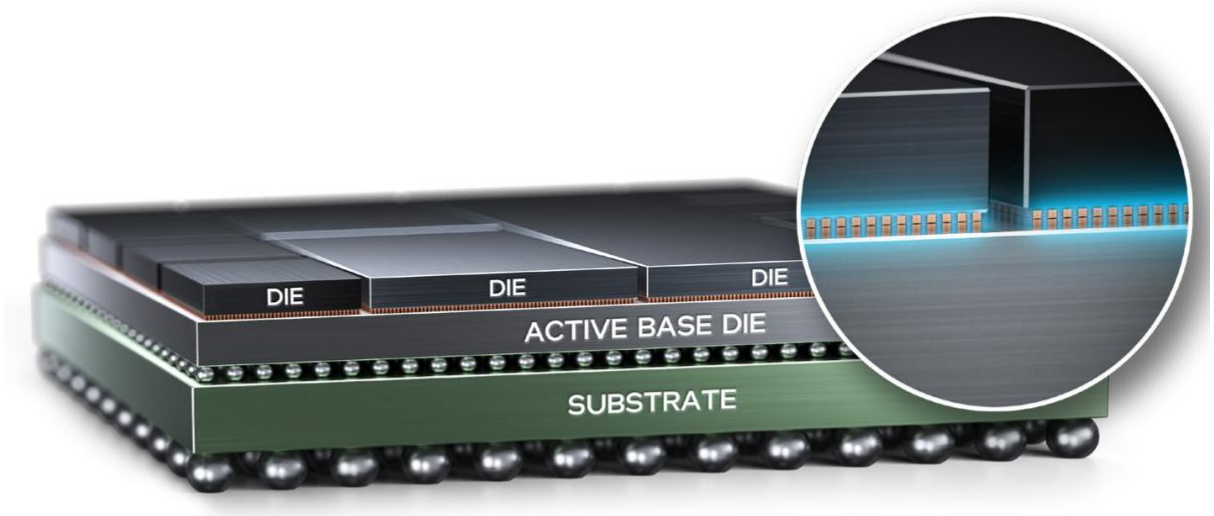


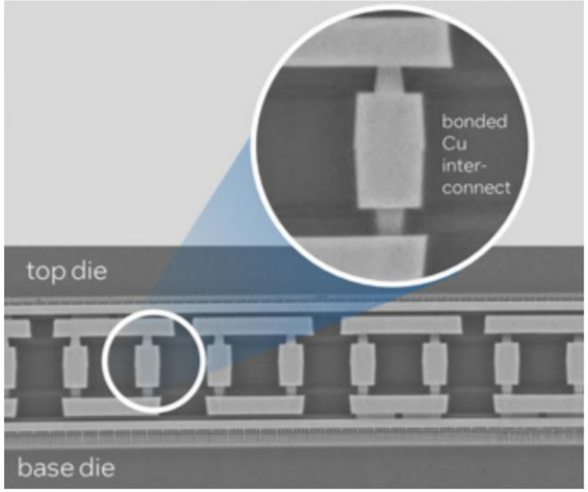
Its main components are:

- Soldered fine-pitch (20-50 µm) micro-bumps
- Si-interposer (base die) with
  - local Cu routing layers
  - TSVs (Through Silicon Vias) for power, ground (réz átvezetők)
  - C4 solder bumps with a much coarser pitch (100-200 µm) than microbumps.
- Organic package substrate with
  - Coarser Cu routing layers
  - power/ground planes
  - embedded decoupling capacitors (beágyazott leválasztó kapacitások)
- BGA Ball grid

It was first implemented in the Lakefield processor in 2019, referred to as Intel's Foveros technology.

The Foveros Direct 3D packaging technology [220]





- It eliminates traditional solder microbumps by utilizing direct Cu-to-Cu hybrid bonding between the chiplets and the base die with a pitch of less than 10 µm.

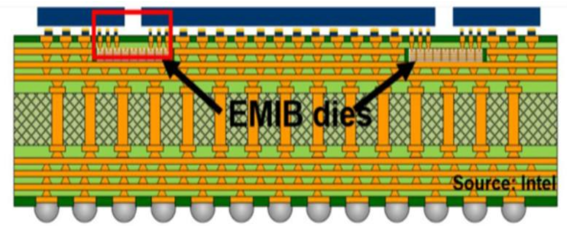
  To achieve direct Cu-to-Cu hybrid bonding, Cu pads (réz érintkezők) are pressed together under heat and pressure, allowing Cu atoms to diffuse across the interface and create a hybrid metallic interconnect.
- This method removes the resistance and parasitic effects of solder bumps, while enabling a much tighter pitch.
- The roles of the base die and the package substrate remain largely the same as in the Foveros 2.5D technology.
- The Foveros Direct 3D packaging will first ibe implemented in Intel's 6+ gen. Clearwater server processor series, which is set to be launched in H1 2026.

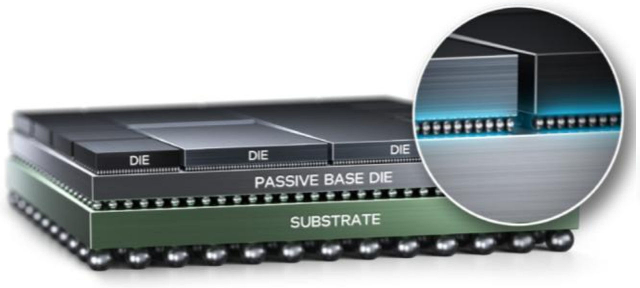
3.4 How are Intel's recent multicore processors packaged? (6)

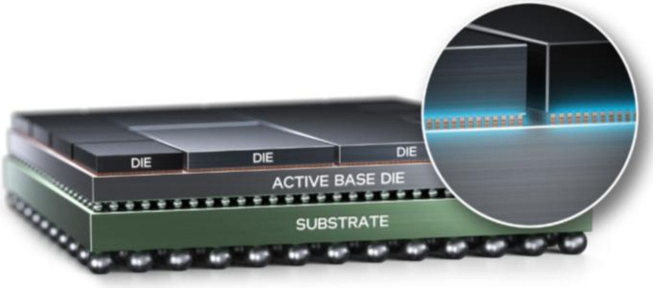
Key features of Intel's recent packaging technologies [221]

Intel's recent packaging technologies and their use

EMIB 2.5D | **Foveros 2.5D** | **Foveros Direct 3D**







	EMI 2.5D	fOVEROS 2.5D	fOVEROS Direct 3D
Bump pitch	55 -45 µm	25 µm	<10 µm
Bump density	770-330/mm2	772-1600/mm2	10000/mm2
C2C power	0.25 pJ/bit	0.15 pJ/bit	0.05 pJ/bit

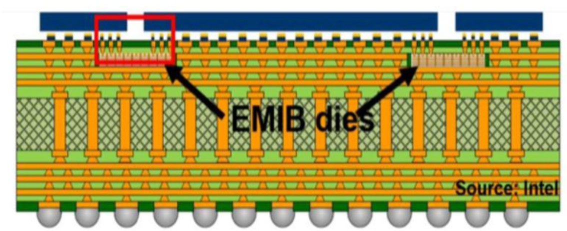
As indicated above, newer packaging technologies provide more dense and more energy-efficient interconnections between slides.

C2C: Chip to Chip

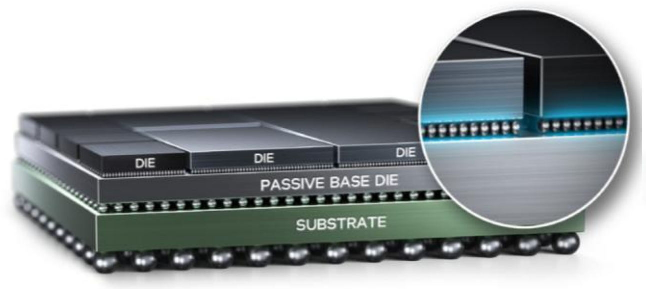
Summing up Intel’s recent packaging technologies

Intel’s recent packaging technologies and their use

EMIB 2.5D



Foveros 2.5D



Foveros Direct 3D

These technologies are utilized to interconnect and package side-by-side placed dies.

EMIB (Embedded Multi-die Interconnect Bridge): In this technology, the dies are soldered to the substrate while they are interconnected using EMIB bridges, which are embedded in an organic package substrate.

The substrate provides additional interconnects, power, and ground, and includes BGA balls.

Foveros 2.5D: This method involves soldering the dies to a passive base die (Si interposer), which facilitates direct interconnections between the dies and provides power and ground connections.

The base die has the same functions as before but provides a finer interconnection structure.

Foveros Direct 3D: This approach replaces soldering the dies to the base die by direct Cu-to-Cu interconnects, which is more dense and more power efficient.

3.5 How has the CPU core width increased?

3.5 How has the CPU core width increased?

- Before discussing this point, let's clarify the concept of the "CPU core width".

- The "CPU core width" usually refers to the number of instruction decoders implemented in the "front-end" of the CPU cores.

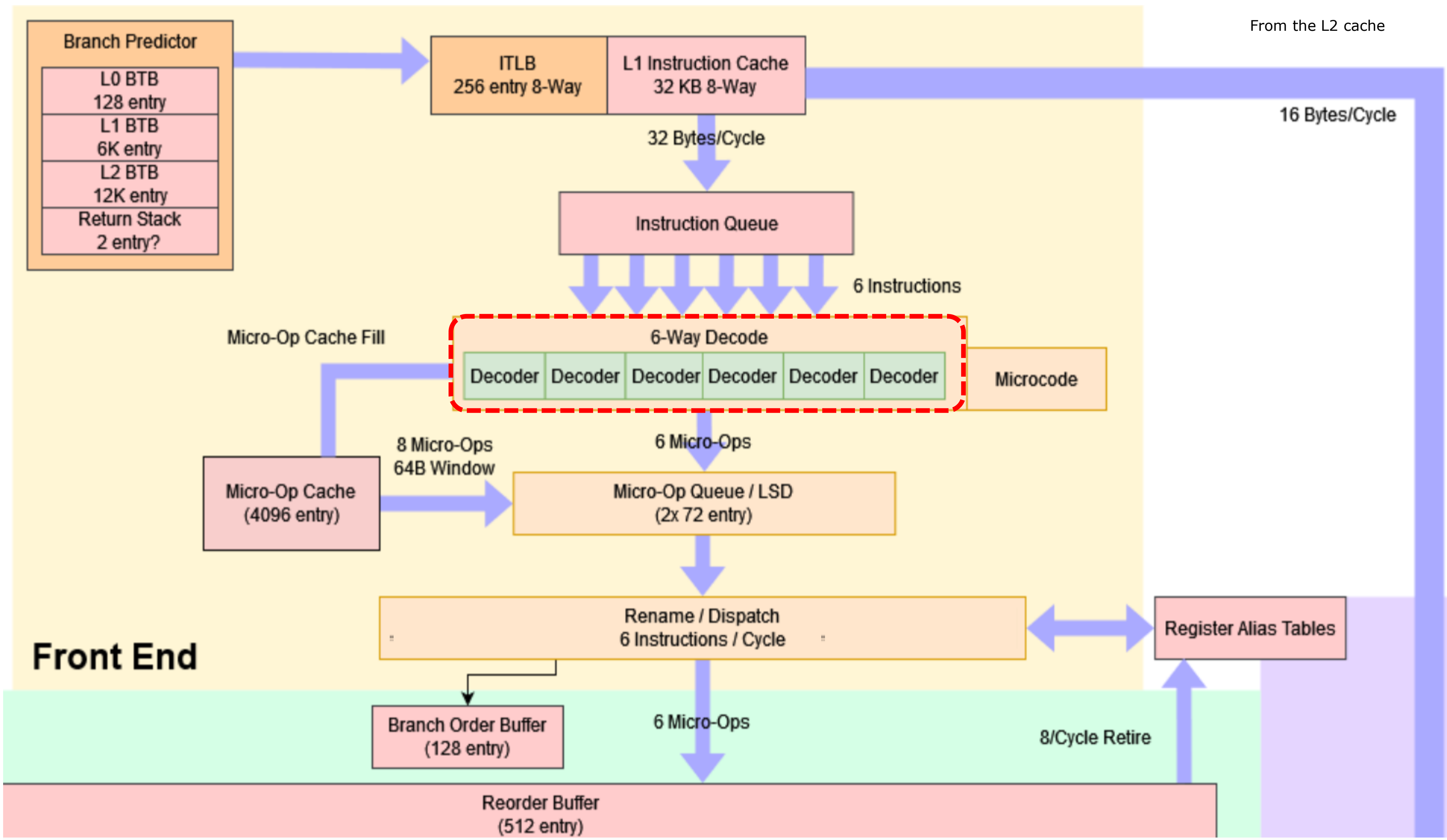
- Remarks on the notion of "CPU core width

  It is important to note that this notion is also referred to as "front-end width" or simply "core width".

- As an example, the following Figure shows that the P-core of Meteor Lake is 6-wide, as it contains 6 decoders.

Front end: előtét fokozatok   Back end: Végrehajtó fokozatok

Example: The 6-wide P-core of client processors in the Meteor Lake line [54]



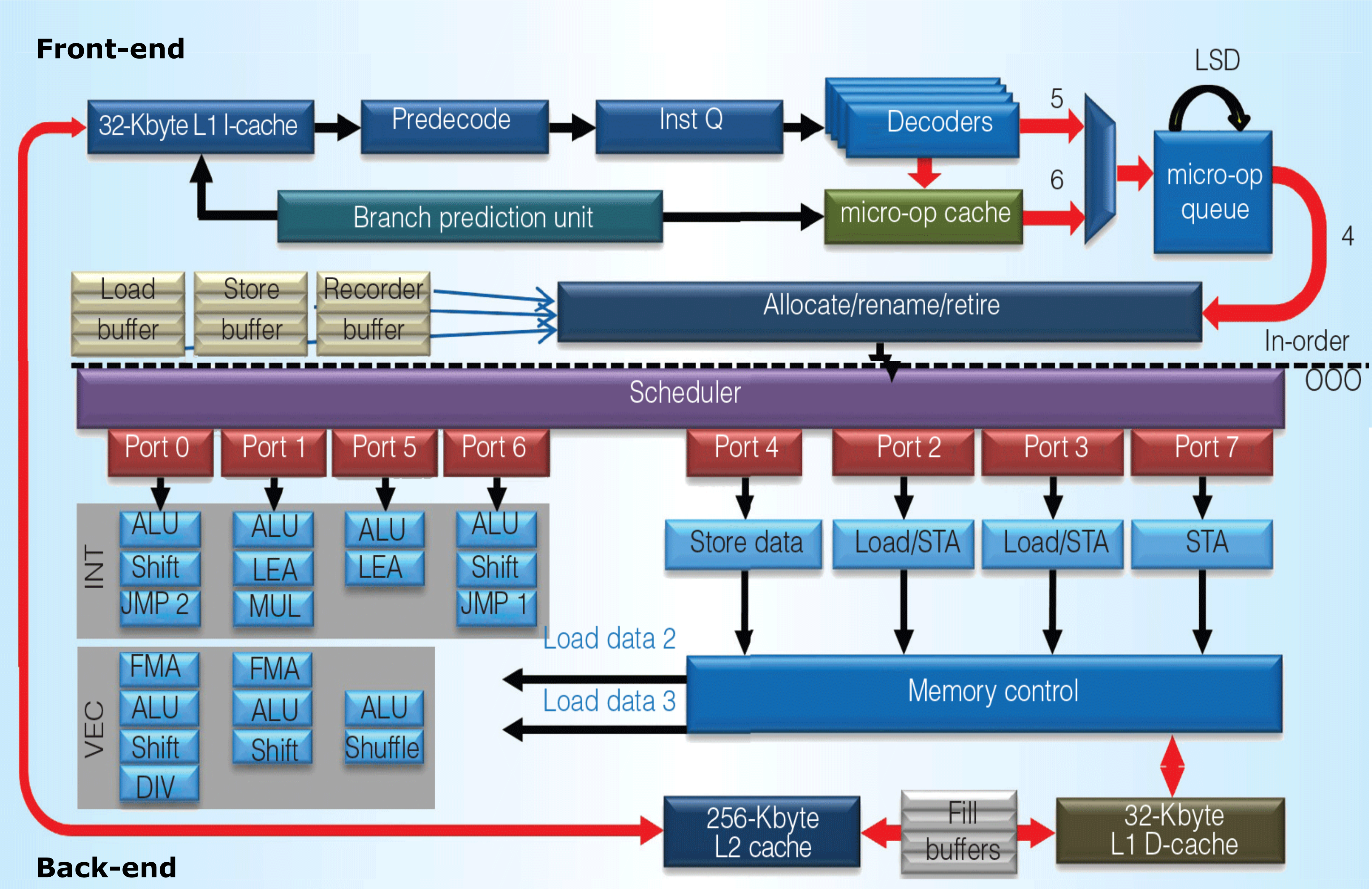
LSD: Loop Stream Detector (Detects loops and inserts decoded micro-op sequences for them.

3.5 How has the CPU core width increased? (3)

Remark on the notions “front-end” and “back-end”

CPU cores can typically be divided into two main parts: the front-end and the back-end, as illustrated in the following Figure.

Block diagram of Intel's 6th gen. Sky Lake core (2015) [145] -1



3.5 How has the CPU core width increased? (5)

The front-end and back-end of CPU cores exemplified by Intel's 6th gen. Sky Lake CPU core (2015) [145] -2

- The front-end of a processor is responsible for preparing instructions for execution.
- It primarily handles instruction fetching, decoding them into microoperations, and performs register renaming.

  The rename process eliminates false data dependencies, such as write-after-write and write-after-load dependencies, enabling more efficient out-of-order execution.
- The micro-op cache stores decoded instructions along with their corresponding micro-op sequences.

  During instruction decoding, the processor checks the micro-op cache to determine whether the current instruction has already been decoded and stored.

  If a match is found, the associated micro-op sequence is directly inserted into the micro-op stream, bypassing the decoding stage.

  This approach significantly reduces decode latency and lowers energy consumption.
- The back-end of the processor core is responsible for executing micro-operations and updating the program state (or program states in the case of multithreading) with the instruction results.
- Instruction results are committed in program order by using a ROB (ReOrder Buffer).

  (In the current example, the ROB's role in updating program state is not illustrated.

3.5 How has the CPU core width increased? (6)

Why is the number of decoders decisive for the performance potential of CPU cores?

- The number of instruction decoders in a core determines the maximum number of microoperations that can be delivered to the core’s back-end per cycle.

  This directly limits the peak single-core (or single-thread) performance.

- While adding a micro-op cache can boost decode bandwidth by about 5 to 15 %, the overall decode bandwidth remains fundamentally constrained by the number of decoders implemented in the core [163].

3.5 How has the CPU core width increased? (7)

Increasing the width of the CPU cores in Intel's and AMD's client processors

Wider cores improve performance, e.g. in terms of IPC (Instructions Per Cycle), therefore both Intel and AMD have strived to widen their cores over time, as shown in the Figure below.

Width of Intel's and AMD's (x86-based) CPU cores

	2-wide	**3-wide**	**4-wide**	**6-wide**	**8-wide**	**9-wide**
Intel	Pentium (1993)	Pentium II (Pentium Pro) 1996 Pentium III (1999) Pentium 4 (2000)	1st gen. Core 2 (2006) until Broadwell (2014)	12th gen. Alder Lake (2021) (big.little configuration)	Lunar Lake (Core Ultra Series 2) (2024) 15th. gen. Arrow Lake (Core Ultra Series 2) (2024) (P-cores of the multi-die big.little configuration)	Lunar Lake (Core Ultra Series 2) (2024) 15th. gen. Arrow Lake (Core Ultra Series 2) (2024) (3x3 E-cores of the multi-die big.little configuration)
AMD		K5 (1996) K7 Athlon (1998) K8 (Hammer)-based designs (2003) K10 (Barcelona/Phenom) based lines (2007) K10.5 (Shanghai etc.) based lines (2008)	K6 (1997) K15 Bulldozer-based lines (2011) Zen-based lines (2017)		Zen 5-based lines (2x4) (2024)	

Figure: Increasing the CPU core widths in Intel's and AMD's client processors

3.6 How has the core count in Intel's client CPUs increased?

3.6 How has the core count in Intel’s client CPUs increased?

Intel has increased the max. core counts of its client processors in three consecutive phases.

These phases are discussed below for your reference.

3.6 How has the core count in Intel's client CPUs increased? (2)

a) Phase 1: Increasing the maximum core count up to four

- Soon after introducing the dual-core Core family in 2006, Intel increased the core count of its client processors to four.

  The first quad-core models, released in 2007, were implemented using dual-dies configurations, followed by monolithic (single-die) designs in 2009.

- However, for nearly a decade, Intel maintained this four-core standard, as most software applications at the time offered limited parallelism and did not benefit significantly from additional cores.

The following Table illustrates this.

3.6 How has the core count in Intel's client CPUs increased? (3)

Increasing the max. core counts from 2 to 4 -1

Gen.	Processor	Techno-logy	Date of intro.	Seg-ment	Max core count	Connecting memory	Mem. speed (up to)	Mem. channels	Socket	MCH/ PCH
1.	Core 2	65 nm	6/2006	M D	2C	Via the NB	DDR2-800	2	LGA 775	945-975
	Core 2 Quad	65 nm	6/2007	D	2x2C		DDR3-1067		LGA 775	3 Series
	Penryn	45 nm	6/2008	D	2x2C		DDR3-1067		LGA 775	4 Series
	2. G. Nehalem (Lynnfield)	45 nm	9/2009	D	4C	Via the processor	DDR3-1333		LGA 1156	5 Series
	Westmere (M: Arrandale) D: Clarkdale/	32 nm	1/2010	M D	2C+G				M:BGA 1288 D:LGA 1156	5 Series
2.	Sandy Bridge	32 nm	1/2011	M D	2C/4C+G		DDR3-1333		M: PGA 988 D: LGA 1155	6 Series
3.	Ivy Bridge	22 nm	4/2012	M D	4C+G		DDR3-1600			7 Series
4.	Haswell	22 nm	6/2013	M D	4C+G		DDR3-1600		M: BGA 1168 D: LGA 1150	D: 8 Ser.
4.	Haswell Refr.	22 nm	5/2014	D	4C+G		DDR3-1666			D: 9 Ser.
5.	Broadwell	14 nm	9/2014	M D	4C+G		DDR3-1866			D: 9 Ser.
6.	Skylake	14 nm	10/2015	M D	4C+G		DDR4-2133		M: BGA 1356 D: LGA 1151	D: 100 Series
7.	Kaby Lake	14 nm	8/2016	M D	4C+G		DDR4-2400			D: 200 Series
8.	Coffee Lake	14 nm	10/2017	M D	6C+G		DDR4-2666		M: BGA 1440 D: LGA 1151	D:300 Series
8.	Amber Lake	14 nm	8/2018	M	2C+G		LPDDR3-2133		BGA 1515	--
8.	Whiskey Lake	14 nm	8/2018	M	4C+G		DDR4-2400		BGA 1528	--
9.	Coffee Lake R.	14 nm	10/2018	D	8C+G		DDR4-2666		LGA 1151	300 Series
10.	Comet Lake	14 nm	8/2019	M D	10C+G		DDR4-2666		M: BGA 1528 D: LGA 1200	D: 400 Series
11.	Rocket Lake	14 nm	3/2021	D	8C+G		DDR4-3200		LGA 1200	

3.6 How has the core count in Intel's client CPUs increased? (4)

b) Phase 2: Increasing the maximum core count to ten -1

- In 2017, AMD launched its Zen-based client processors without integrated GPUs, initially with 8 cores, and later expanding its maximum core count to 16.
- This aggressive scaling challenged Intel's dominance in the client processor market.
- In response, beginning in 2017, Intel gradually increased the core count of its client processors first to 6, then to 8, and subsequently to 10, as indicated in the following two Tables.

3.6 How has the core count in Intel's client CPUs increased? (5)

Phase 2: Increasing the maximum core count to 10 -2

Gen.	Processor	Techno-logy	Date of intro.	Seg-ment	Max core count	Connecting memory	Mem. speed (up to)	Mem. channels	Socket	MCH/ PCH
1.	Core 2	65 nm	6/2006	M D	2C	Via the NB	DDR2-800	2	LGA 775	945-975
	Core 2 Quad	65 nm	6/2007	D	2x2C		DDR3-1067		LGA 775	3 Series
	Penryn	45 nm	6/2008	D	2x2C		DDR3-1067		LGA 775	4 Series
	2. G. Nehalem (Lynnfield)	45 nm	9/2009	D	4C	Via the processor	DDR3-1333		LGA 1156	5 Series
	Westmere (M: Arrandale) D: Clarkdale/	32 nm	1/2010	M D	2C+G				M:BGA 1288 D:LGA 1156	5 Series
2.	Sandy Bridge	32 nm	1/2011	M D	2C/4C+G		DDR3-1333		M: PGA 988 D: LGA 1155	6 Series
3.	Ivy Bridge	22 nm	4/2012	M D	4C+G		DDR3-1600			7 Series
4.	Haswell	22 nm	6/2013	M D	4C+G		DDR3-1600		M: BGA 1168 D: LGA 1150	D: 8 Ser.
4.	Haswell Refr.	22 nm	5/2014	D	4C+G		DDR3-1666			D: 9 Ser.
5.	Broadwell	14 nm	9/2014	M D	4C+G		DDR3-1866			D: 9 Ser.
6.	Skylake	14 nm	10/2015	M D	4C+G		DDR4-2133		M: BGA 1356 D: LGA 1151	D: 100 Series
7.	Kaby Lake	14 nm	8/2016	M D	4C+G		DDR4-2400			D: 200 Series
8.	Coffee Lake	14 nm	10/2017	M D	6C+G		DDR4-2666		M: BGA 1440 D: LGA 1151	D:300 Series
8.	Amber Lake	14 nm	8/2018	M	2C+G		LPDDR3-2133		BGA 1515	--
8.	Whiskey Lake	14 nm	8/2018	M	4C+G		DDR4-2400		BGA 1528	--
9.	Coffee Lake R.	14 nm	10/2018	D	8C+G		DDR4-2666		LGA 1151	300 Series
10.	Comet Lake	14 nm	8/2019	M D	10C+G		DDR4-2666		M: BGA 1528 D: LGA 1200	D: 400 Series
11.	Rocket Lake	14 nm	3/2021	D	8C+G		DDR4-3200		LGA 1200	

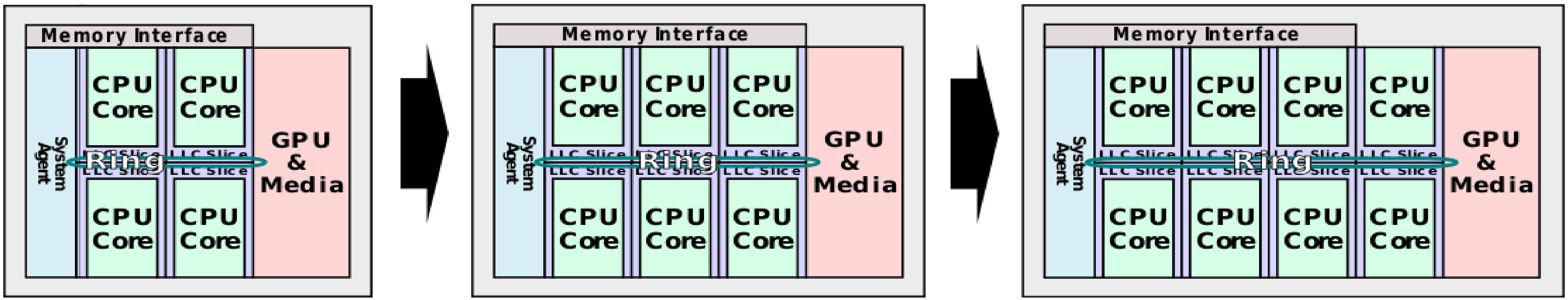
3.6 How has the core count in Intel's client CPUs increased? (6)

Phase 2: Increasing the maximum core count to 10 -3

	Processor	Techno-logy	Date of intro.	Seg-ment	Max. core count	Mem. speed (up to)	Mem. channels	Socket	PCH
8.	Cannon Lake	10 nm	5/2018	M:U	2C	DDR4-2400		BGA 1440	--
10.	Ice Lake	10 nm	7/2019	M: G	4C+G	DDR4-3200 LPDDR4- 3733		BGA 1526	--
11.	Tiger Lake	10 nm	09/2020 01/2021 05/2021	M: G M: H M: H	4C+G 4C+G 8C+G	LPDDR4x-4267 DDR4-3200		BGA 1449 BGA 1449 BGA 1787	--
11.	Tiger Lake	10 nm	01/2021	D	8C+G	DDR4-3200		LGA 1700	500 Series
12.	Alder Lake	Intel 7	11/2021	D	8+8C+G	DDR5-4800		D: LGA 1700	600 Series
12.	Alder Lake	Intel 7	02/2022	M:H,P M: U	6+8C+G 2+8+G	DDR5-4800 LPDDR5-5200 LPDDR4x-4267		BGA 1744	--
12.	Alder Lake	Intel 7	05/2022	M: HX	8+8+G	DDR5-4800		BGA: 1781	600 Series
13	Raptor Lake	Intel 7	10/2022	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
13.	Raptor Lake	Intel 7	01/2023	M:HX	8+16C+G	DDR5-5600	2x64- bit	BGA 1964	700 Series
13.	Raptor Lake	Intel 7	01/2023	M:H,P M:U	6+8C+G 2+8C8C	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA 1744	--
14	Raptor Lake Refresh	Intel 7	10/2023	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
14	Raptor Lake Refresh	Intel 7	01/2024	M:HX	8+16C+G	DDR5-5600 DDR4-3200		BGA 1964	700 Series
14.	Raptor Lake Refresh	Intel 7	01/2024	M:U	2+8C+G	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA-1744	--
14	Meteor Lake	Intel 4	12/2023	M:H M:U-15	6+8+2C+G 2+8+2+G	DDR5-5600 LPDDR5X-7467		BGA 2049	--
14	Meteor Lake	Intel 4	12/2023	M:U-9	2+8+2+G	LPDDR5X-6400		BGA 2551	--

Examples: Increasing core counts in Intel's Core-based client processors [24]

2nd Sandy Bridge | **8th Coffee Lake** | **9th Coffee Lake Refresh**



(2011)
Up to 4 cores

(2017)
Up to 6 cores

(2018)
Up to 8 cores

3.6 How has the core count in Intel's client CPUs increased? (8)

c) Phase 3: Increasing the maximum core count over 10 in Intel's big.little lines -1

- AMD's gaming DT processors traditionally do not include GPUs.
- In 2019, their Zen 2-based generation introduced 16 core models, intensifying competition and prompting Intel to push its client processor core count beyond 10.
- In response, Intel launched its 12th gen. Alder Lake line in 2021, featuring a hybrid architecture with up to 8 performance cores (P) and 8 efficiency cores (E).

  This marked a significant shift in Intel's design philosophy.
- The momentum continued with the 13th gen. Raptor Lake line, which expanded the core configuration to a maximum of 8+16 cores, as shown in the preceding Table.

Please note that the 14th gen. Meteor Lake line, introduced in 2023, has 2 additional cores on the SOC die, to allow switching off the CPU die during low loads.

Phase 3: Increasing the maximum core count over 10 in Intel's monolithic big.little lines

	Processor	Techno-logy	Date of intro.	Seg-ment	Max. core count	Mem. speed (up to)	Mem. channels	Socket	PCH
8.	Cannon Lake	10 nm	5/2018	M:U	2C	DDR4-2400		BGA 1440	--
10.	Ice Lake	10 nm	7/2019	M: G	4C+G	DDR4-3200 LPDDR4- 3733		BGA 1526	--
11.	Tiger Lake	10 nm	09/2020 01/2021 05/2021	M: G M: H M: H	4C+G 4C+G 8C+G	LPDDR4x-4267 DDR4-3200		BGA 1449 BGA 1449 BGA 1787	--
11.	Tiger Lake	10 nm	01/2021	D	8C+G	DDR4-3200		LGA 1700	500 Series
12.	Alder Lake	Intel 7	11/2021	D	8+8C+G	DDR5-4800		D: LGA 1700	600 Series
12.	Alder Lake	Intel 7	02/2022	M:H,P M: U	6+8C+G 2+8+G	DDR5-4800 LPDDR5-5200 LPDDR4x-4267		BGA 1744	--
12.	Alder Lake	Intel 7	05/2022	M: HX	8+8+G	DDR5-4800	2x64- bit	BGA: 1781	600 Series
13	Raptor Lake	Intel 7	10/2022	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
13.	Raptor Lake	Intel 7	01/2023	M:HX	8+16C+G	DDR5-5600		BGA 1964	700 Series
13.	Raptor Lake	Intel 7	01/2023	M:H,P M:U	6+8C+G 2+8C8C	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA 1744	--
14	Raptor Lake Refresh	Intel 7	10/2023	D	8+16C+G	DDR5-5600		LGA 1700	700 Series
14	Raptor Lake Refresh	Intel 7	01/2024	M:HX	8+16C+G	DDR5-5600 DDR4-3200		BGA 1964	700 Series
14.	Raptor Lake Refresh	Intel 7	01/2024	M:U	2+8C+G	DDR5-5200 LPDDR5x-6400 LPDDR4x-4267		BGA-1744	--

3.6 How has the core count in Intel's client CPUs increased? (10)

Phase 3: Increasing the maximum core count over 10 in Intel's hybrid big.little lines -3

W	Processor	Techno-logy	Date of intro.	Seg-ment	Max. core count	Mem. speed (up to)	Mem. channels	Socket	PCH
14 CUS1	Meteor Lake	Intel 4 TSMC N5/N6	12/2023	M:H M:U-15W	6+8+2C+G 2+8+2C+G	DDR5-5600 LPDDR5X-7467	2x64-bit	BGA 2049	--
14 CUS1	Meteor Lake	Intel 4 TSMC N5/N6	12/2023	M:U-9W	2+8+2C+G	LPDDR5X-6400		BGA 2551	--
14 CUS1	Meteor Lake	Intel 4 TSMC N5/N6	04/2024	M: UL	2+8C+G	LPDDR5X-7467		BGA 2049	--
CUS2	Lunar	TSMC N3B/N6	09/2024	M:V	4+4C+G	LPDDR5X-8533		BGA 2833	--
CUS2	Arrow Lake	TSMC N3B/N5P/N6	10/2024	D: G	8+16C+G	DDR5-5600 CUDIMM-6400		LGA1851	--
CUS2	Arrow Lake	TSMC N3B/N5P/N6	01/2025	M: U M: H M: HX	2+8+2C+G 6+8+2C+G 8+16C+G8	DDR5-5600 CUDIMM-6400		BGA 2114	--
CU3	Panther Lake	Intel 1.8A/3, TSMC N3E/N6	01/2026	M: H	4+8+4+G	DDR5-7200 LPDDR5X-9600		BGA 2540	-

CUS: Core Ultra Series

3.7 How has the memory subsystem been enhanced?

3.7 How has the memory subsystem been enhanced?

The memory subsystem covers several issues, that we will discuss subsequently.

- 3.7.1 Evolution of the memory connection
- 3.7.2 At which rate have memory speeds been increased?
- 3.7.3 Introduction of laptop-oriented new memory types
- 3.7.4 Introduction of high-speed CUDIMM memories
- 3.7.5 Introducing in–package integrated main memory

3.7.1 Evolution of the memory connection

- 3.7.1 Evolution of the memory connection
- 3.7.2 At what rate have memory speeds been increased?
- 3.7.3 Introduction of laptop-oriented new memory types
- 3.7.4 Introducing on-package integrated main memory

3.7.1 Evolution of the memory connection

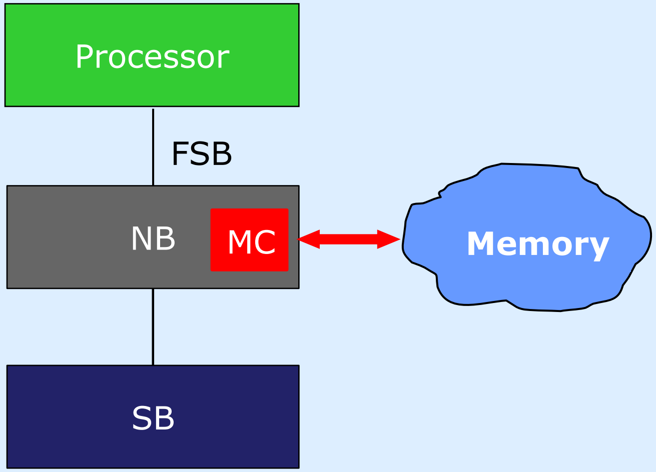
For connecting memory to the processor, there are two options, as depicted below.

Options for connecting memory to the processor

Connecting memory through the NB to the processor

The MC is integrated into the NB.

Data is transmitted via the FSB between the processor and memory.

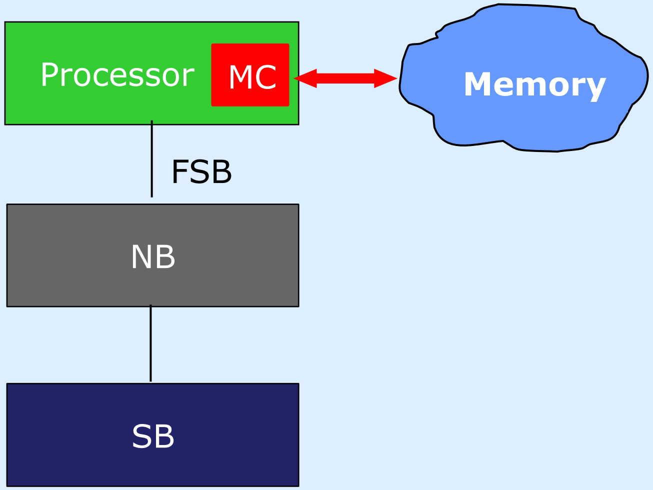


Up to the Penryn (2008)

Connecting memory directly to the processor

The MC is integrated onto the processor die.

Data is transmitted directly between the processor and memory.



From the Nehalem on (2009)

MC: Memory Controller   NB: North Bridge SB: South Bridge

3.7.1 Evolution of the memory connection (2)

Benefits of connecting memory directly to the processor

- Connecting memory directly to the processor offers notable advantages, primarily by reducing memory latency.
- Another significant advantage - especially in multiprocessor server architectures - is the substantial increase in memory bandwidth.

Namely, when memory is connected via the Northbridge (NB), the FSB must handle traffic for all processors.

This creates a bottleneck, as the available memory bandwidth becomes limited by the FSB's transfer rate, as illustrated in the following Figure.

3.7.1 Evolution of the memory connection (3)

Memory bandwidth restrictions in a 4S server when the memory is connected through the NB and a single FSB (based on [55]) -1

When the FSB is shared among all four processors, the average per-core bandwidth is one-fourth of the total FSB bandwidth, as indicated in the Figure below.

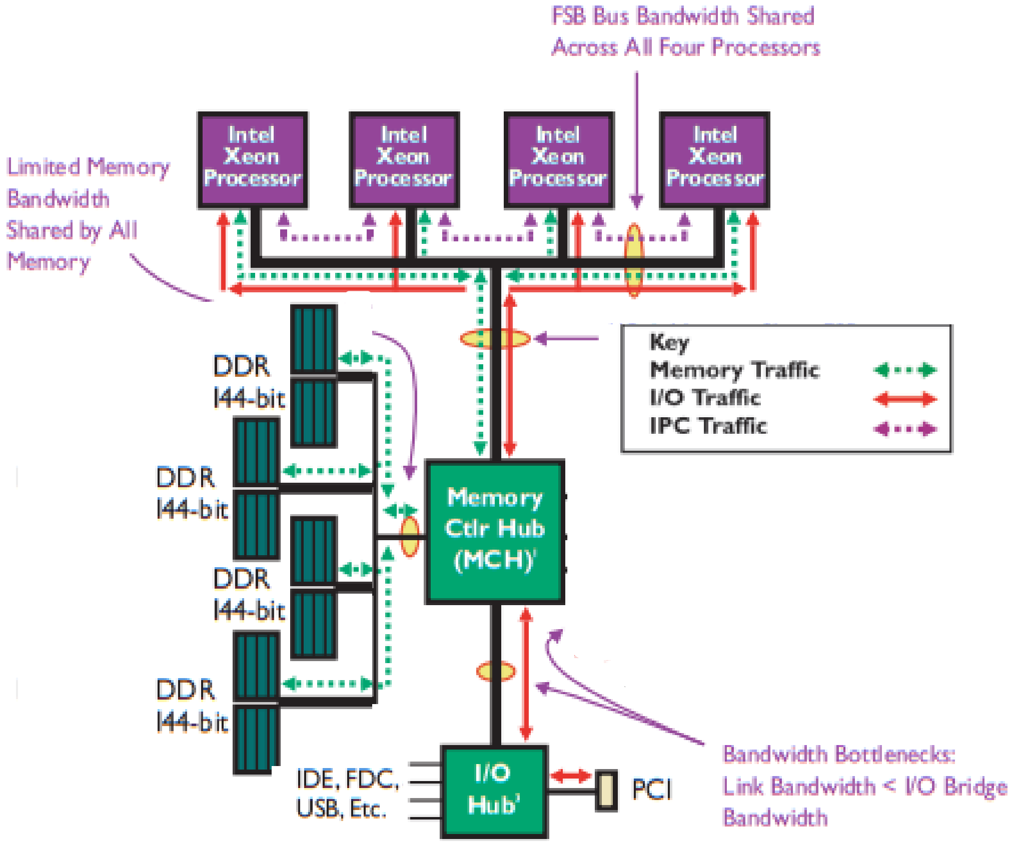
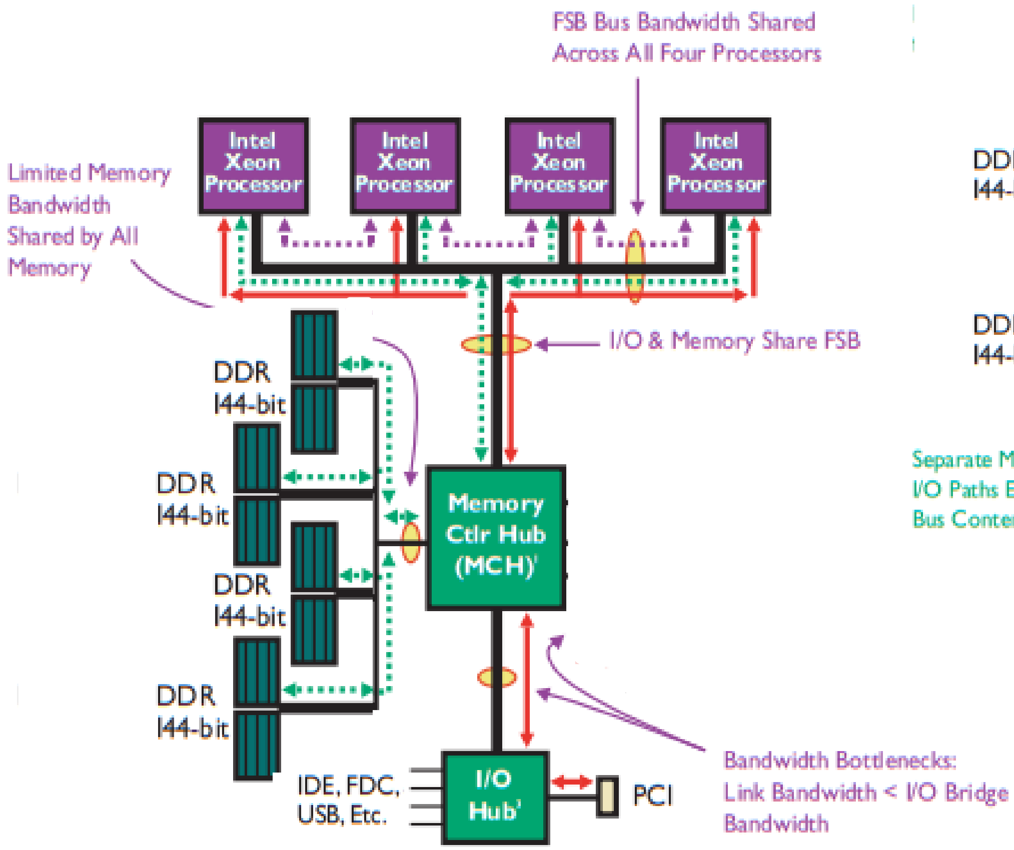


Figure: Connecting memory through the NB and a single FSB in a 4S server

4S: 4-Socket FSB: Front-Side Bus (Rendszerbusz)

Eliminating the memory bandwidth restrictions discussed in 4S servers by directly connecting memory to the processors (based on [55])

When memory is directly connected to the processors, the FSB does not limit the memory bandwidth anymore in multi-processor servers, and each processor gets the full bandwidth of the memory channel connected to it, as shown in the Figure below.



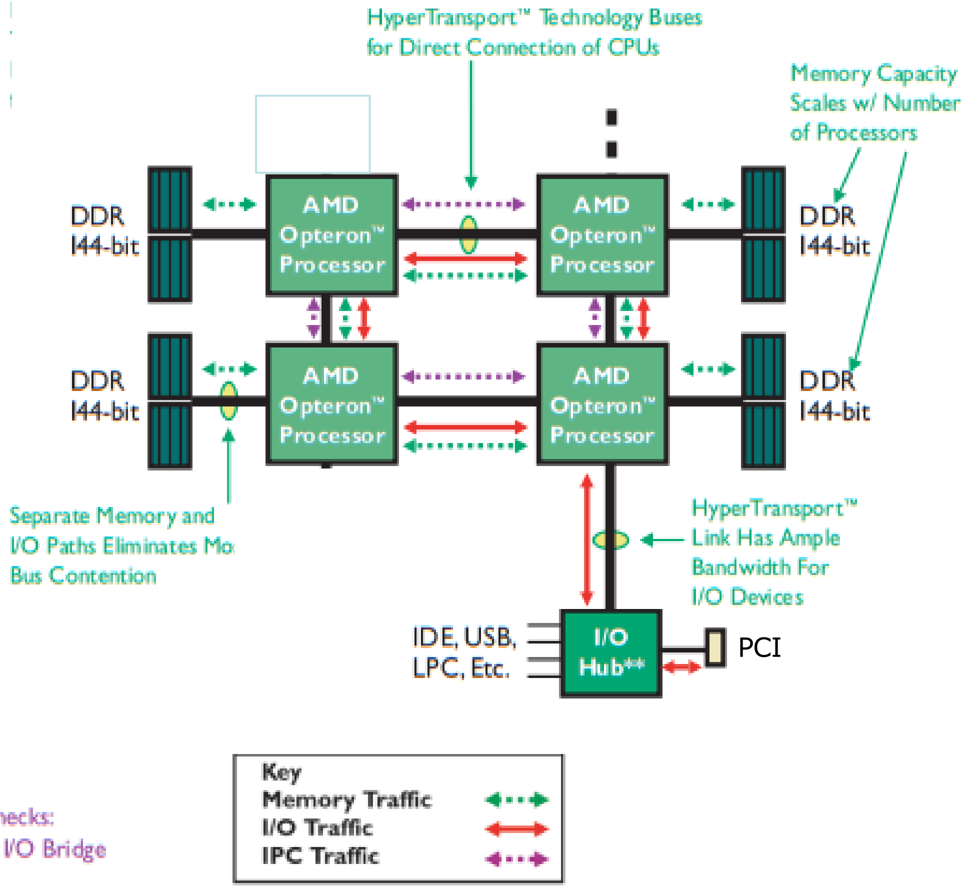


Figure: Connecting memory via the FSB

Figure: Connecting memory directly to the processors

The transition to direct memory connections in processors of various vendors

Due to its benefits, processor vendors gradually shifted to direct memory connection, as depicted in the Table below.

Options for connecting memory toprocessors

	Connecting memory through the NB to the processor	**Connecting memory directly to the processor**
Examples	HP PA-7000 (1991)	*HP PA-7100 (1992)* *HP PA-7150 (1994)* *and subsequent PA models*
	UltraSPARC II (1C) (~1997)	*UltraSPARC III (2001)* *and all subsequent Sun models*
	AMD's K7 lines (1C) (1999-2003)	*Opteron server lines (2C) (2003)* *and all subsequent AMD lines*
	POWER4 (2C) (2001)	*POWER5 (2C) (2005)* *and subsequent POWER families*
	Core 2 Duo line (2C) (2006) *and all preceding Intel lines* *Core 2 Quad line (2x2C) (2006/2007)* *Penryn line (2x2C) (2008)*	*Nehalem lines (4) (2008)* *2. gen. client-oriented Nehalem line* *and all subsequent Intel lines*
	Montecito (2C) (2006)	*Tukwila (4C) (2010)*

3.7.1 Evolution of the memory connection (6)

Comparing Intel's decision to connect memory directly to the processor with similar developments from other firms

- As shown in the previous Figure, Intel began connecting memory directly to the processors with its server-oriented Nehalem line in 2008, followed by the client-oriented 2nd gen. Nehalem family in 2009.
- It is worth noting that Intel was relatively late in adopting direct memory connection; other companies transitioned earlier, for example, AMD implemented direct memory memory connections approximately five years prior.

- The Table below illustrates Intel's shift toward a direct memory connection.

Intel's transition to direct memory connection in their Core-line

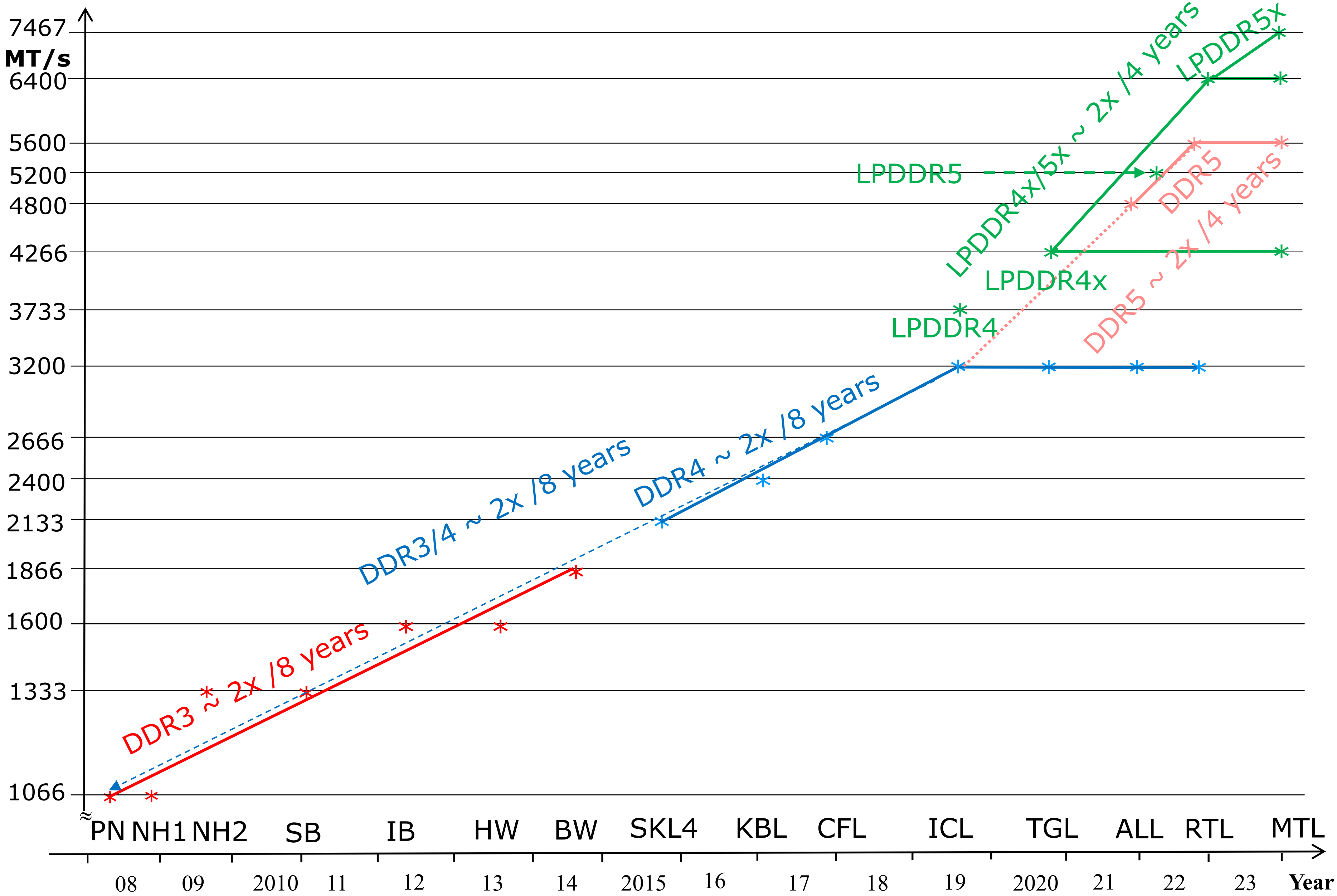
Gen.	Processor	Techno-logy	Date of intro.	Seg-ment	Max core count	Connecting memory	Mem. speed (up to)	Mem. channels	Socket	MCH/ PCH
1.	Core 2	65 nm	6/2006	M D	2C	Via the NB	DDR2-800	2	LGA 775	945-975
	Core 2 Quad	65 nm	6/2007	D	2x2C		DDR3-1067		LGA 775	3 Series
	Penryn	45 nm	6/2008	D	2x2C		DDR3-1067		LGA 775	4 Series
	2. G. Nehalem (Lynnfield)	45 nm	9/2009	D	4C	Via the processor	DDR3-1333		LGA 1156	5 Series
	Westmere (M: Arrandale) D: Clarkdale/	32 nm	1/2010	M D	2C+G				M:BGA 1288 D:LGA 1156	5 Series
2.	Sandy Bridge	32 nm	1/2011	M D	2C/4C+G		DDR3-1333		M: PGA 988 D: LGA 1155	6 Series
3.	Ivy Bridge	22 nm	4/2012	M D	4C+G		DDR3-1600			7 Series
4.	Haswell	22 nm	6/2013	M D	4C+G		DDR3-1600		M: BGA 1168 D: LGA 1150	D: 8 Ser.
4.	Haswell Refr.	22 nm	5/2014	D	4C+G		DDR3-1666			D: 9 Ser.
5.	Broadwell	14 nm	9/2014	M D	4C+G		DDR3-1866			D: 9 Ser.
6.	Skylake	14 nm	10/2015	M D	4C+G		DDR4-2133		M: BGA 1356 D: LGA 1151	D: 100 Series
7.	Kaby Lake	14 nm	8/2016	M D	4C+G		DDR4-2400			D: 200 Series
8.	Coffee Lake	14 nm	10/2017	M D	6C+G		DDR4-2666		M: BGA 1440 D: LGA 1151	D:300 Series
8.	Amber Lake	14 nm	8/2018	M	2C+G		LPDDR3-2133		BGA 1515	--
8.	Whiskey Lake	14 nm	8/2018	M	4C+G		DDR4-2400		BGA 1528	--
9.	Coffee Lake R.	14 nm	10/2018	D	8C+G		DDR4-2666		LGA 1151	300 Series
10.	Comet Lake	14 nm	8/2019	M D	10C+G		DDR4-2666		M: BGA 1528 D: LGA 1200	D: 400 Series
11.	Rocket Lake	14 nm	3/2021	D	8C+G		DDR4-3200		LGA 1200	

3.7.2 At what rate have memory speeds been increased?

3.7.2 At what rate have memory speeds been increased?

- Memory technology has steadily evolved over time, but at what rate?
- The following Figure provides insight into this question for the client processors of Intel's Core family.
- It shows the progression of maximum memory transfer rates across memory generations using a logarithmic scale to highlight its exponential growth.

At what rate have memory speeds been raised in Intel's client processors? -1



At what rate have memory speeds been raised in Intel's client processors? -2

- The graph above indicates that the transfer rate of DDR memories within the same technology rises exponentially.
- As shown,

  DDR3 and DDR4 memories typically require around 8 years to double their transfer rates, while DDR5 and LPDDRx memories only need about 4 years for doubling.

- Here, we note that regarding the transfer rates of DDR5 and LPDDR memories, currently, only a limited number of data points are currently available.

  Therefore, the related statement should be taken with caution.

- Additionally, it is worth mentioning that Intel's Alder Lake (ALL) and Raptor Lake (RTL) processors support both the DDR4 and DDR5 memory types.

  Notably, both processor families have the same DDR4 transfer rate of DDR4-3200.

LPDDR: Low Power DDR

3.7.3 Introduction of laptop-oriented new memory types

3.7.3 Introduction of laptop-oriented new memory types

In recent years, Intel has introduced three new memory types, primarily for its laptop processors, which are:

a) the LPDDR and LPDDRx memories and
b) the LPCAMM2 memory and
c) the CUDIMM memory.

In this Section, we will describe these new memory types.

Remark

Remember that Intel refers to laptop processors as laptop processors.

LPCAMM: pronounced as ɛl.piːˈkæm.
CUDIMM: pronounced as kjuː dɪm.

3.7.3 Introduction of laptop-oriented new memory types (2)

a) What are LPDDR and LPDDRx memories, and why are they important?

- As their names suggest, LPDDR (Low Power DDR) and its successor, LPDDRx memories are designed to consume significantly less power than standard DDR memories, making them ideal for laptop and embedded applications.

  This energy efficiency is primarily achieved through reduced operating voltages, which lower both the dynamic and static dissipation.

- Despite their lower power requirements, LPDDR memories deliver the same data transfer rates to conventional DDR counterparts, ensuring high performance without compromising battery life.
- LPDDRx memories have an even more advanced memory design than their LPDDR predecessors, offering even higher data rates while consuming less power compared to standard DDR memories.
- Note that both LPDDR and LPDDx memories are typically packaged using BGA sockets that are soldered directly to the mainboard.

  This design choice enhances compactness but eliminates upgrading and replacing.

- The Table below provides a summary of the main features of LPDDR and LPDDRx memories.
- Remark

  Additionally, there were more advanced LPDDR memories designed by several vendors referred to as LPDDRE memories.

  They provide higher transfer rates than traditional LPDDR memories (see next Table).

Types and main features of subsequent LPDDR/LPDDRx generations [56] -1

LPDDR	1	2	3	4	4X	5	5X
Memory array clock (MHz)	200	200	200	200	266.7	400	533
Data width	x16/x32	x16/x32	x16/x32	x16			
I/O bus clock frequency (MHz)	200	400	800	1600	2133	3200	4267
Peak Data transfer rate for LPDDR (MT/s)	400	800	1600	3200	4267	6400	8533
Peak Data transfer rate for LPDDRE (MT/s)	533	1066	2133	4267			
Supply voltage(s)	1.8 V	1.2 V	1.2 V	1.1 V	0.6 V, 1.1 V	0.5 V, 1.05 V	0.5 V, 1.05 V
Introduction	2009	2009	2012	2014	2017	2019	2021

- LPDDR3/LPDDR4 memory modules operate at significantly lower voltages than traditional DDR memories (DDR: 2.5 V, DDR2: 1.8 V, DDR3: 1.5 V, DDR4: 1,2 V).
- Note that despite their lower power consumption, LPDDR and standard DDR memory generations offer the same maximum data rates.
- LPDDRx memories introduce further advancements, including the use of dual voltages, which enable even higher transfer rates than earlier LPDDR designs.
- Notably, LPDDR5 memories switched to dual voltage designs as well.

Principle of the LPDDR4X memory (released by Samsung in 01/2017) [56]

- The LPDDR4X design introduces a key innovation by separating the output driver voltage (VDDQV) from the supply voltage (VDDQ), reducing VDDQV from 1.1 V to just 0.6 V, as illustrated in the Figure below.

  This results in both higher transfer rates and greater power savings, as the next Figure shows.
- However, the new design comes with a trade-off, it needs an additional power rail.

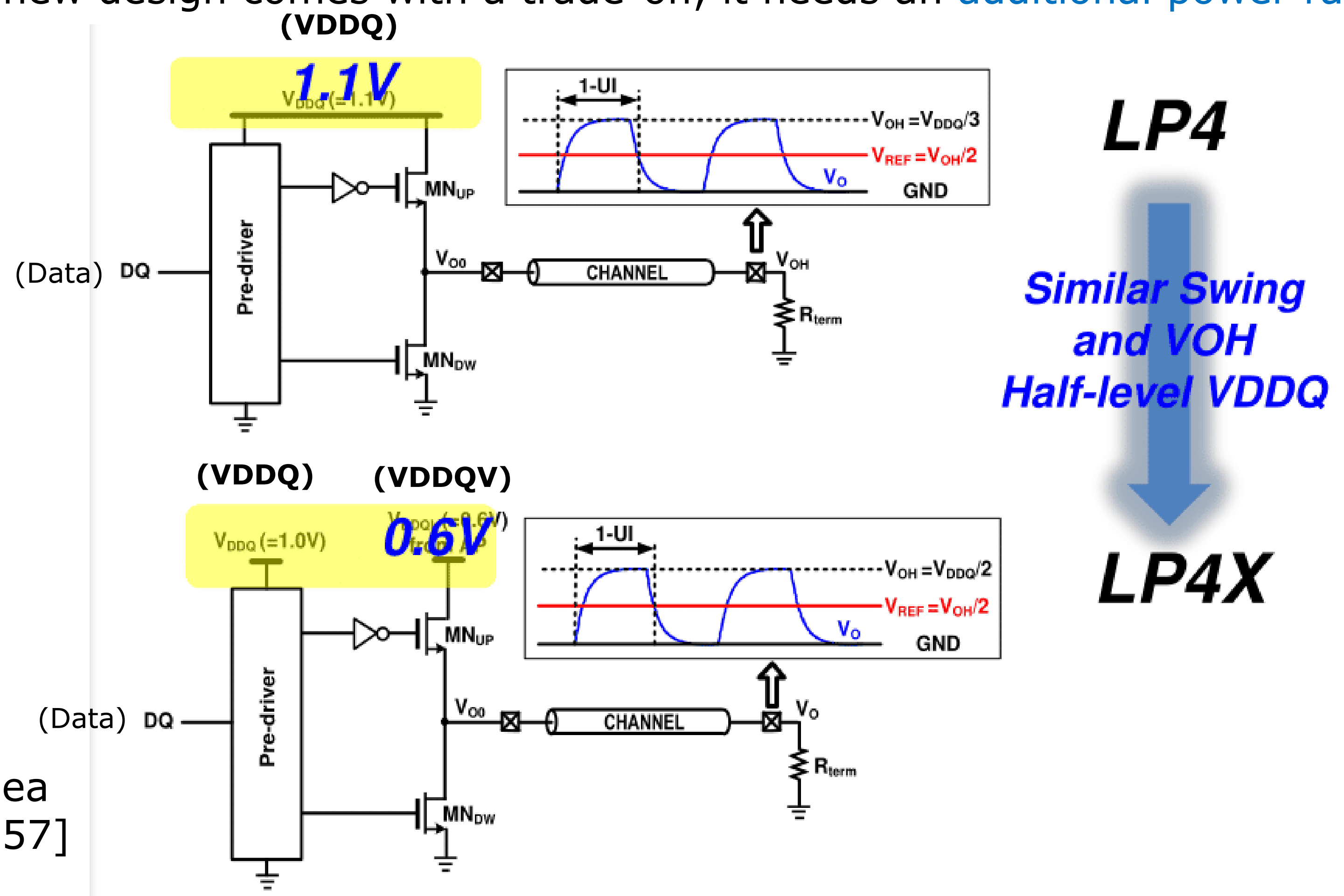


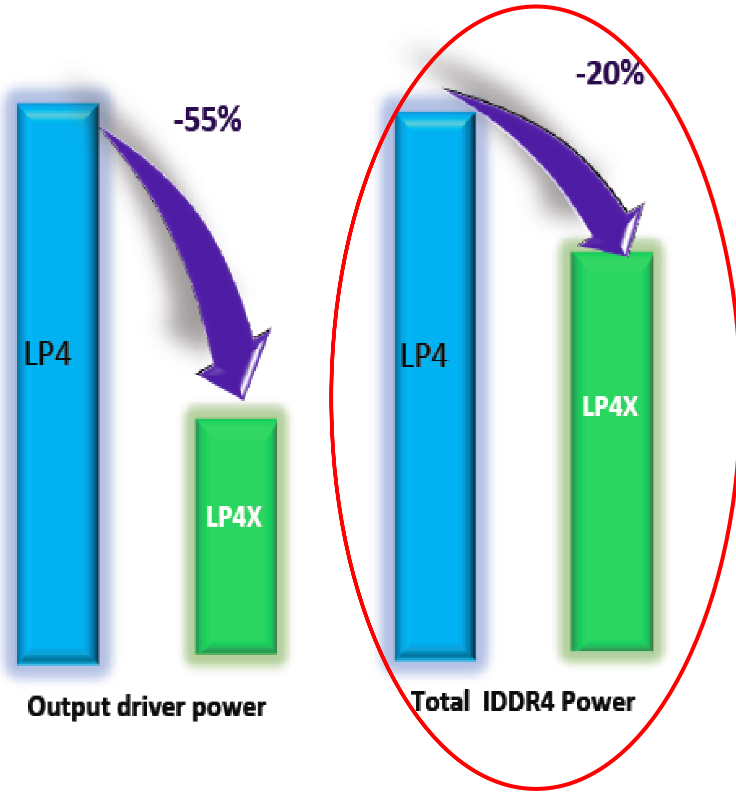
Figure: The idea of LPDDR4X [57]

Power consumption of LPDDR4x memories compared to LPDDR memories [58]

LP4/LP4x Spec Comparison

Spec	LPDDR4	LPDDR4X
Function	Same	
Speed*	3200/3733/4266	3200/3733/4266
AC Timing	Same	
VDD1/VDD2	Same(VDD1=1.8V/VDD2=1.1V)	
VDDQ	1.1V	0.6V
Power Eff. (mW/GB/s)	1X	0.8X

Power reduction by LPDDR4x



As shown in the Figure, LPDDR4x memories consume approximately 20 % less power than LPDDR4 memories.

3.7.3 Introduction of laptop-oriented new memory types (6)

Standardization of the LPDDR and LPDDR4X memories [56]

- Initially, Micron and Samsung introduced low-power memory solutions for laptop processors, branding them as Mobile DDR memories.

  By 2009, this naming was changed to LPDDR1, marking the beginning of a new memory technology.

  Following the success of the early implementations of this memory type, JEDEC released formal specifications starting with LPDDR2 in 2009.

  In the following years, JEDEC released the subsequent LPDDR standards, such as the LPDDR3 to LPDDR5 specifications.

Recently, in 07/2015 JEDEC released the LPDDR6 standard as well.

The specified peak data transfer rate is 14.4 Gb/s and the sub-channel width is x12.

- In 2016, Samsung unveiled its LPDDR4x design, which introduced dual operating voltages, to achieve both lower power consumption and higher speed.
- JEDEC later standardized this innovation with the LPDDR4x specifications released in 2017, followed by the LPDDR5x standard in 2021.

JEDEC (dʒɛdɛk) : It is the leading organization for setting open standards for the microelectronics industry.

Use of LPDDR memories in client processors

- LPDDR4 memories were initially developed for smartphones and tablets, where low power consumption is critical.

  Over time, its efficiency led to broader adoption, including into client processors.

- More recently, both LPDDR and LPDDRx memories have been widely utilized in Intel's client processors, as indicated in Section 3.7.2 before.

  This shift marks the growing demand for energy-efficient, high-performance memory solutions across computing platforms.

b) The LPCAMM memories [59]

- LPCAMM stands for Low-Power Compression Attached Memory Module, a new memory technology standardized by JEDEC for energy efficient computing.
- Its distinctive feature lies in its electrical interface, LPCAMM modules establish electrical contacts through compression rather than insertion into a socket or soldering.
- This approach enables a compact, replaceable design in contrast to the BGA socketed memories.
- So far, two generations of LPCAMM memories have been released:
  - The 1st gen. LPCAMM memory module is a single-channel LPDDR5 design, standardized by JEDEC in 03/2023.
  - The more advanced 2nd gen. LPCAMM2 memory module supports dual-channel LPDDR5 or LPDDR5x memory, it was standardized in 12/2023.
- Next, we will give a brief overview of the 2nd gen. LPDDR2 memory module.

JEDEC: It is the leading body for establishing microelectronic standards.

The LPCAMM2 memory module [59]

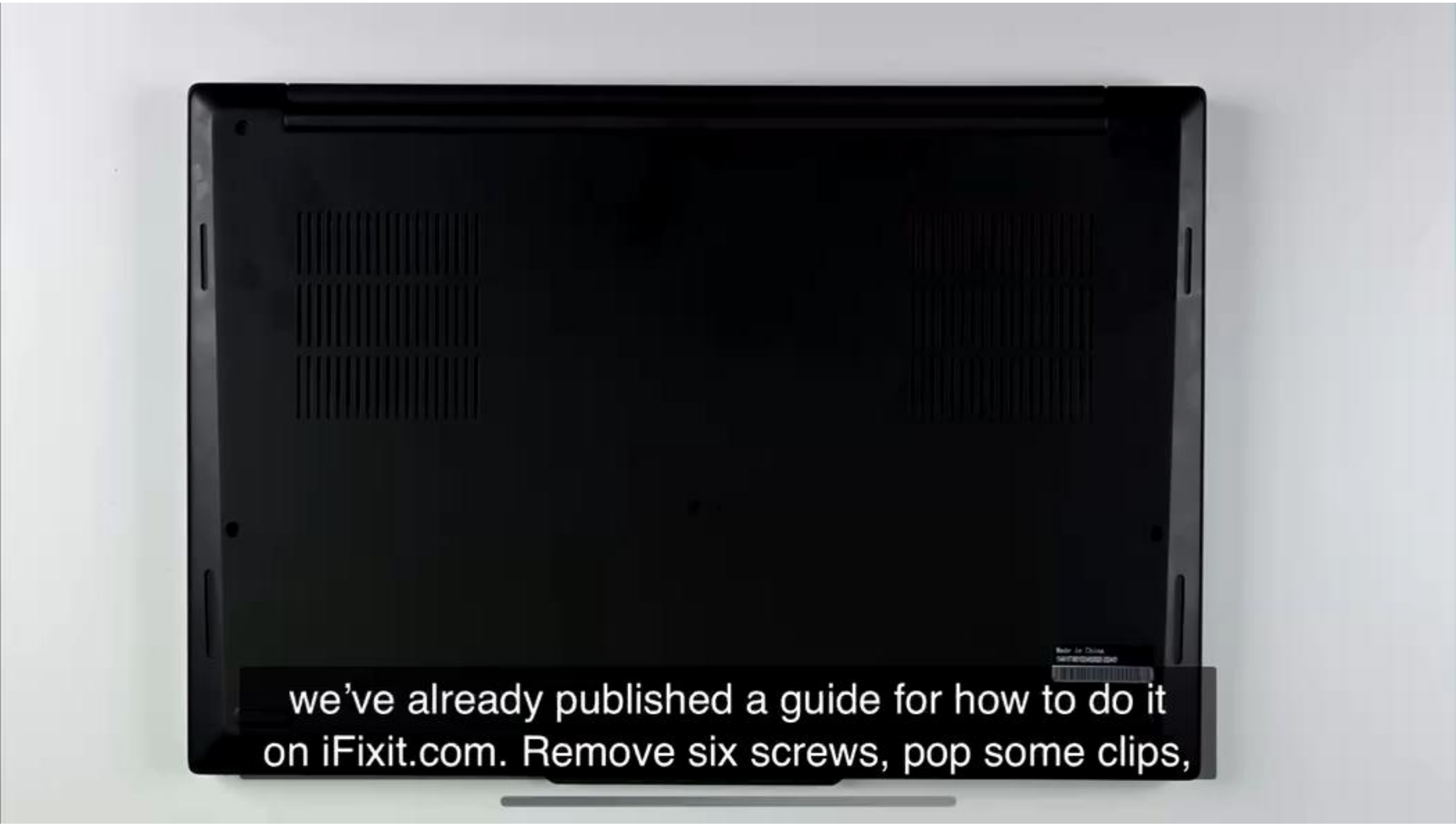
- LPCAMM2 memory modules provide dual 64-bit LPDDR5/x or DDR5 memory channels. LPCAMM places the memory chips on thin, replaceable circuit boards (as seen in the Figure below).

(Összenyomással csatlakoztatott)

Figure: LPCAMM2 memory module [59]

- Unlike traditional memory modules that are either inserted into sockets or soldered directly onto the mainboard, LPCAMM memory technology establishes electrical contacts through compression, as shown in the following video.

Demonstration of removing and replacing an LPCAMM2 memory module [124]



3.7.3 Introduction of laptop-oriented new memory types (11)

Use of the LPCAMM memory technology

- The LPCAMM2 memory made its debut in 2024 with the Lenovo (ThinkPad P1 Gen 7) laptop which provides 64 GB of LPDDR5x memory.
- As of Q4 2024, several companies, including Micron, Samsung, and Hynix, are offering or preparing LPCAMM2 memories with capacities of 32 GB or 64 GB.

c) CUDIMM memories

- CUDIMM means Clocked, Unbuffered DDR5 DIMM memory modules.
- They provide higher speeds than the conventional DDR5 memory by integrating a clock driver (CKD) directly onto the memory module, as shown below.
- This component reshapes both the timing and voltage of the incoming clock signal, significantly improving signal integrity.

  The new layout allows a doubling of the clock frequency delivered by the memory controller (QCK is twice as high as DCK).

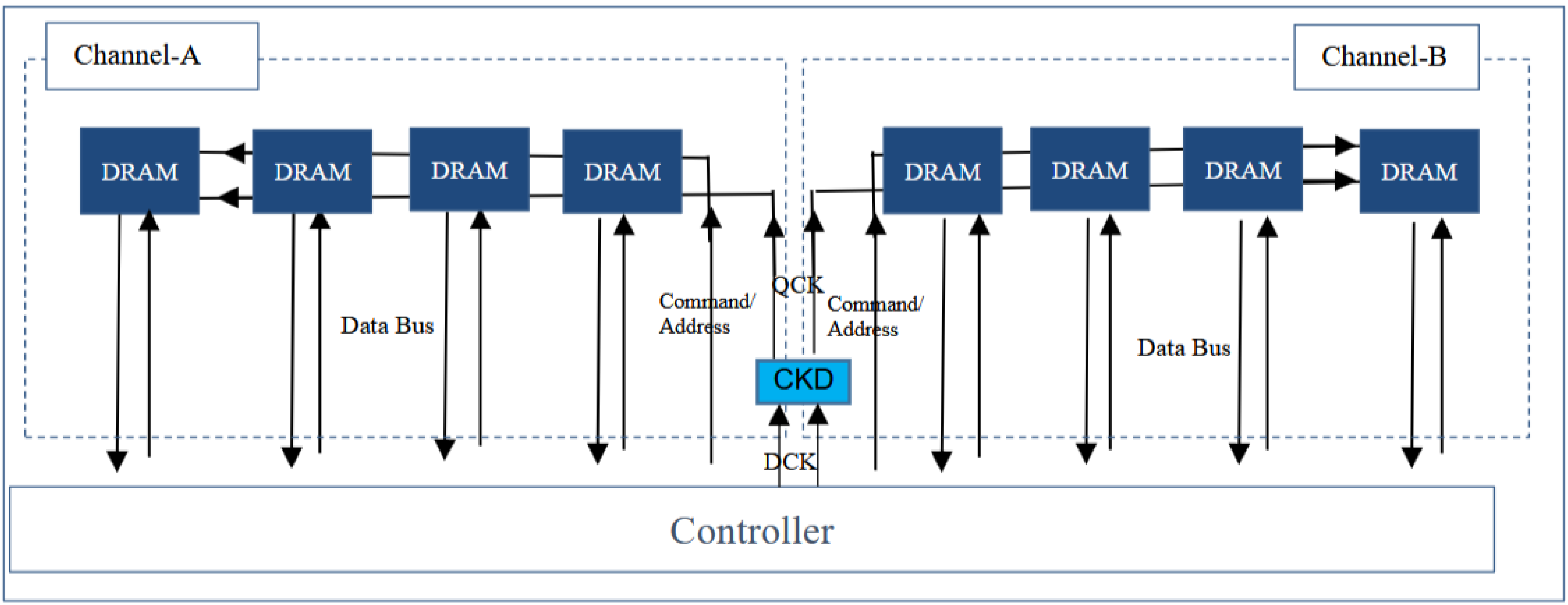
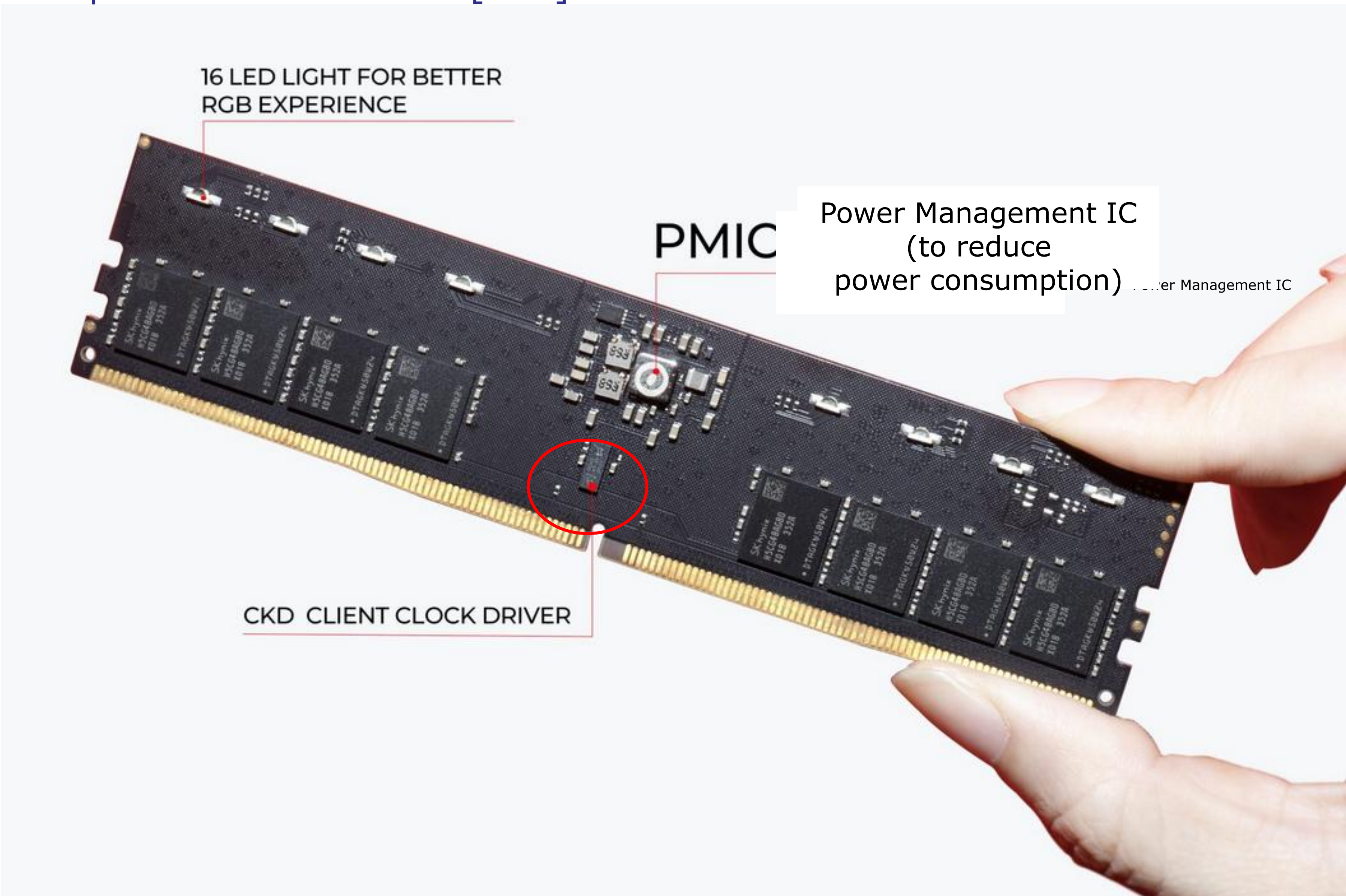


Figure: Example topology of a CUDIMM memory [146] JEDEC Standard No. 323

Example CUDIMM module [147]



3.7.3 Introduction of laptop-oriented new memory types (14)

Standardization of CUDIMM memories

- In early 2024, JEDEC standardized DDR5 CUDIMM memories.

  These modules are drop-in compatible with existing DDR5 UDIMMs, using the same 288-pin socket and operating at 1.1.V supply voltage.
- The standard specifies memory capacities of 16/24/32 or 64 GB.
- CUDIMM memory modules feature two operation modes.
  - In one mode, they double (and potentially quadruple in future designs) the clock frequency received from the memory controller.
  - In the other, they simply bypass the clock frequently received.
- This dual-mode capability allows CUDIMMs to deliver improved performance while maintaining compatibility with existing DDR5 platforms.

DIMM: Dual-In-Line Memory Module

UDIMM: Unbuffered DIMM

JEDEC: (Joint Electron Device Engineering Council) is a semiconductor standardization body.

Emergence and use of CUDIMM memories

- First CUDIMM memories appeared on the market from various vendors, such as Micron (Crucial), Kingston, and others with speeds up to DDR5-9600 by the end of 2024.
- As of 01/2025, several OEMs offer mainboards with 2x16/2x24 or 2x 32GB CUDIMM memories operating at speed grades of up to DDR5-9600 for
  - Intel's 14th gen. Arrow Lake (2024) and
  - AMD's Zen 4-based Ryzen 8000G DT APU series (2024) as well as Zen 5-based Ryzen 9000X (2024) gaming DT processors.

Remarks on memory types targeting server processors -1

- Server processors must support large memory capacities, which requires connecting multiple memory modules per channel.

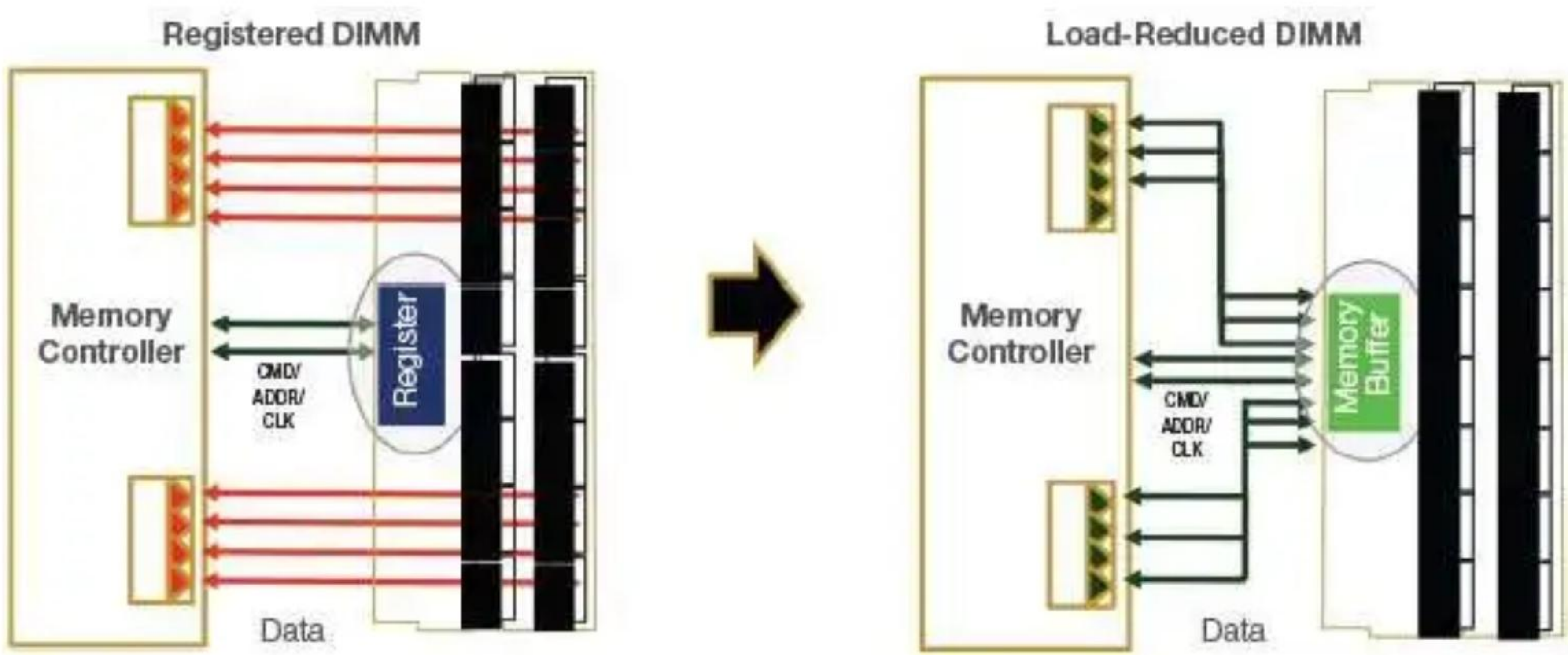
  However, parallel connected memory modules introduces high electrical load, especially on the clock, address and control signal lines of memory controllers.

  Excessive load can lead to signal degradation and timing errors which cause memory errors.

- RDIMM and LRDIMM are memory types especially developed to mitigate these issues by incorporating buffering mechanisms to preserve signal integrity.

RDIMM Registered DIMM (Buffering of clock, command, and address signals

Contrasting the layouts of RDIMM and LRDIMM memories [173]



- RDIMM memory modules include a dedicated component on the module (called Register in the Figure), which buffers the command, address, and clock signals arriving from the memory controller. to the DRAM chips.

  However, RDIMMs do not buffer data signals, data flows directly between the MC and the DRAM chips.

- In contrast, LRDIMM memory modules extend buffering to the data signals as well, as indicated by the Memory Buffer component in the Figure.

Comparing the use of various buffering techniques implemented in memories of client and server processors

DIMM Type	Clock driver	Buffering of clock, command and address signals	Buffering of data signals	Typical use
UDIMM	❌ No	❌ No	❌ No	Clients, workstations
CUDIMM	✅ Yes	❌ No	❌ No	Clients
RDIMM	❌ No	✅ Yes	❌ No	Standard servers
LRDIMM	❌ No	✅ Yes	✅ Yes	High-capacity servers

UDIMM: Unbuffered DIMM

3.7.4 Introducing on-package integrated main memory

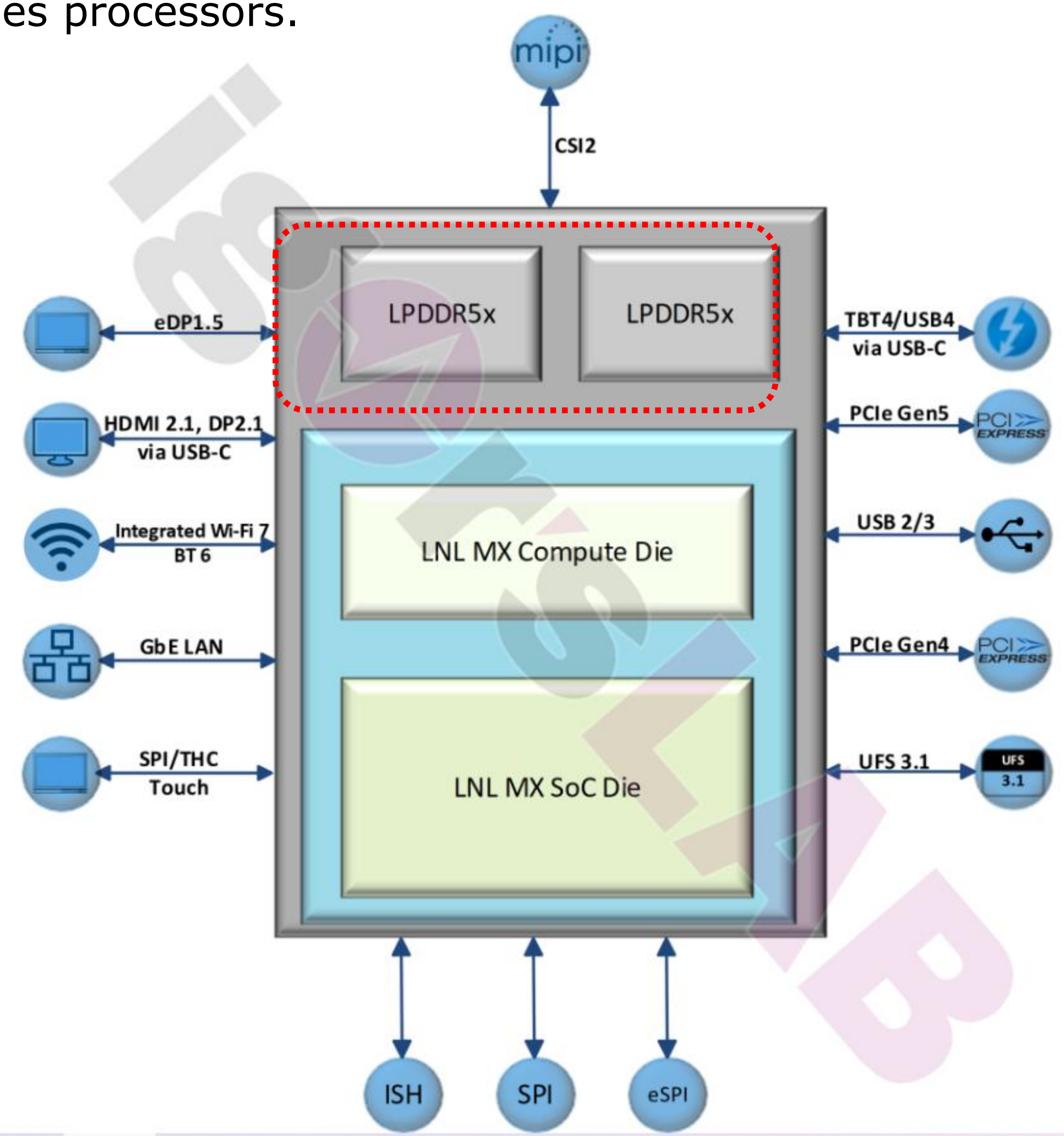
3.7.4 Introducing on-package integrated main memory

- In 2024 Intel released its Lunar Lake laptop processor, designed to compete with ARM-based Snapdragon X series processors.

A key innovation of the Lunar Lake line of processors is the integration of 2x8 GB or 2x16 GB LPDDR5x-8533 main memory into the package, as, illustrated on the right.

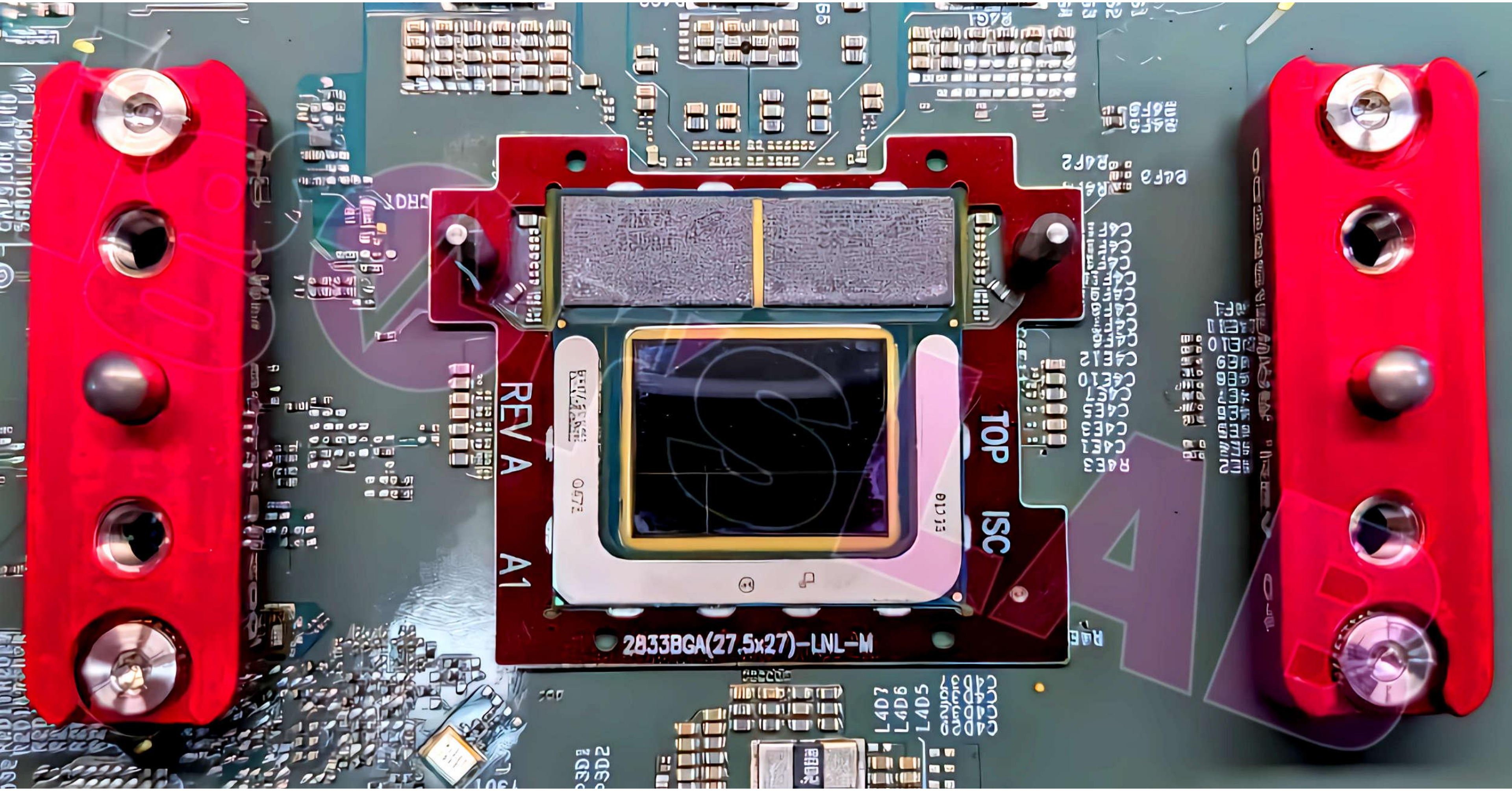
On-Package integrated memories offer lower memory access latencies, but they prevent the ability to replace or expand the main memory.

[157].



3.7.4 Introducing on-package integrated main memory (2)

Intel’s Lunar Lake processor with dual on-package LPDDR5x memory dies [164]



3.7.5 Introducing on-package integrated main memory (3)

Remark 1

Apple's M-series processors also came with on-package-integrated main memories.

Apple's M1 processor, launched in 2020, comprises three on-package integrated dies: the CPU die, and two low-power LPDDR4x-4266 memory dies, each of 8 GB size, as depicted in the Figure below.

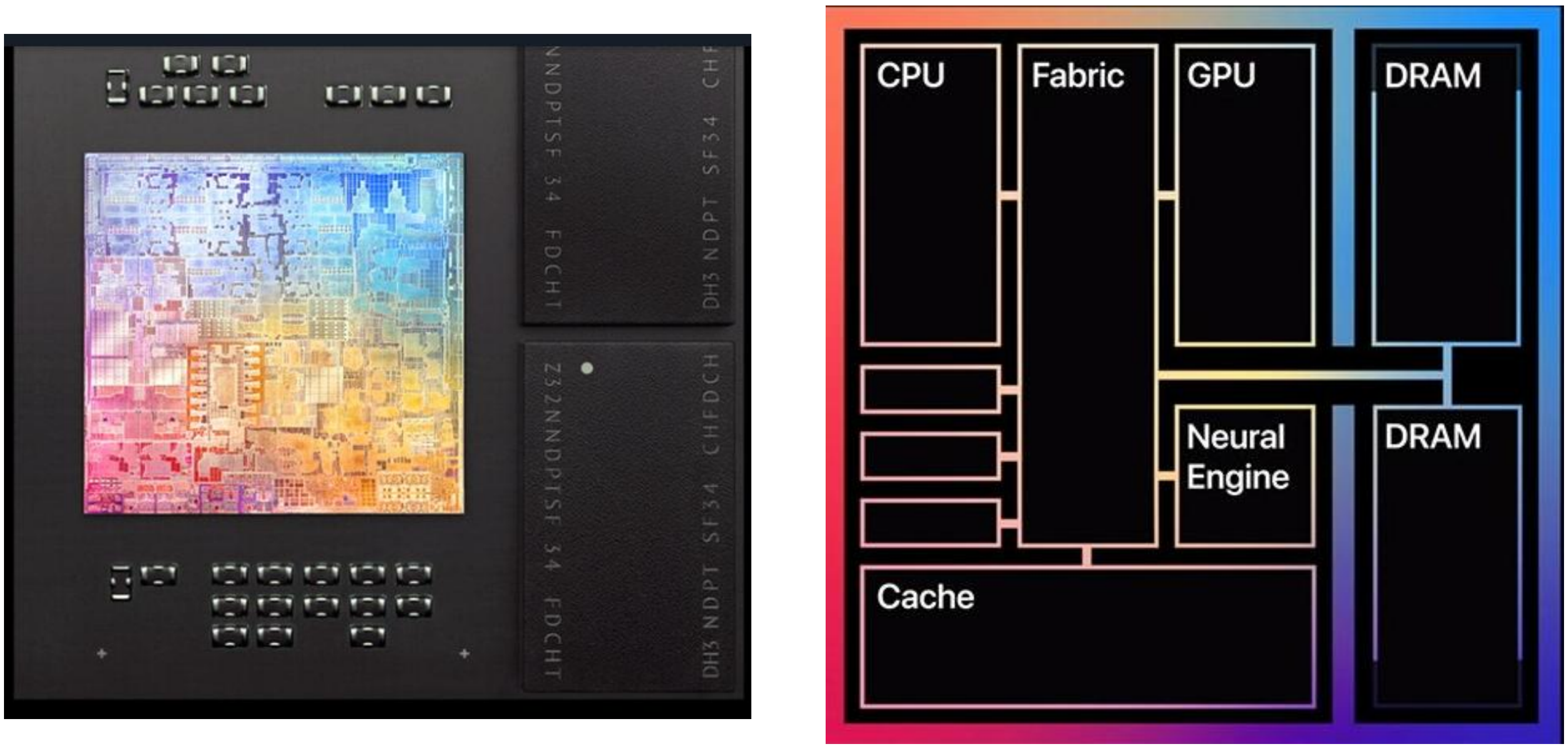


Figure: Layout of Apple's M1 chip (2020) [132]

Remark 2

- Smartphone processors utilize another type of on-package integrated main memories, known as PoP (Package-on-Package) memories.
- This technology involves stacking dies vertically and properly interconnecting their solder balls, as illustrated in the Figure below.

Figure: Possible PoP structure for implementing a baseband or application processor and two memory dies [133]

- The first PoPs emerged in cell phones around 2004 [133].
- Apple's original iPhone was an early example of using the PoP technology.

3.7.5 Introducing on-package integrated main memory (5)

Remark: Introduction of on-package integrated PCHs in Intel's Core-based "laptop" lines

- Intel introduced on-package-integrated PCHs in their 4th gen. "laptop" Haswell series in 2013, as shown in the Figure below.
- Subsequently, all of Intel's laptop-oriented processors (characterized by low TDP values of less than 50 W), implement in-package integrated PCHs, with a few exceptions.

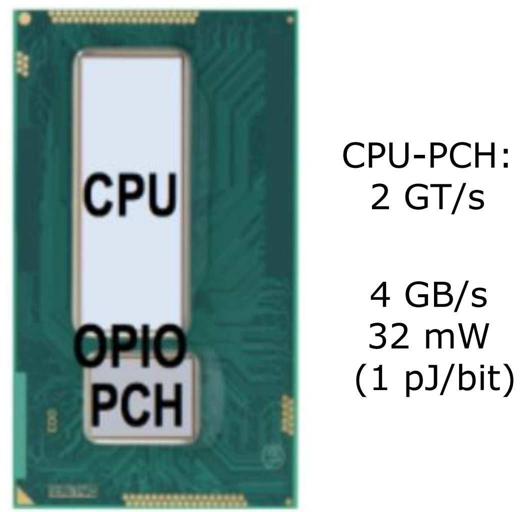


Figure: On-package integrated PCH

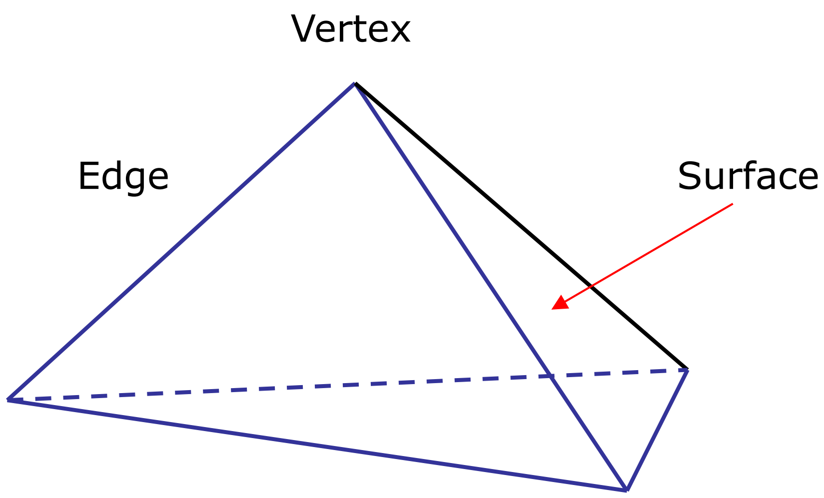
- In this solution, the CPU die, and the PCH die are interconnected by an IO interface called OPIO (On-Package I/O).
- The OPIO interface allows for high-bandwidth (4 GB/s) low-power (1 pJ/bit) communication between the two dies [148].
- The two dies are enclosed by a Multi-Chip Packaging (MCP).

PCH: Platform Control Hub   MCP:Multi-Chip Package

3.8 Basics of 3D graphics processing

3.8 Basics of 3D graphics processing

In 3D graphics processing, objects are represented by a set of triangles, as indicated in the Figure below.

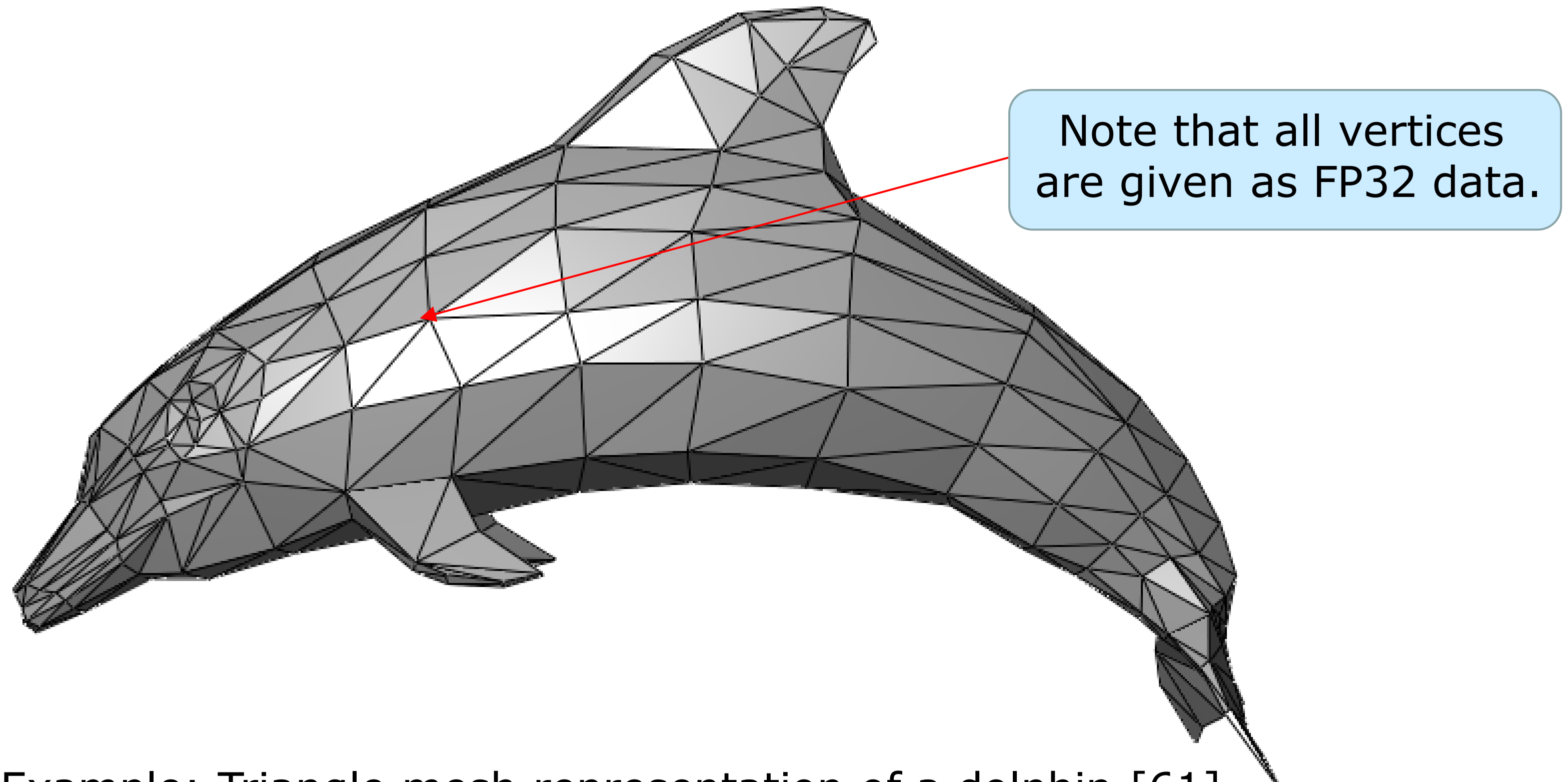


Vertices in graphics

- have three spatial coordinates (represented by FP32 data), and
- supplementary information necessary to render the object, such as
  - color (RGB)
  - texture
  - reflectance properties
  - etc.

Triangle mesh representation of objects in graphics processing

Graphics software and hardware operate more efficiently on a group of triangles, called triangle meshes rather than on individual triangles, so graphics objects are typically represented by triangle meshes.



Example: Triangle mesh representation of a dolphin [61]

3.8 Basics of 3D graphics processing (3)

Shaders (in graphics)

- The term "shader" refers to graphics routines that run on the GPU's graphics cores and perform tasks such as processing vertices or pixels of 2D/3D objects.

3.8 Basics of 3D graphics processing (4)

The main types of shaders and their basic tasks

There are three main types of shaders, as shown below.

Main types of shaders

Vertex shaders	**Pixel shaders (Fragment shaders)**	**Geometry shaders**
Transforms the 3D-position of each vertex in the virtual space to 2D coordinates, at which it will appear on the screen	Calculates the color of the pixels	Can add or remove vertices from a mesh of vertices

- We note that there are further shader types as well, like compute shaders, ray tracing shaders, but they are beyond the scope of our Lecture.
- Graphics APIs (like Microsoft's DirectX or OpenGL) assume the triangle representation of graphics objects.

Shaders: Graphics programs running on the GPU hardware

3.8 Basics of 3D graphics processing (5)

Shader languages

- Shaders are written using shader languages and will be run on the GPU.
- Examples of the most common shader languages are:
  - HLSL (High-Level Shader Language):
    Used primarily with DirectX.
    Developed by Microsoft.
    Its syntax is similar to C/C++.
  - GLSL (OpenGL Shading Language):
    Used with OpenGL.
    Developed by the Khronos Group.
    Also has a C-like syntax.
  - Cg (C for Graphics):
    Developed by NVIDIA.
    Similar to HLSL and GLSL.

Historical remark

- In early graphics processing, in the beginning of the 2000s' different shaders run on different parts of the graphics hardware, as seen below on the left.
- However, since about 2006, all shaders have been running on the same hardware. which is the GPU, instead of using dedicated HW units for each shader, as shown below on the right.

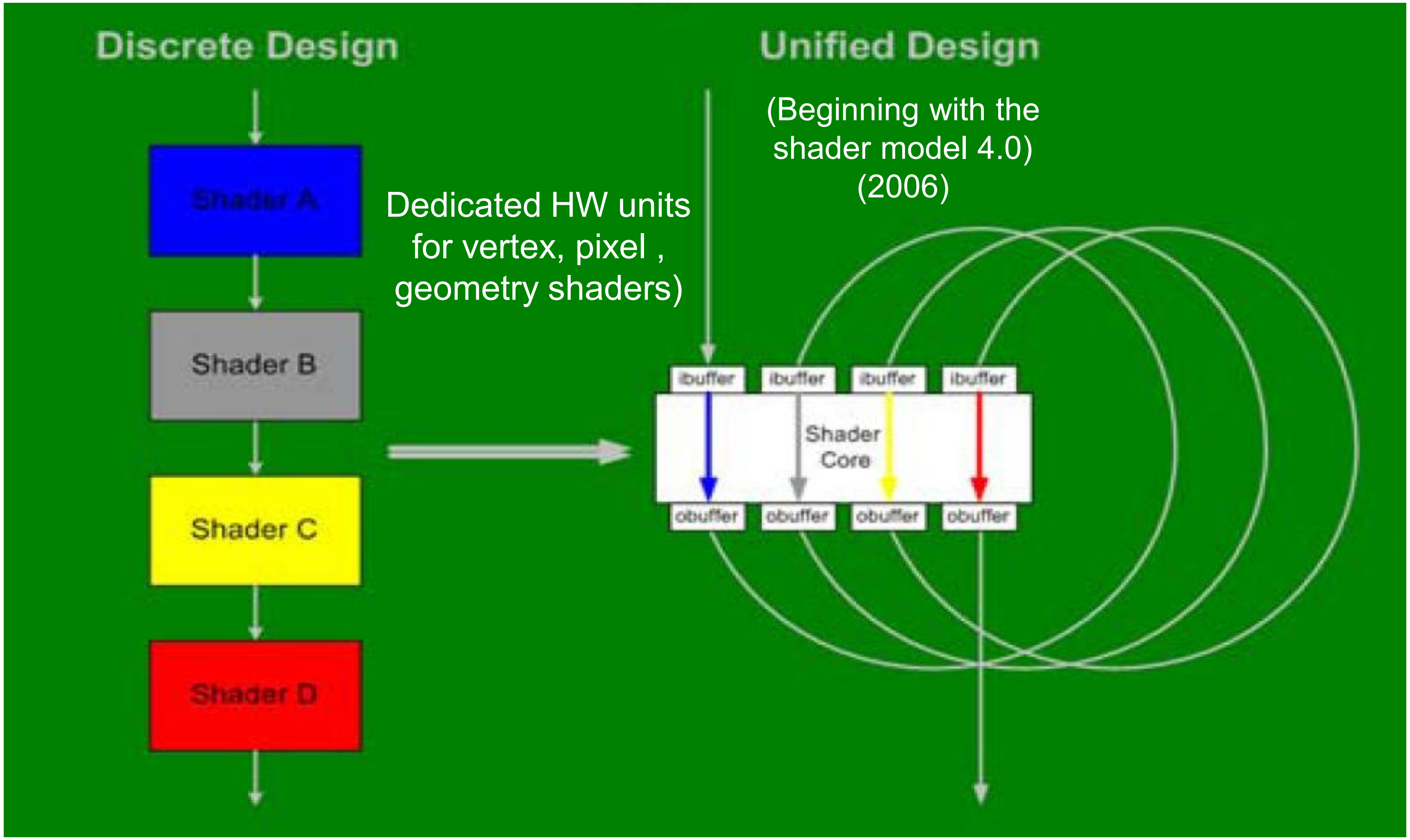


Figure: Basic shader designs [62]

Remark on graphics processing

- Note that due to the huge number of vertices to be processed for objects, graphics computations need a vast number of FP32 execution resources.

- This is why integrated GPUs can execute a few thousand FP32 operations per cycle, while recent graphic cards have tens of thousands of FP32 operations per cycle.
- E.g. NVIDIA's GeForce RTC 5090 can perform 21760 FP32 MAC operations per cycle, that is 43520 FP32 operations per cycle.

MAC: Multiply and Accumulate (axb +c) operation.

4. Case examples for recent client processors

- 4.1 Case example 1: Intel’s Core Ultra Serie 2 (Lunar Lake) laptop family (2024)
- 4.2 Case example 2: Intel’s Core Ultra (Panther Lake) laptop family (2026)

4.1 Case example 1: Intel's Core Ultra Serie 2 (Lunar Lake) laptop family (2024)

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (2024)

- The Lunar Lake processor is designed to compete with Qualcomm's ARM-based low-power, high-performance Snapdragon X processor optimized for Windows on ARM PCs.
- Its key features are as follows:
  - heterogeneous, dual-die big.little design,
    the tiles are: the Compute tile and the Platform controller tile.
    The Compute tile is manufactured using TSMC's N3B process, while the Platform controller tile is fabricated using TSMC's N6 technology.
  - 4 P-cores and 4 LPE-cores (Low-Power E-cores),
    8-wide P-cores and 3x3-wide E-cores.
    no multithreading,
  - On-package 16 or 32 GB DDR5x-8522 DRAM providing low latency, low-power memory,
  - NPU 4 with up to 48 TOPS,
  - Xe2 GPU (8 cores, 8 MB L2)
  - TDP: 17-30 W

Remark

In the Lunar Lake processor, Intel designates its E-cores as LPE (Low-Power E-cores) to highlight their improved power efficiency compared to previous E-cores.

In future processors (such as the Panther Lake, expected in Q1/2026), Intel plans to implement both LPE-cores and traditional E-cores alongside P-cores.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (2)

Key features of Intel's Core Ultra 200V laptop processors (2024) [158]

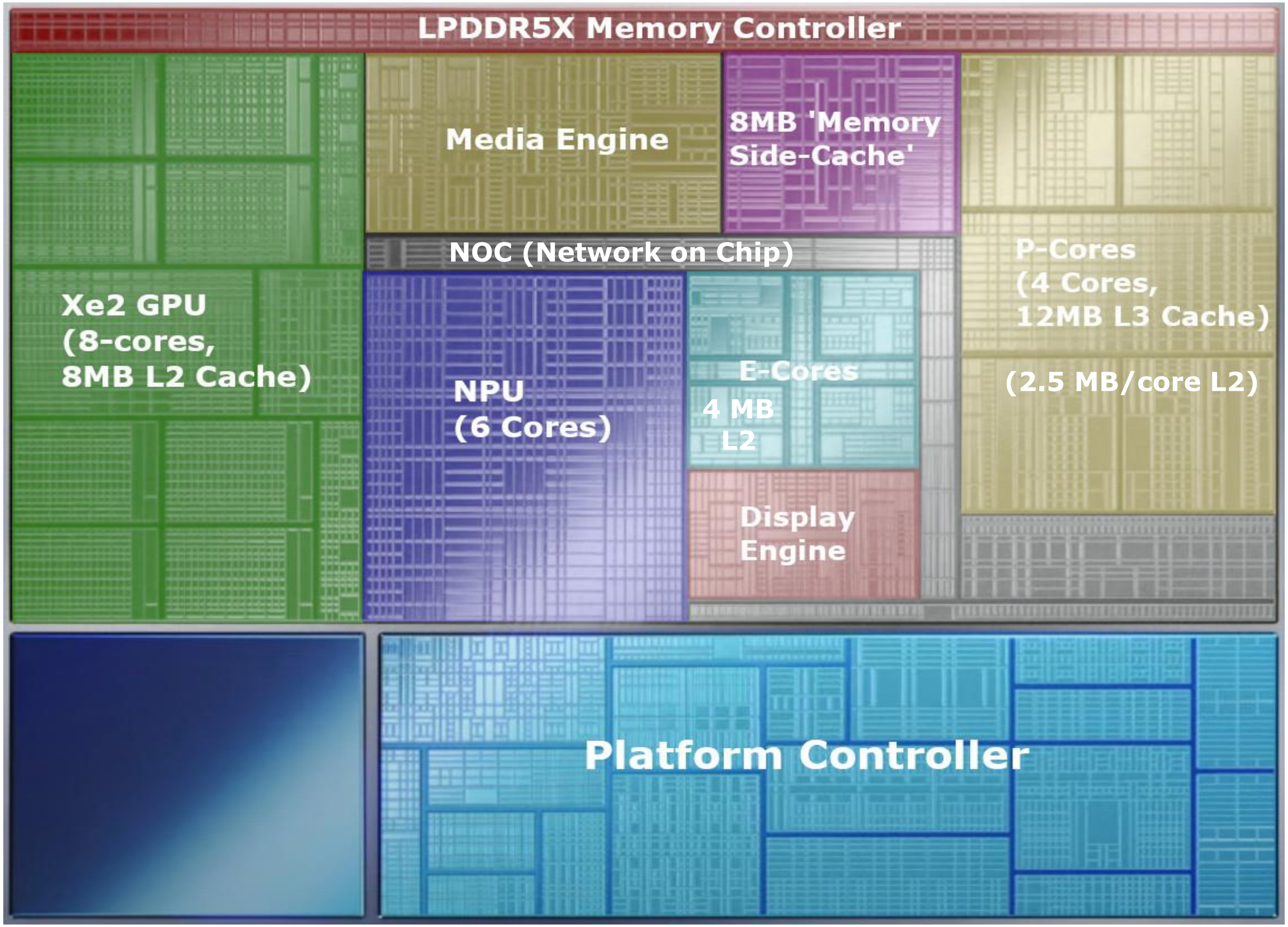
Bran-ding	SKU	Cores (threads)		Clock rate (GHz)			Arc Graphics		NPU (TOPS)	Smart cache[i]	RAM	TDP			Release date
		P	LPE	Base	Turbo P	Turbo LPE	Xe cores (XVEs)	Max. freq. (GHz)				Base	Turbo	cTDP	
Core Ultra 9	**288V**	4 (4)	4 (4)	3.3	5.1	3.7	8 (64)	2.05	48	12 MB	32 GB	30 W	37 W	17-37 W	Sep. 24, 2024
Core Ultra 7	**268V**			2.2	5.0	3.7	8 (64)	2.0	48	12 MB	32 GB	17 W		8-37 W	
	266V										16 GB				
	258V				4.8			1.95	47		32 GB				
	256V										16 GB				
Core Ultra 5	**238V**			2.1	4.7	3.5	7 (56)	1.85	40	8 MB	32 GB	17 W		8-37 W	
	236V										16 GB				
	228V				4.5						32 GB				
	226V										16 GB				

Note the lack of multithreading.

The physical implementation of the Lunar Lake processors [135]

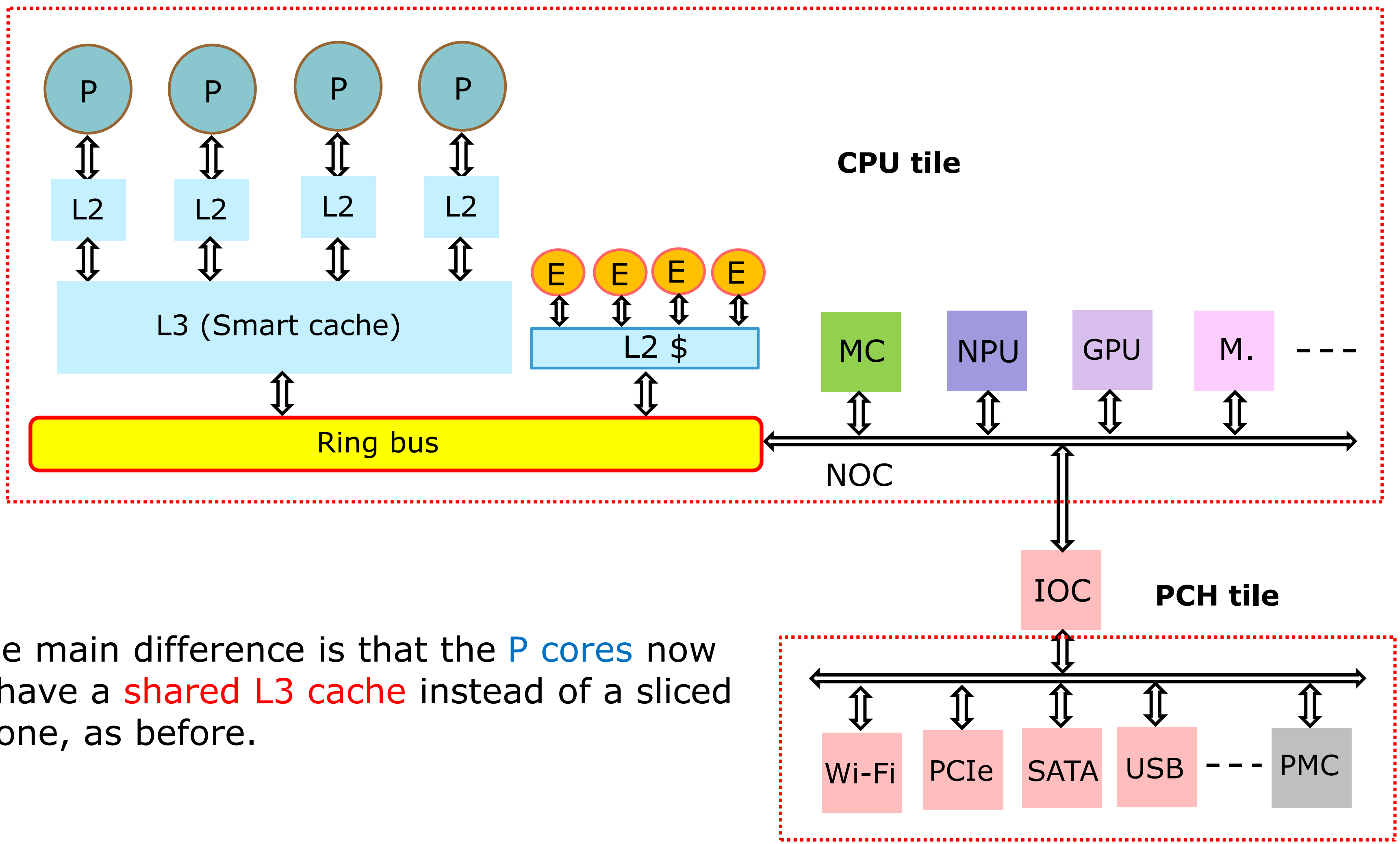
As shown, Intel's Lunar Lake has a multi-die big.little architecture, like Meteor Lake.

The annotated die plot of the Lunar Lake processor without the dual memory dies [149]



Assumed block diagram of the Core Ultra 200V Lunar Lake laptop processors -1

This architecture resembles the 14th gen. Meteor Lake's processor, but it is implemented using only two tiles and has a few differences, as shown below.



The main difference is that the P cores now have a shared L3 cache instead of a sliced one, as before.

Figure: Assumed block diagram of the Core Ultra 200V Lunar Lake laptop processors

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (6)

Remark

- The CPU tile of the Lunar Lake processor includes a memory-side cache (system level cache) that is 8 MB in size.
- This cache is accessible to all agents of the processor, including the CPU, GPU, and NPU.

  However, there is no reference to how this cache is integrated into the block diagram, so it is not indicated there.

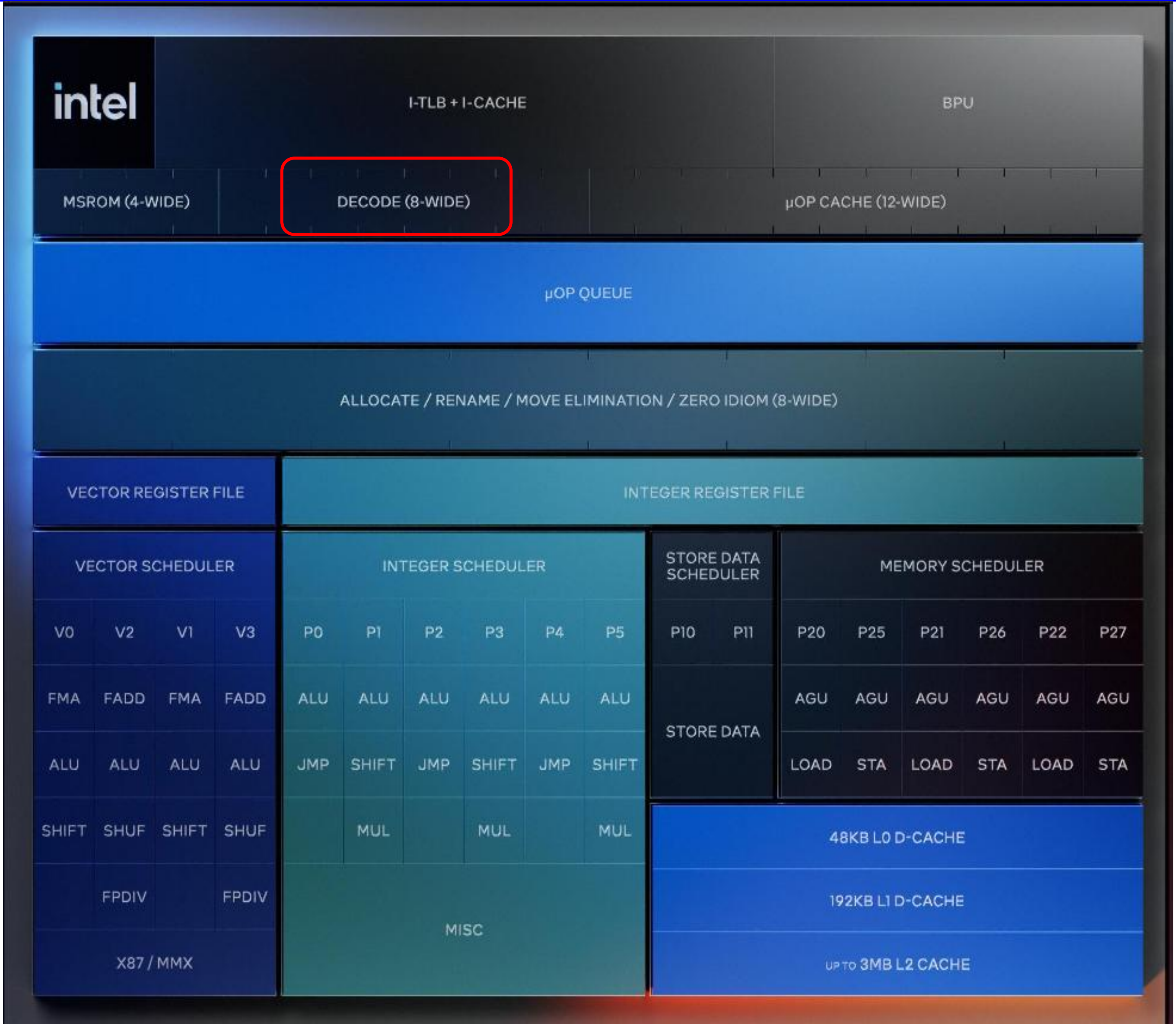
The P-core
(Lyon core)
[150]

8-wide decode
+14 % perf.
over
Redwood core
(Meteor Lake)

No multithreading

12 MB shared L3
cache

16.67 MHz
clock intervals
in the CPPC
for more
efficiency



4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (8)

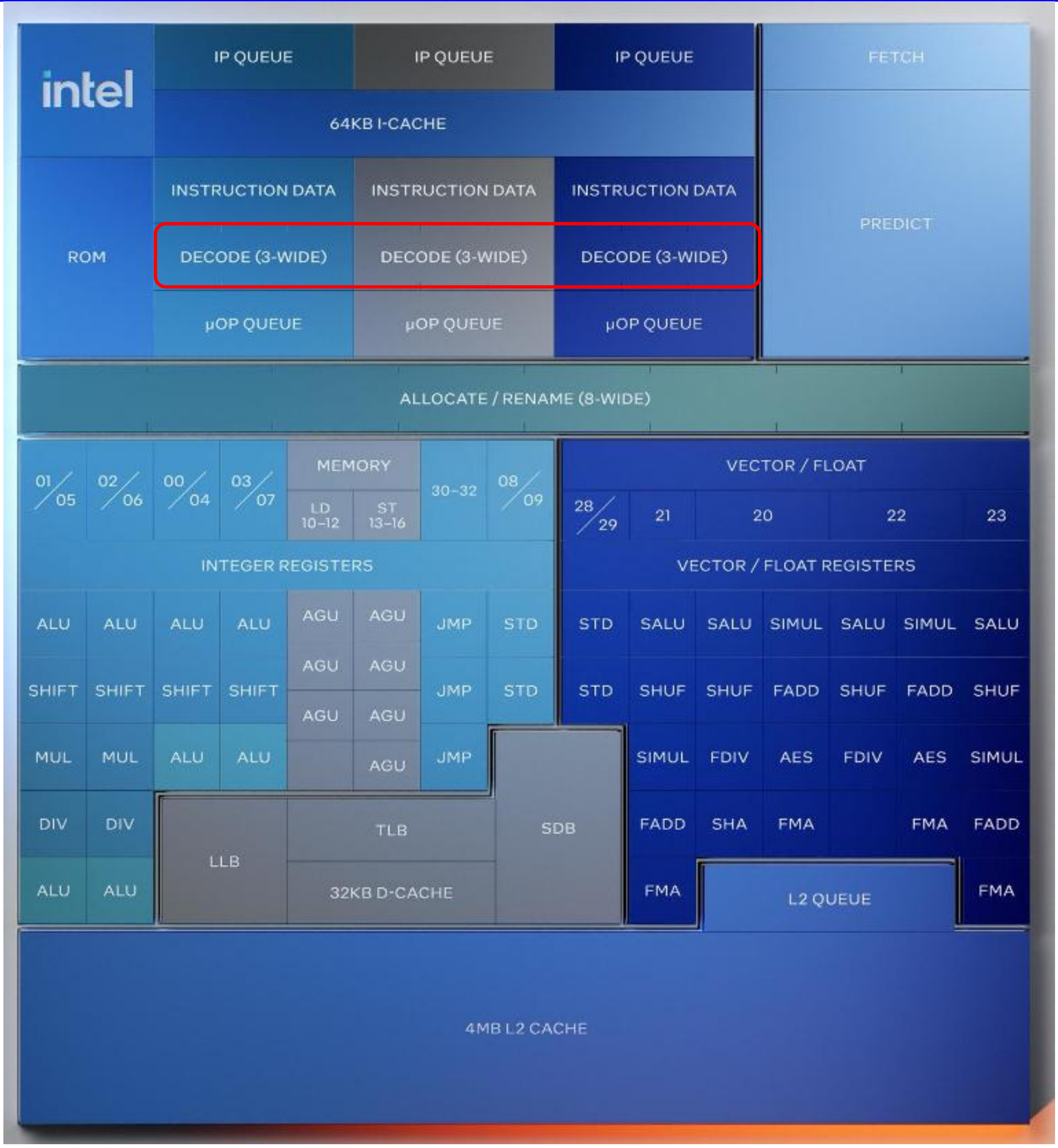
The LPE core (Skymont core) [150]

It is the same core as found in the Arrow Lake S DT processor

3x3 wide decode

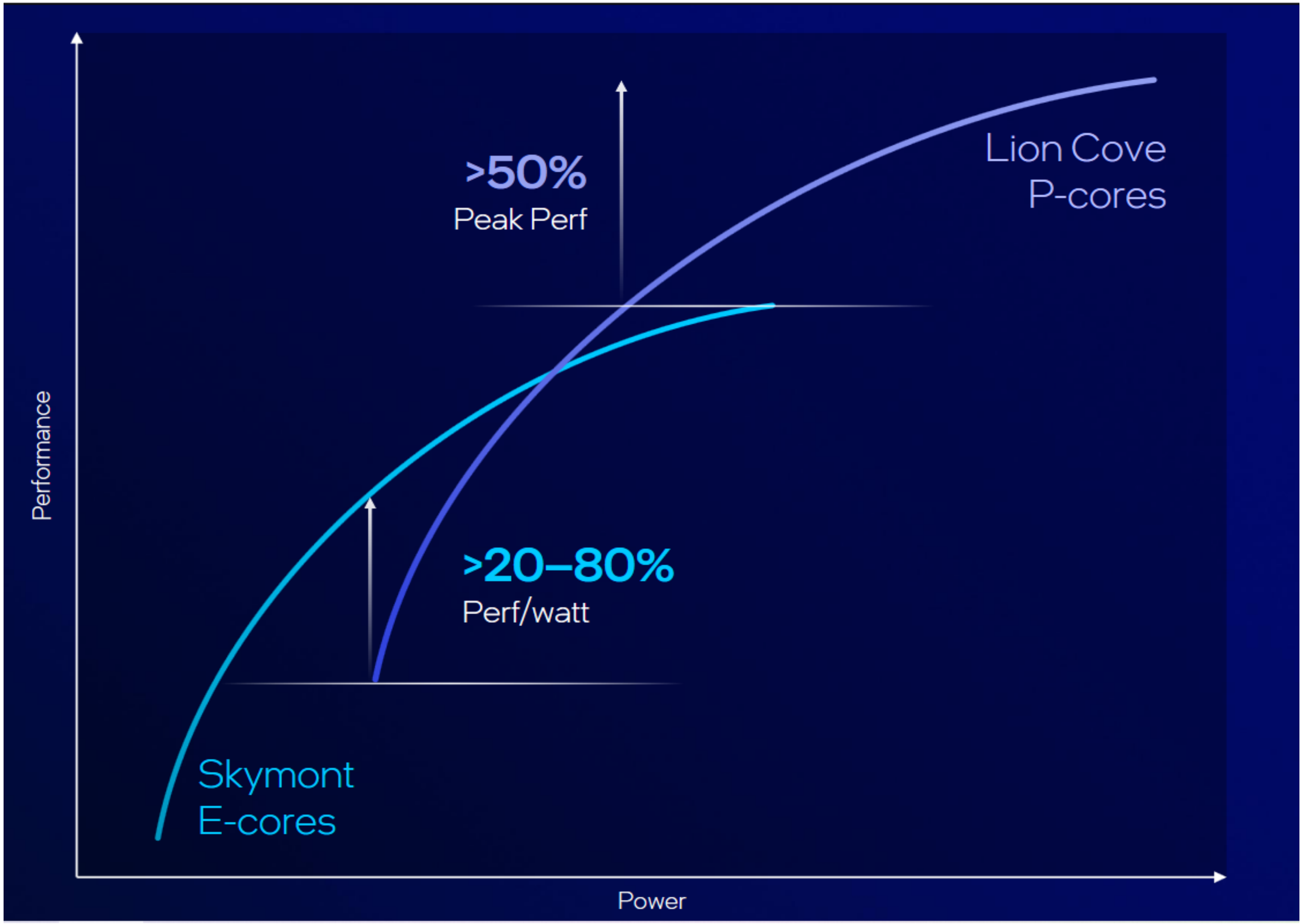
1.38x Meteor Lake LP-E SPECrate2017_int_base est /

1.68x Meteor Lake LP-E SPECrate2017_fp_base est /



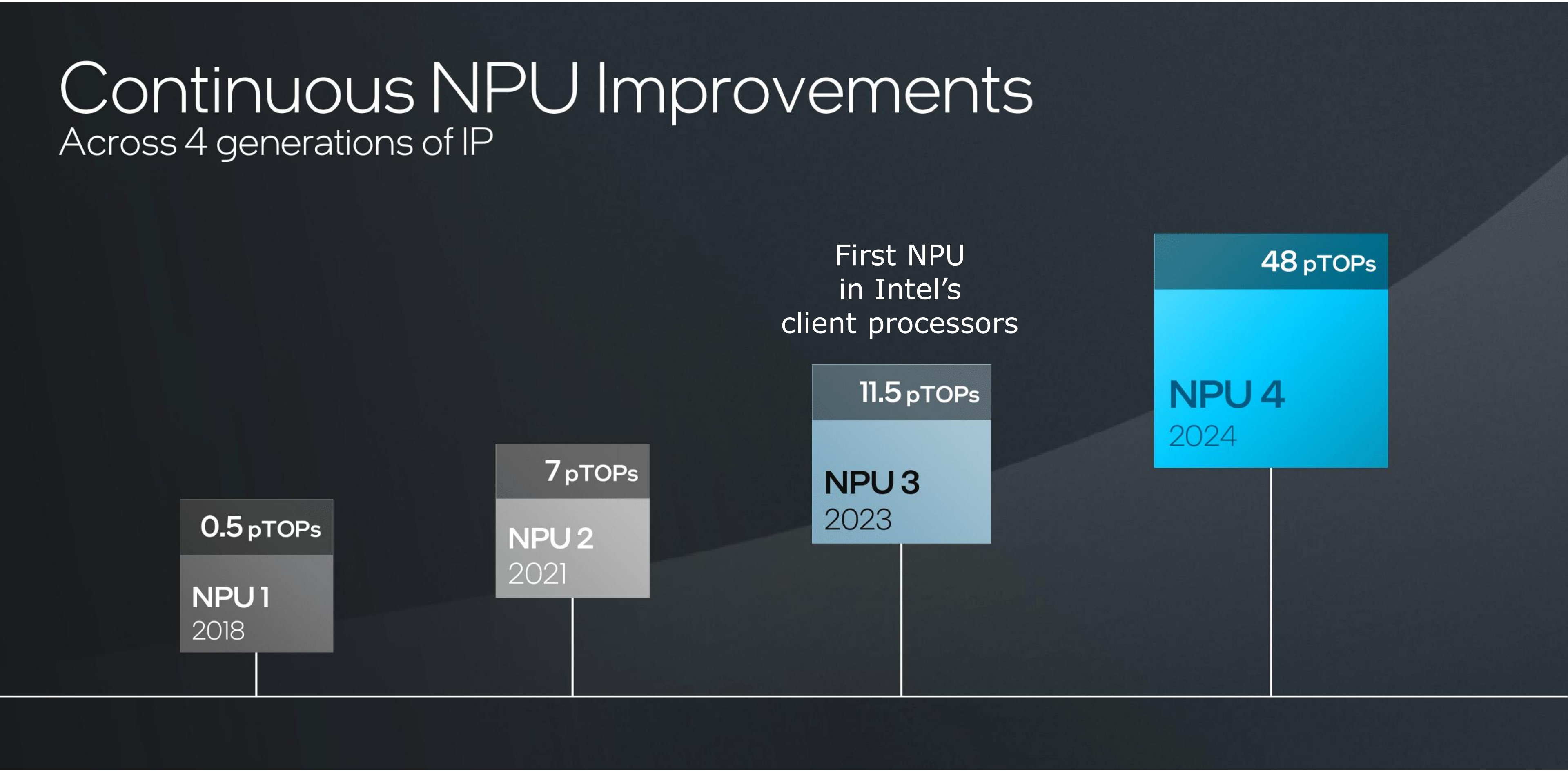
Note that Lunar Lakes E cores offer significantly higher performance than the E cores in the Meteor Lake processor

The efficiency graphs of Lunar Lake's P- and E-cores [150]



For lighter workloads, the E-cores, while for more intensive ones the P-cores are used.

Evolution of the NPUs in Intel's Core family over time [151]

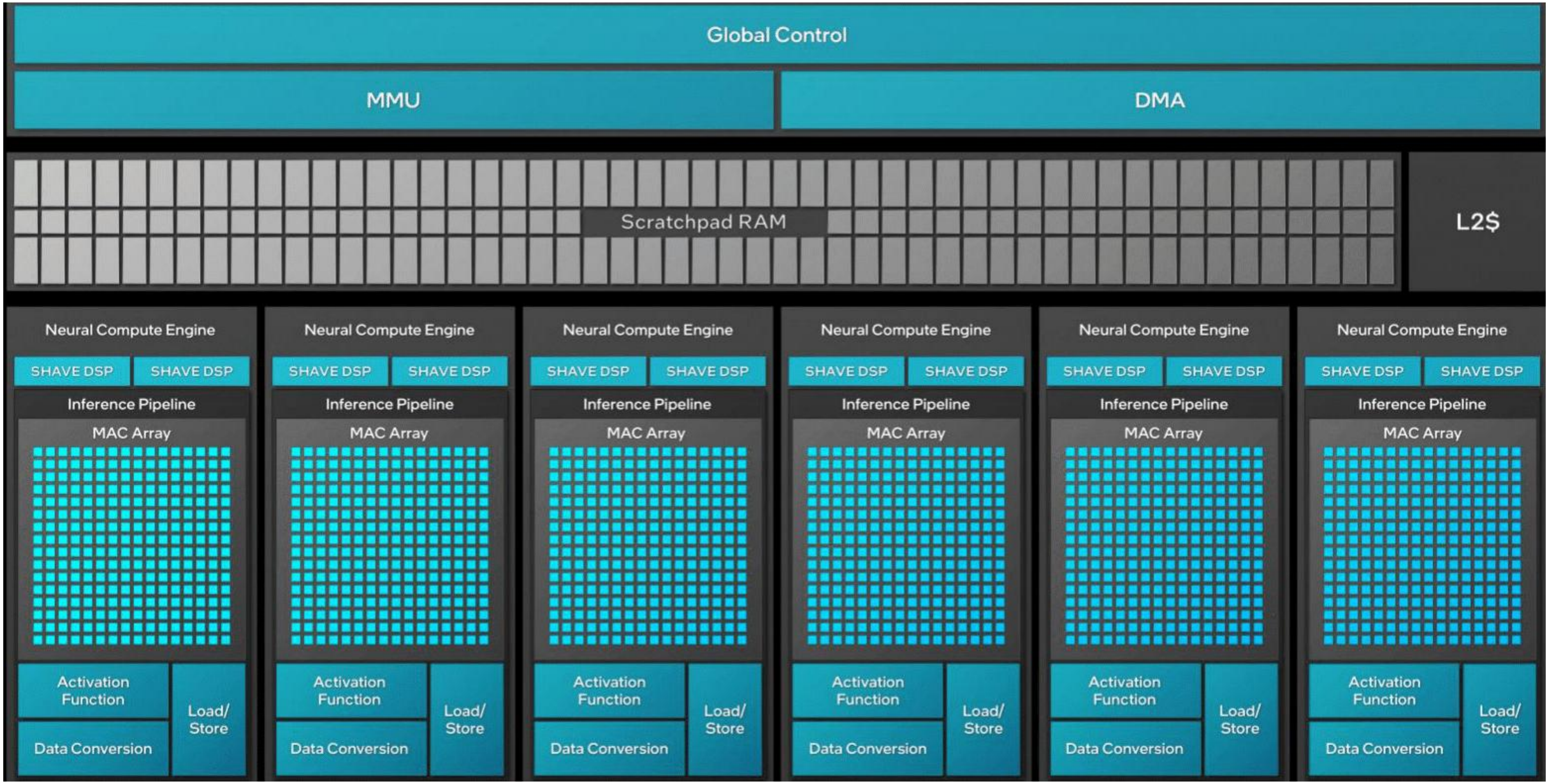


14th gen.
Meteor Lake

Core Ultra 200V
For Copilot+ PCs

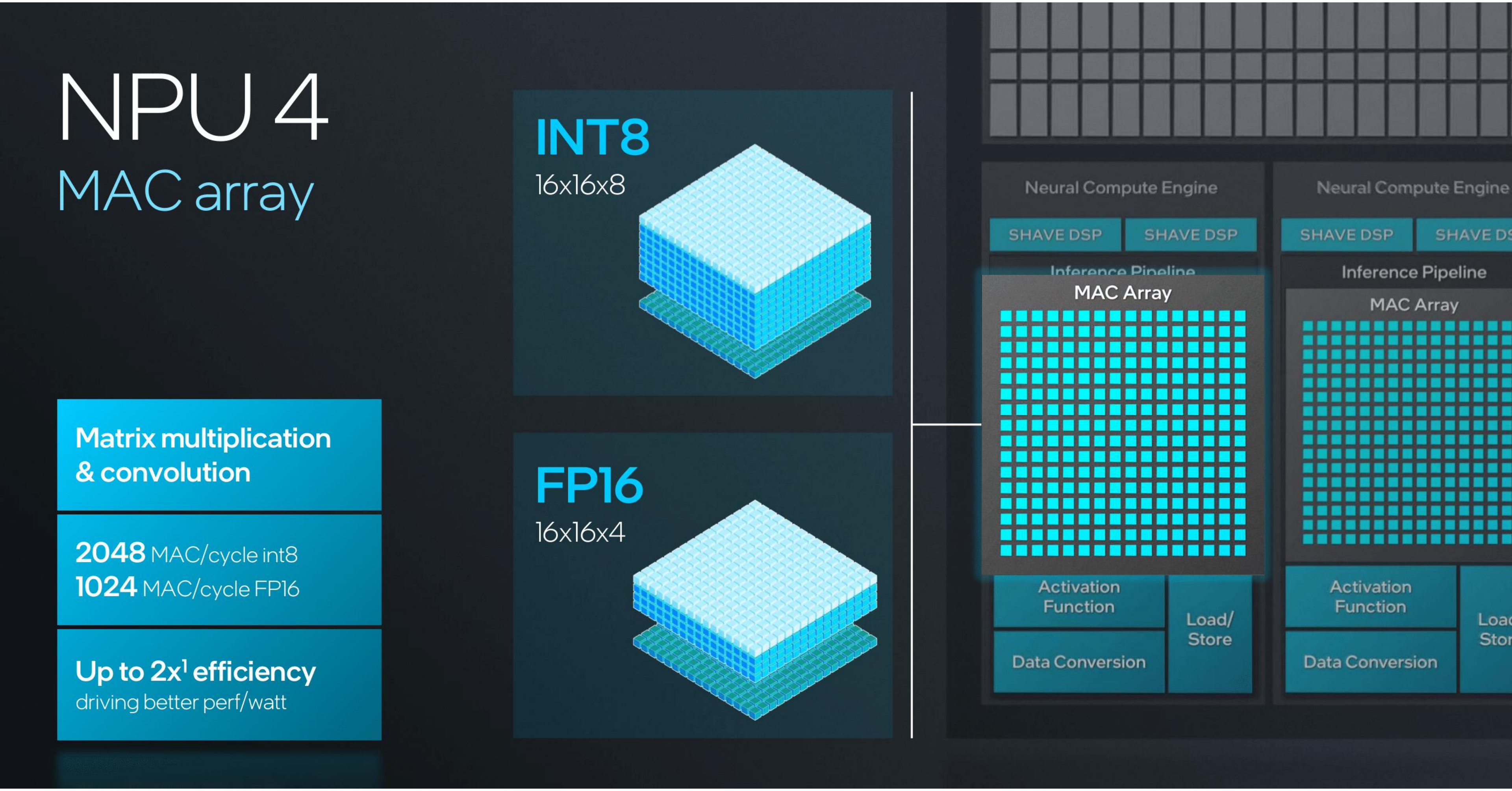
TOPS: Tera Operations Per Second (10^{12})

The central part of the NPU unit of the Lunar Lake processor -1 [151]



- It includes 8x512 MAC Processing Engines (MPEs), but above only 6 are indicated.
- Each MPE is capable of performing four INT8 or two FP16 operations per cycle [152].
- It follows that the NPU can execute 8x512x4 = 16 K INT8 operations per cycle.

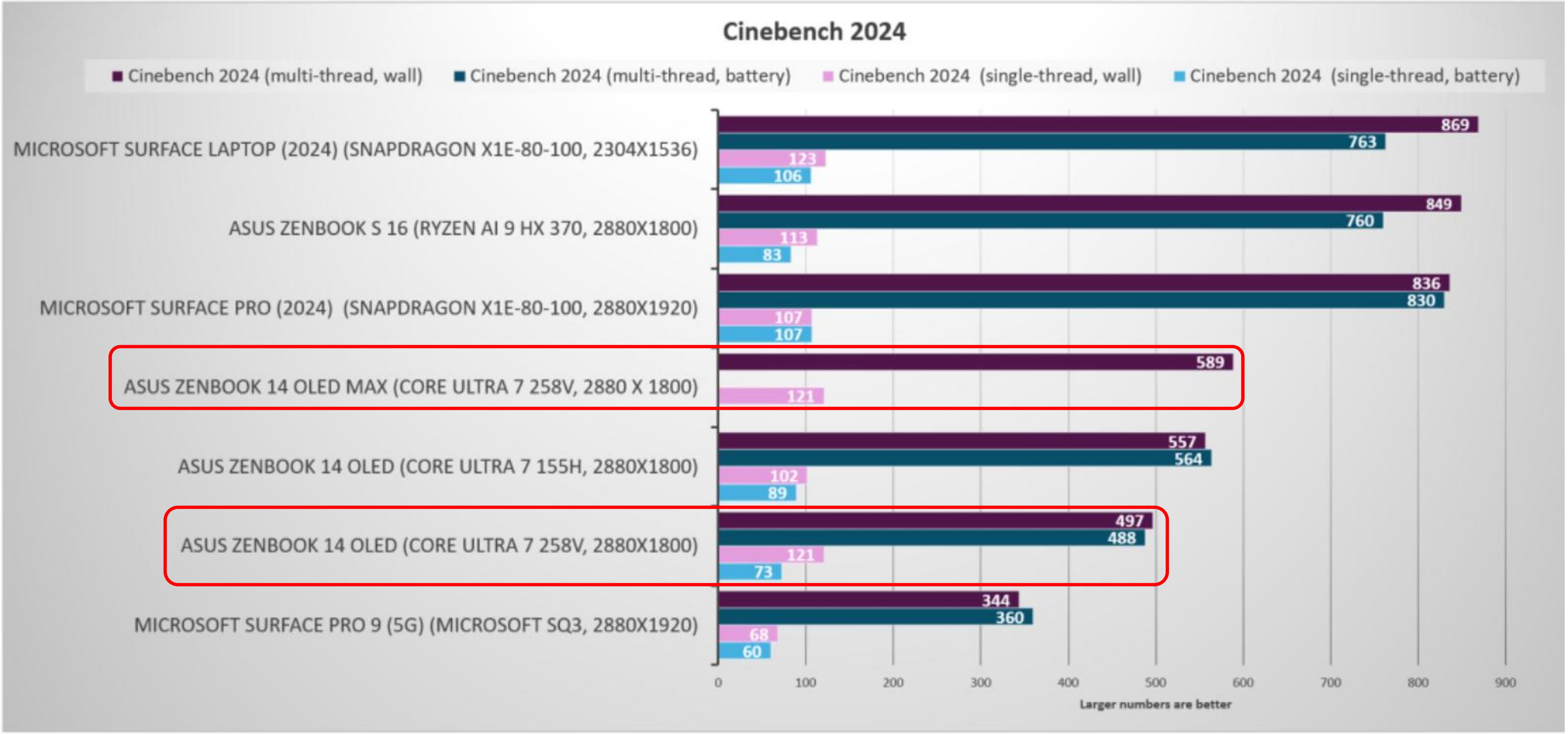
The central part of the NPU unit of the Lunar Lake processor -2 [151]



Assuming an NPU 4 clock cycle of 300 MHz, the NPU performance would be calculated as 16 K operations x 300 MHz, resulting in 48 TOPS (relating to FP8 operations).

Ahmed Y., Introducing Copilot+ PCs

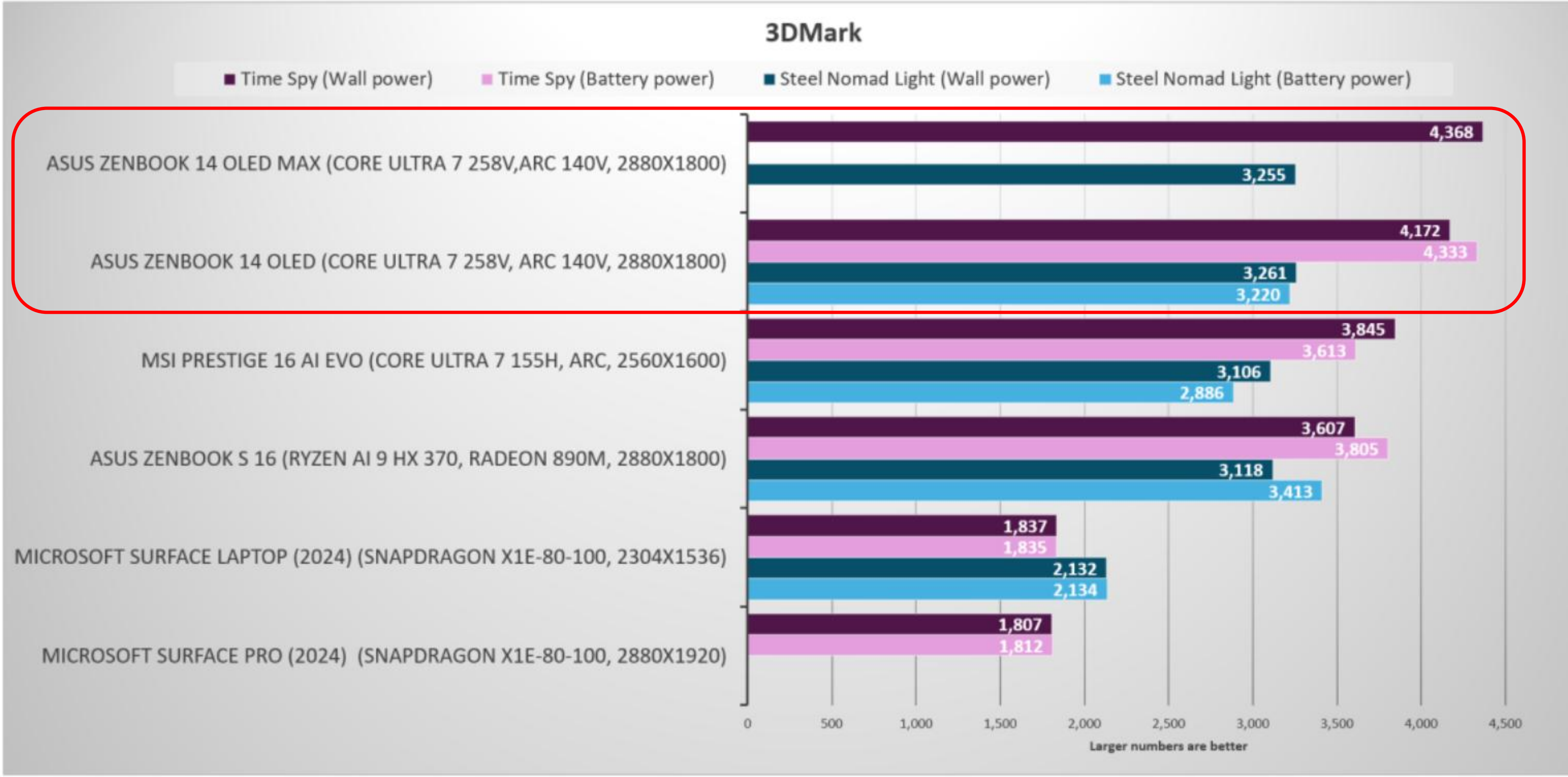
Comparing the CPU performance of advanced laptop processors [165]



As shown in the 4+4 core Figure, the Lunar Lake processor (Ultra 7 258V) has similar ST CPU performance compared to the competing Qualcomm Snapdragon X1E 80-100 and AMD Ryzen AI9 HX370 processors, but it provides lower MT performance.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (15)

Comparing the GPU performance of advanced laptop processors [165]



As shown in the Figure, the Lunar Lake processor (Ultra 7 258V) demonstrates higher GPU performance compared to the AMD Ryzen AI 9 HX370 processor as well as significantly higher performance compared to Qualcomm's Snapdragon X1E 80-100 processor.

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (16)

Comparing battery-life performance of advanced laptop processors [165]

- This test was conducted using UL's Procyon battery-life test, which simulates working on the laptop by cycling through Office applications until the battery is down.
- The results of the tests are as follows:
  - **Intel Lunar Lake:** 17 hours, 7 minutes
  - **Qualcomm Snapdragon X Elite:** 16 hours, 20 minutes
  - **AMD Ryzen AI 300:** 10 hours, 42 minutes
  - **Intel Meteor Lake:** 10 hours, 35 minutes
- The above results demonstrate that Intel's Lunar Lake processor offers the longest battery life, slightly outpacing Qualcomm's Snapdragon X Elite processors.

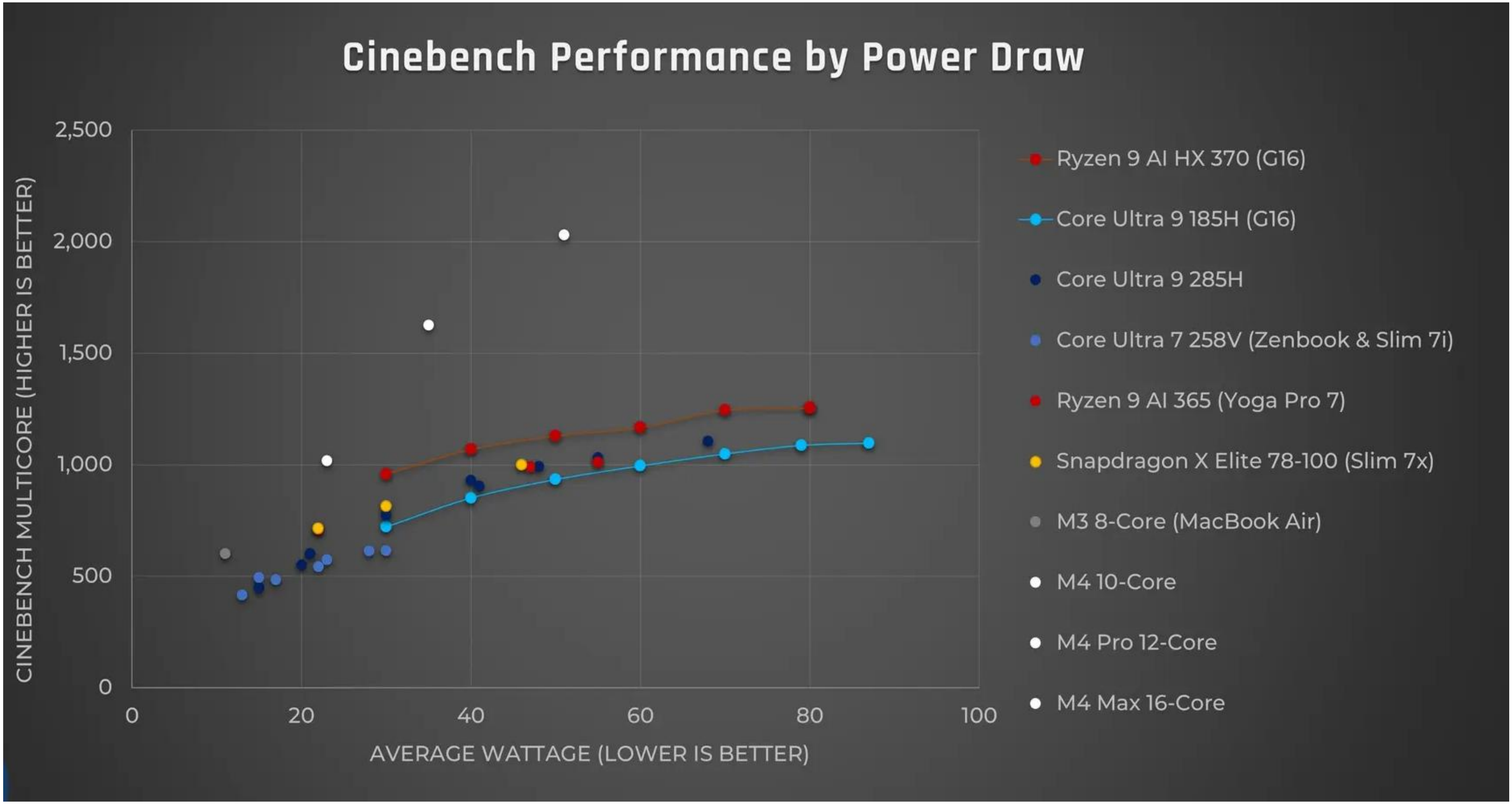
  Both of these processors have significantly longer battery lives compared to AMD's Ryzen AI 300 and Intel's Meteor Lake processors.

The test configurations were:

the Asus ZenBook S 16 (with a Ryzen AI 300 chip inside and a 78 Wh battery),
the 13.8-inch Surface Laptop 7th Edition (with a Snapdragon X Elite chip and a 54 Wh battery),
the Asus ZenBook 14 OLED (with a first-gen Core Ultra Core 155H and a 75 Wh battery) versus the current Asus ZenBook OLED (with its Lunar Lake chip and a 73 Wh battery).

4.1 Case example 1: Intel's Core Ultra Series 2 (Lunar Lake) laptop family (17)

Efficiency of laptop processors in terms of Cinebench 24 Multi-Core/W [171]



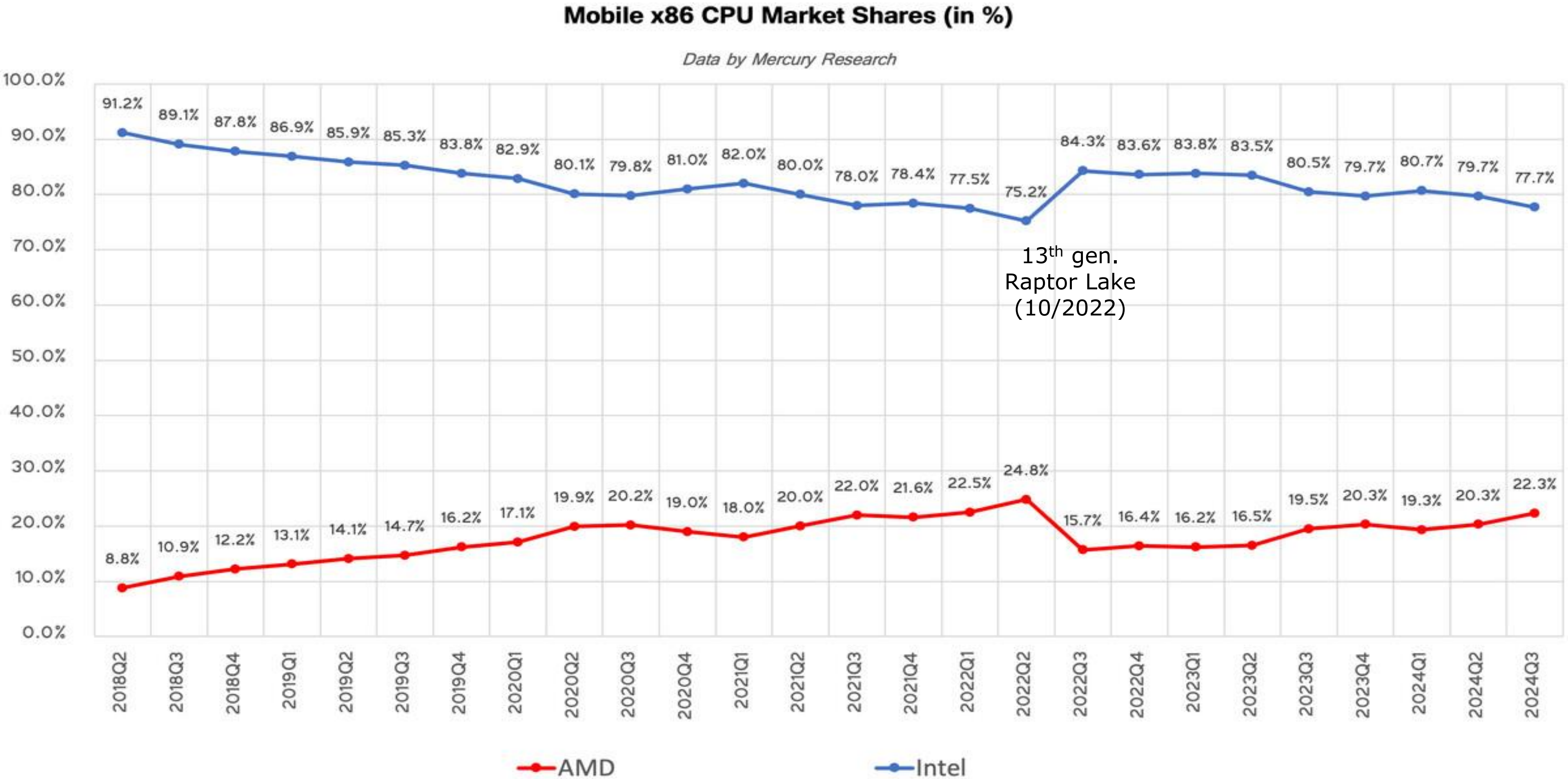
As shown in the Figure, Apple's M4 processors are the most efficient, followed by AMD's Ryzen 9 AI, Qualcomm's Snapdragon X Elite and finally by Intel's Core Ultra lines.

Revealed vulnerability of recent Intel processor lines [131].

- In 11/2023, Google released a bulletin reporting a serious vulnerability in several processor lines of Intel, including its 10th gen. Ice Lake, 12th gen. Alder Lake and 13th gen. Raptor Lake lines.
- Potentially, this vulnerability can lead to denial of service or disclosure of sensitive data.
- Intel acknowledged it and fixed it with microcode patches.
- As a result of this vulnerability, Intel experienced a significant loss of market share.

vulnerability: sérülékenység

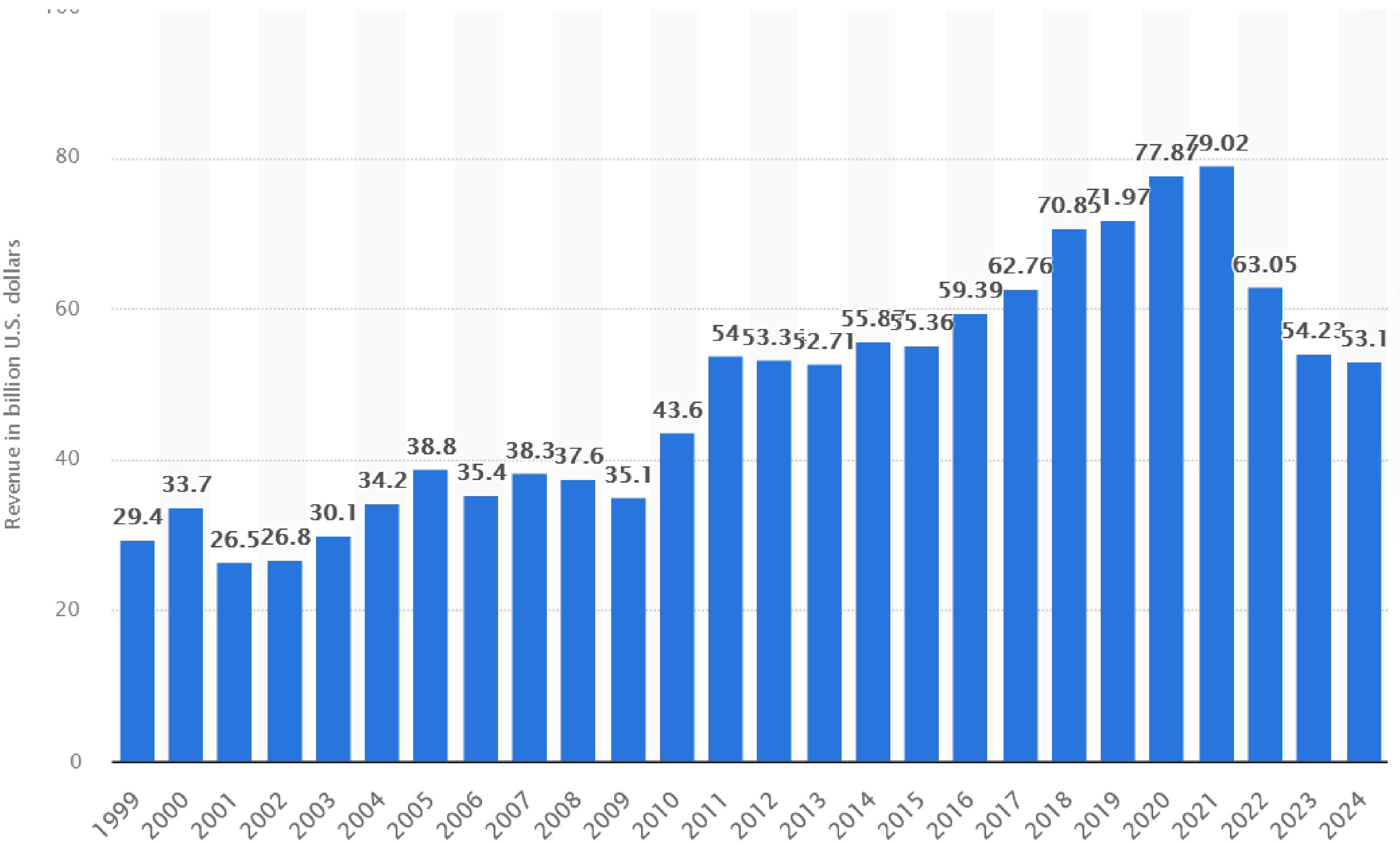
Global laptop processor revenue market share: Intel vs. AMD [153]



The Figure above indicates Intel's dominance in the global market revenue share for client processors, although to a gradually lesser extent.

Please note in the Figure how significantly the appearance of Intel's Raptor Lake line modified the revenue share between the two competing vendors.

Intel's revenues from 1999 to 2024 (in billion USD) [154]



4.2 Case example 2: Intel’s Core Ultra (Panther Lake) laptop family (2026)

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (2)

The evolution of Intel's advanced multi die processor families [225]

All of the client processor series mentioned above are heterogeneous multi-die designs.

The 4th gen. Sapphire Rapids and 5th gen. Emerald Rapids server processors are homogeneous multi-die designs; in contrast, the 6th gen. Granite Rapids and Sierra Forest server series are heterogeneous layouts.

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (3)

Intel's Core Ultra Series 3 (Panther Lake) laptop family (2026) -2 [225]

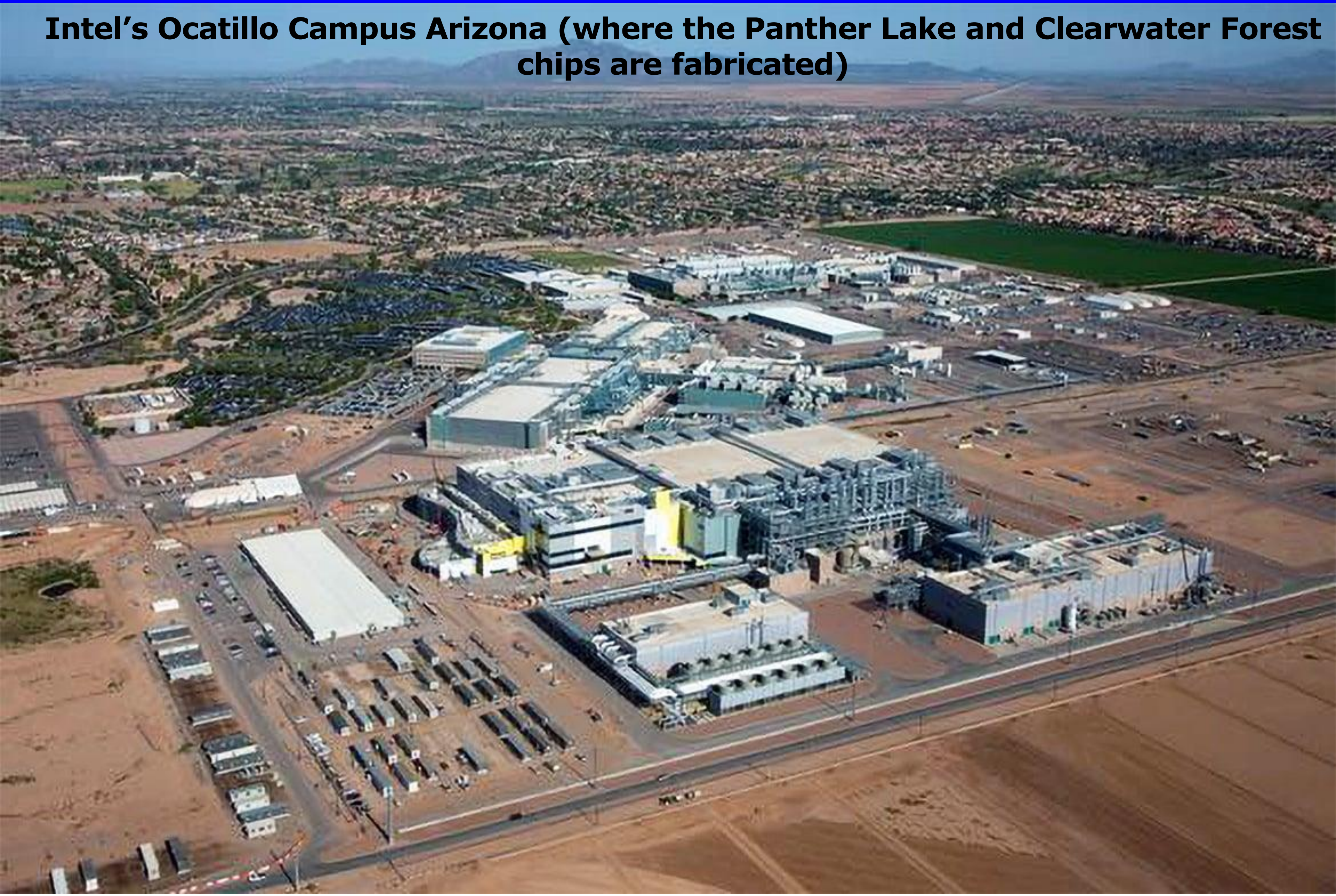
- Compared to the Lunar Lake line, the Panther Lake line offers up to 16 cores, composed of three core types, while the Lunar Lake line has 8 cores made up of two core types.

  These tiles are:
  - the Compute tile (Intel 1.8A)
  - the Graphics tile (Intel3/TSMS N3E) and
  - the SOC tile (TSMC N6)

- Additionally, the Panther Lake line has a higher TDP of 25 W rather than 17 W for the Lunar Lake line (except for the highest performance model, which has a TDP value of 30 W).
- The Panther Lake line is Intel's first 18A product, along with its server-oriented 6+ gen. Clearwater Forest server processor line, which is scheduled for launch in H1 2026.

4.2 Case example 2: Intel’s Core Ultra Series 3 (Panther Lake) laptop family (4)

Intel’s Ocatillo Campus Arizona (where the Panther Lake and Clearwater Forest chips are fabricated)



https://hothardware.com/photo-gallery/article/3567?image=big_panther-lake-chip-configs-2-16-core.jpg&tag=&p=1

Intel's Core Ultra Series 3 (Panther Lake) laptop family (2026) [226] -2

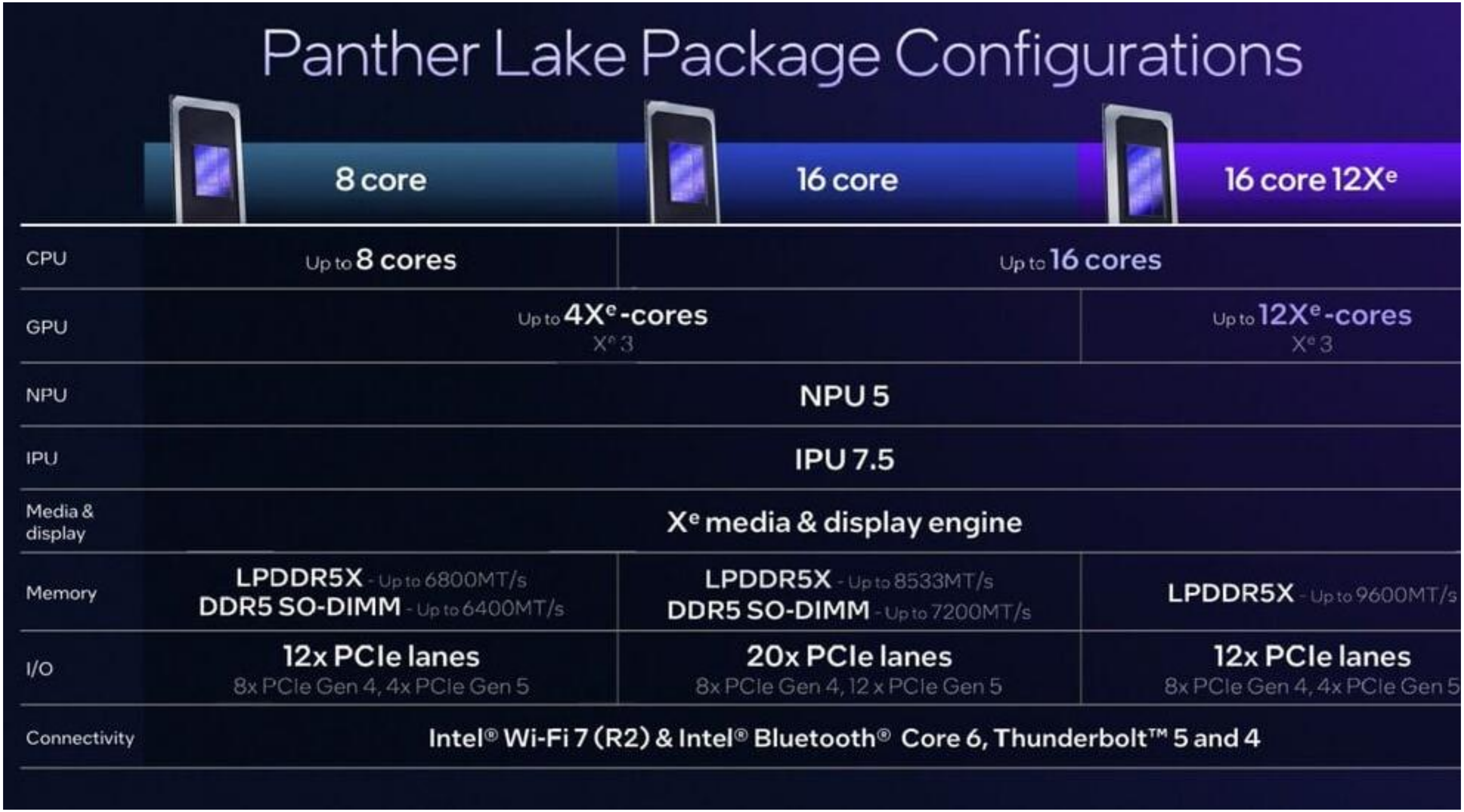
All the P, E, and LPE Panther Lake cores are new designs, as seen below.

Figure: Intel's heterogeneous big.little client processors [226]

Basic layout of the Panther Lake processors [226]

- As the Figure shows, Panther Lake processors include three dies, referred to as: the Compute tile, the GPU tile, and the Platform controller tile.
- These tiles are packaged using the Foveros 2.5D technology.

Three main configurations of the Panther Lake processors [226]



Panther Lake Package Configurations

	8 core	16 core	16 core 12Xe
CPU	Up to 8 cores	Up to 16 cores	
GPU	Up to 4Xe-cores Xe3		Up to 12Xe-cores Xe3
NPU	NPU 5		
IPU	IPU 7.5		
Media & display	Xe media & display engine		
Memory	LPDDR5X - Up to 6800MT/s DDR5 SO-DIMM - Up to 6400MT/s	LPDDR5X - Up to 8533MT/s DDR5 SO-DIMM - Up to 7200MT/s	LPDDR5X - Up to 9600MT/s
I/O	12x PCIe lanes 8x PCIe Gen 4, 4x PCIe Gen 5	20x PCIe lanes 8x PCIe Gen 4, 12 x PCIe Gen 5	12x PCIe lanes 8x PCIe Gen 4, 4x PCIe Gen 5
Connectivity	Intel® Wi-Fi 7 (R2) & Intel® Bluetooth® Core 6, Thunderbolt™ 5 and 4		

Next, we will outline these basic configurations.

IPU: Image Processing Unit for handling camera, sensor, and multimedia imaging tasks.

Main components of the 8-core layout [226]

4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (8)

Main components of the 16-core layout [226]

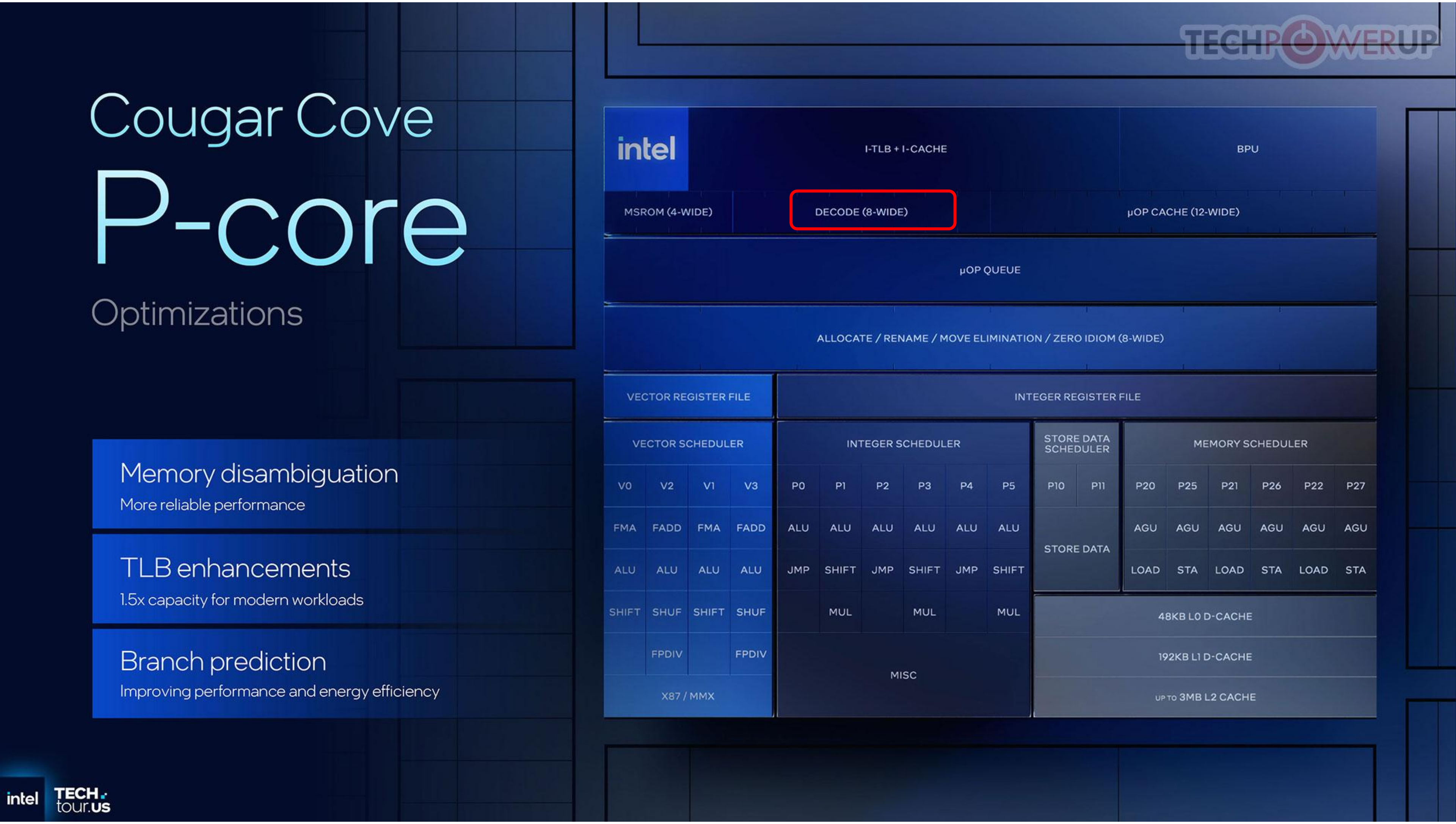
Main components of the 16-core 12X layout [226]

This configuration has 12 Xe cores and 12 ray tracing cores, compared to 4 and 4 cores of the previously discussed versions.

Main features of Intel’s Core Ultra Series 3 (Panther Lake) processor models [226]

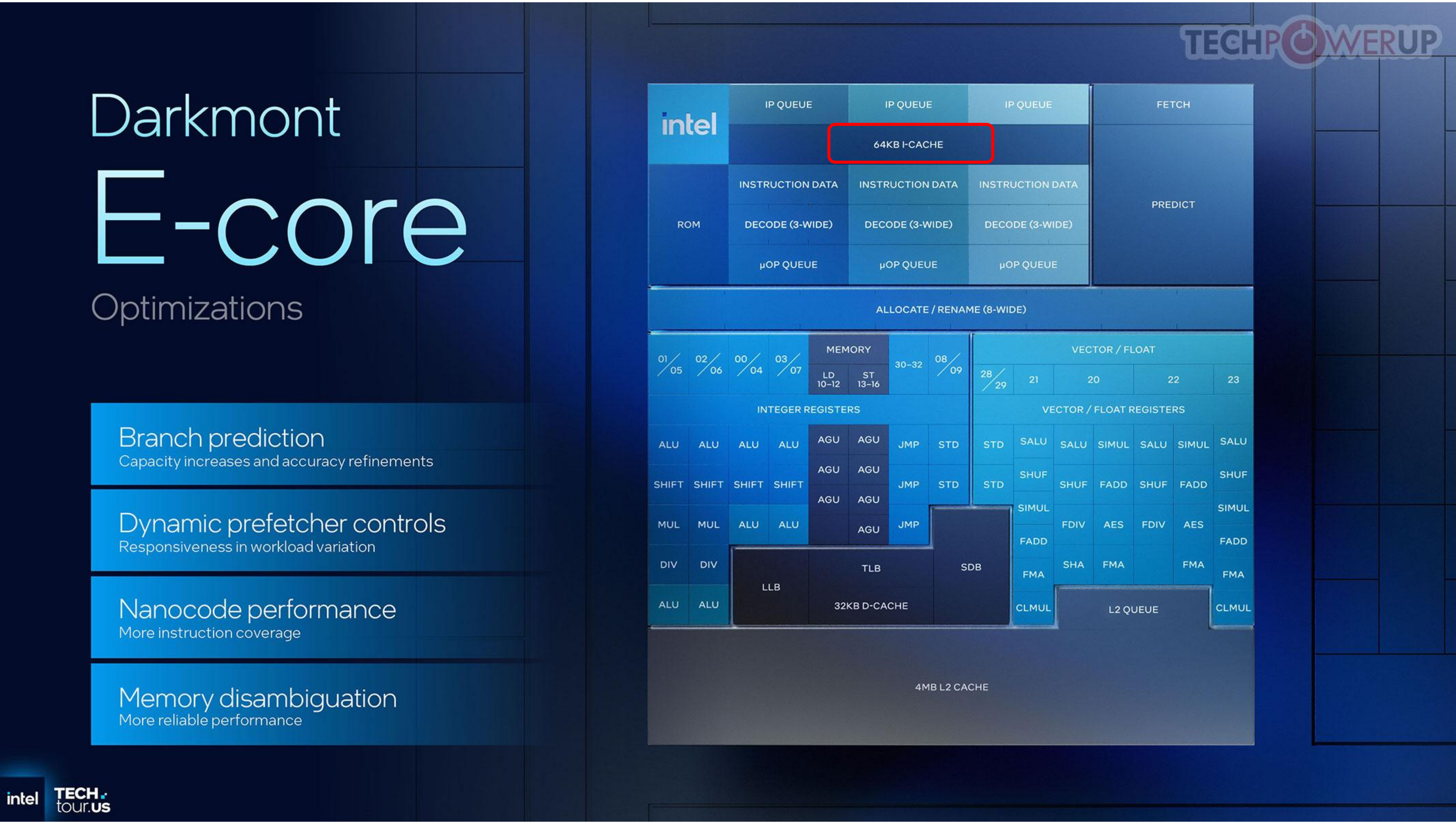
	CPU			NPU	Memory		Power	
Processor Number	Total Cores & Threads	P-core Max Turbo Freq (GHz)	Intel® Smart Cache LLC (MB)	TOPS	Maximum Memory Speed (MT/s)	Maximum Memory Capacity (GB)	Processor Base Power	Maximum Turbo Power (W)
Intel® Core™ Ultra X9 388H	16	5.1	18	50	LP5/X 9600	96 (LP5/x)	25W	65[7],80[7]
Intel® Core™ Ultra 9 386H	16	4.9	18	50	LP5/X 8533 DDR5 7200	96 (LP5/x) 128 (DDR5)		65[7],80[7]
Intel® Core™ Ultra X7 368H	16	5.0	18	50	LP5/X 9600	96 (LP5/x)		65[7],80[7]
Intel® Core™ Ultra 7 366H	16	4.8	18	50	LP5/X 8533 DDR5 7200	96 (LP5/x) 128 (DDR5)		65[7],80[7]
Intel® Core™ Ultra 7 365	8	4.8	12	49	LP5/X 6800 DDR5 6400			55
Intel® Core™ Ultra X7 358H	16	4.8	18	50	LP5/X9600	96 (LP5/x)		65[7],80[7]
Intel® Core™ Ultra 7 356H	16	4.7	18	50	LP5/X 8533 DDR5 7200	96 (LP5/x) 128 (DDR5)		65[7],80[7]
Intel® Core™ Ultra 7 355	8	4.7	12	49	LP5/X 6800 DDR5 6400			55
Intel® Core™ Ultra 5 338H	12	4.7	18	47	LP5/X 8533	96 (LP5/x)	25W	65[7],80[7]
Intel® Core™ Ultra 5 336H	12	4.6	18	47	LP5/X 8533 DDR5 7200	96 (LP5/x) 128 (DDR5)		65[7],80[7]
Intel® Core™ Ultra 5 335	8	4.6	12	47	LP5/X 6800 DDR5 6400			55
Intel® Core™ Ultra 5 325	8	4.5	12	47				55
Intel® Core™ Ultra 5 332	6	4.4	12	46				55
Intel® Core™ Ultra 5 322	6	4.4	12	46				55

The 8-wide P-core (Cougar Cove core) of Panther Lake laptop processors [226]



4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (12)

The 3x3-wide E-core (Darkmont core) of Panther Lake laptop processors [226]



Note that both core types above share the same core width as the corresponding cores of the Lunar Lake laptop and Raptor Lake desktop lines.

Assumed block diagram of the Core Ultra Series 3 (Panther Lake) processors -1

This architecture has been implemented across three tiles, as shown below.

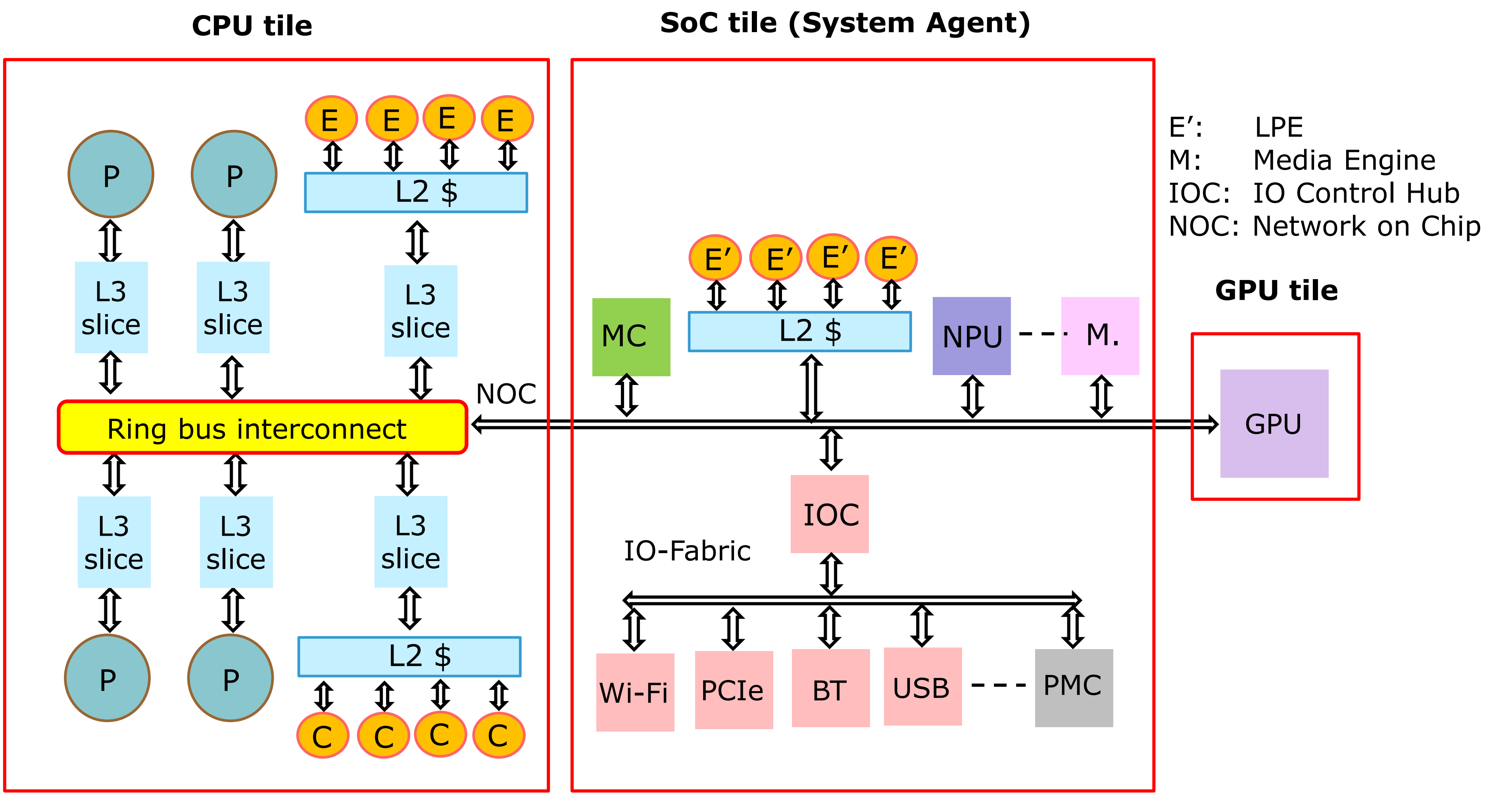


Figure: Assumed block diagram of the Core Ultra Series 3 (Panther Lake) processors

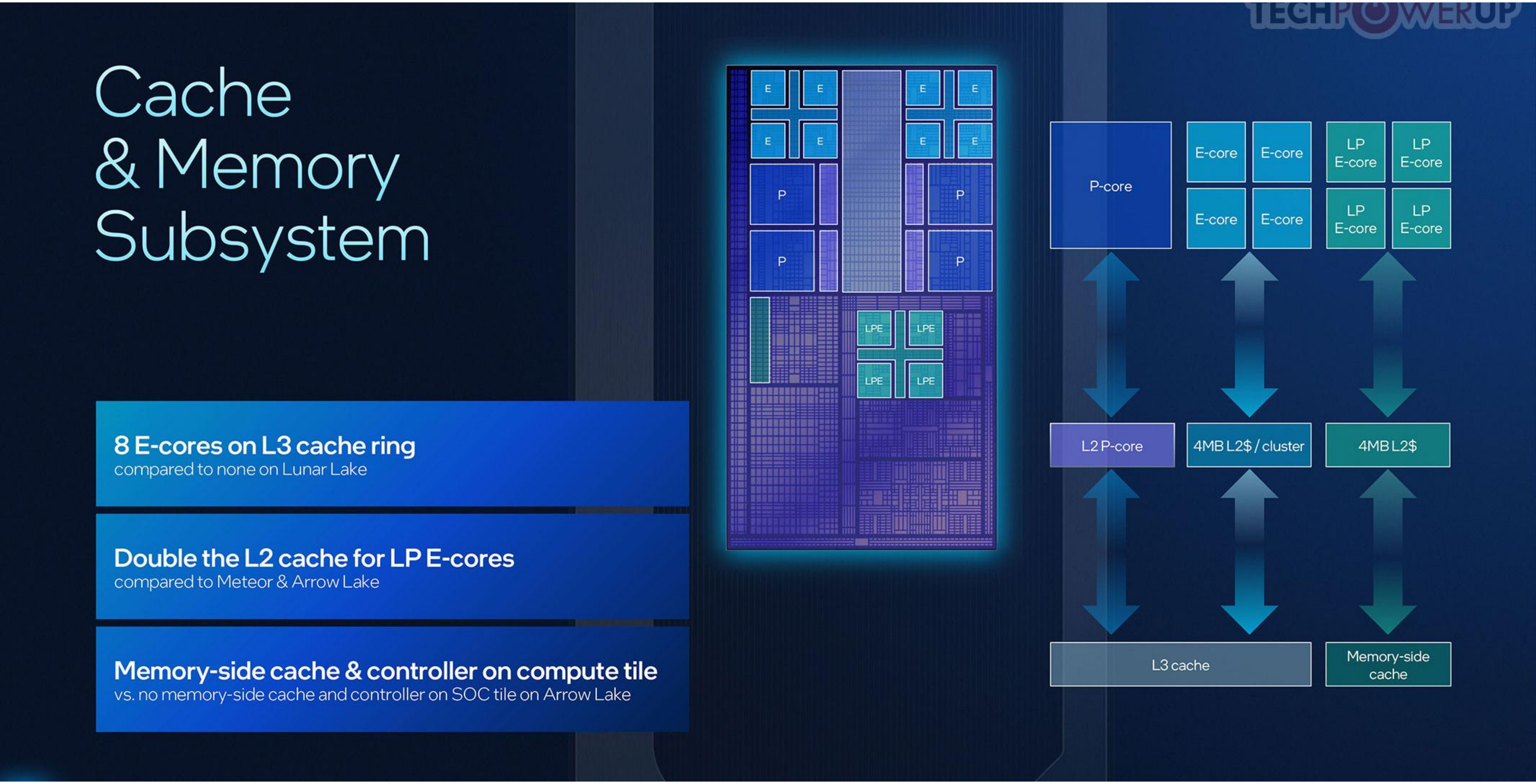
Assumed block diagram of the Core Ultra Series 3 (Panther Lake) processors -2

- Intel has not published the architecture of the Panther Lake processor.

  However, available details of this processor suggest that its microarchitecture is similar to that of Meteor Lake processors.
- The corresponding block diagram is shown in the preceding Figure.

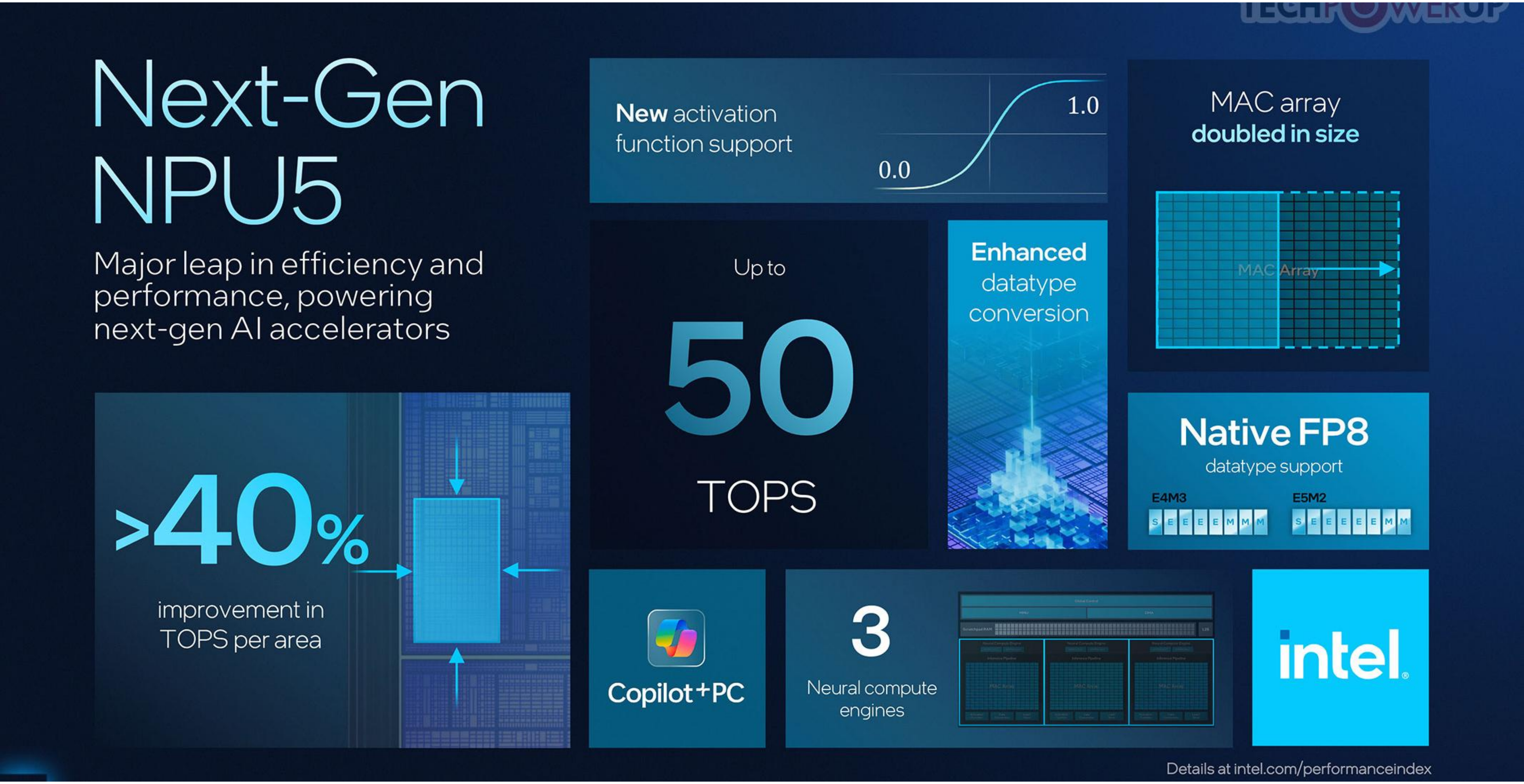
4.2 Case example 2: Intel's Core Ultra Series 3 (Panther Lake) laptop family (15)

The cache architecture of the Panther Lake processors [226]



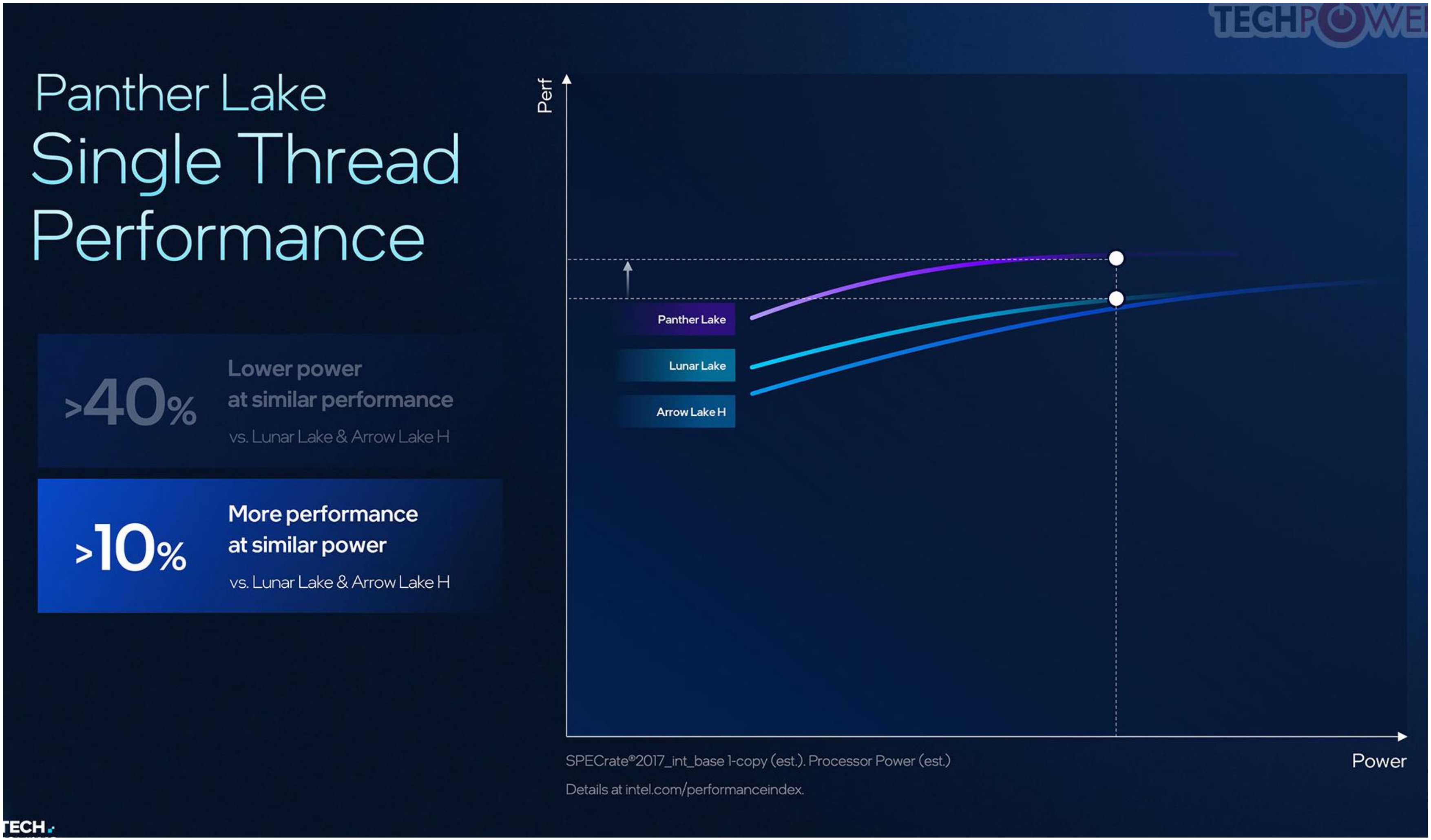
- As shown above, the P cores have separate L2 caches and L3 cache segments.
- Every 4 E-cores are supported by a shared L2 cache and an L3 cache segment, whereas the 4 LPE cores only have a shared L2 cache.

The AI performance of the Panther Lake line [226]



As shown above, the Panther Lake Line offers 50 TOPS AI performance 10 TOPS more than the Lunar Lake line.

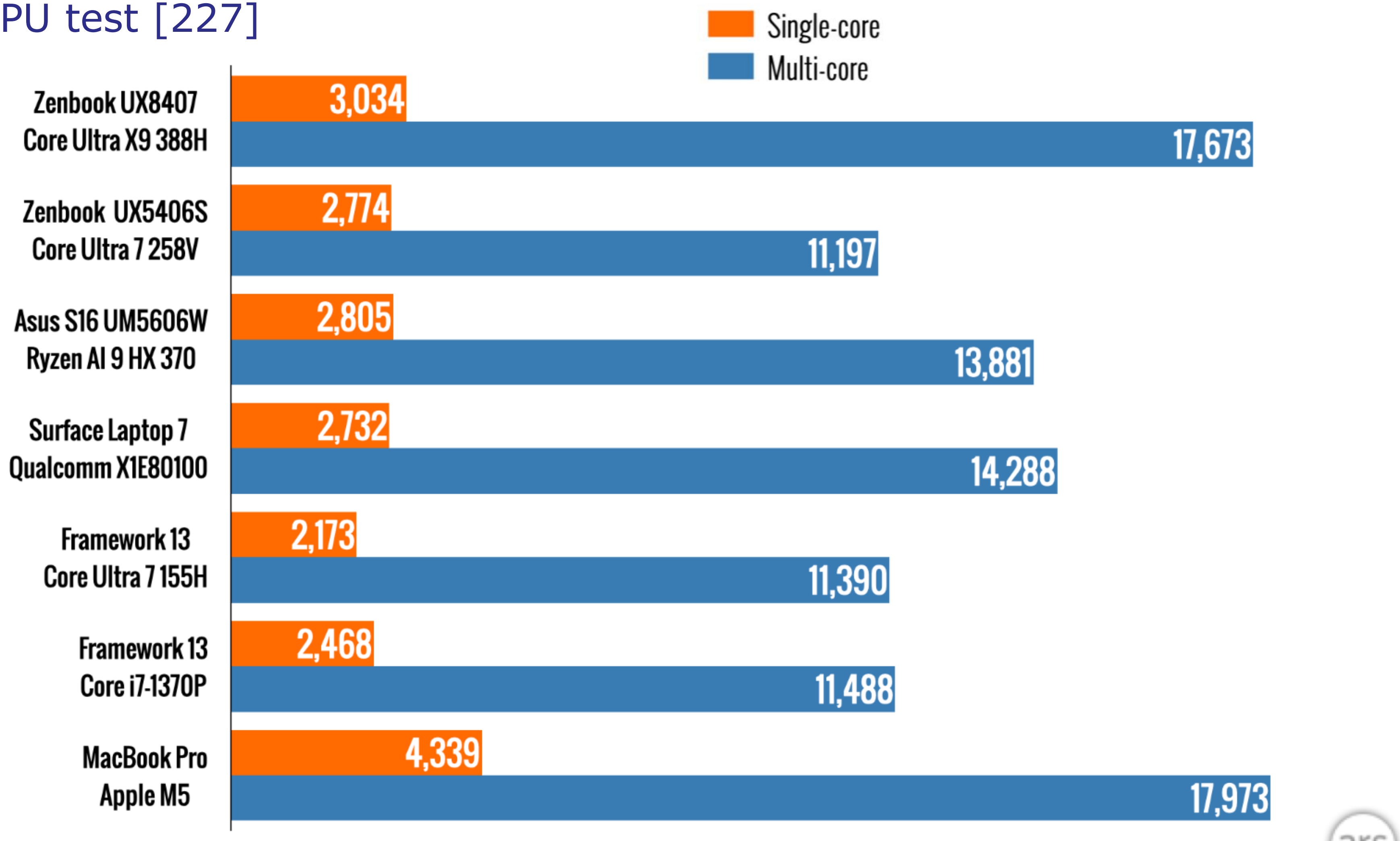
Comparison of single-thread SPECInt2007 performance of Intel’s 2nd and 3rd Series Core Ulta processors [2026)]



Panther Lakes’ performance and power consumption compared to competing processors

Next, we will compare the performance and power consumption of Panther Lake processors against the competition using a few relevant benchmarks.

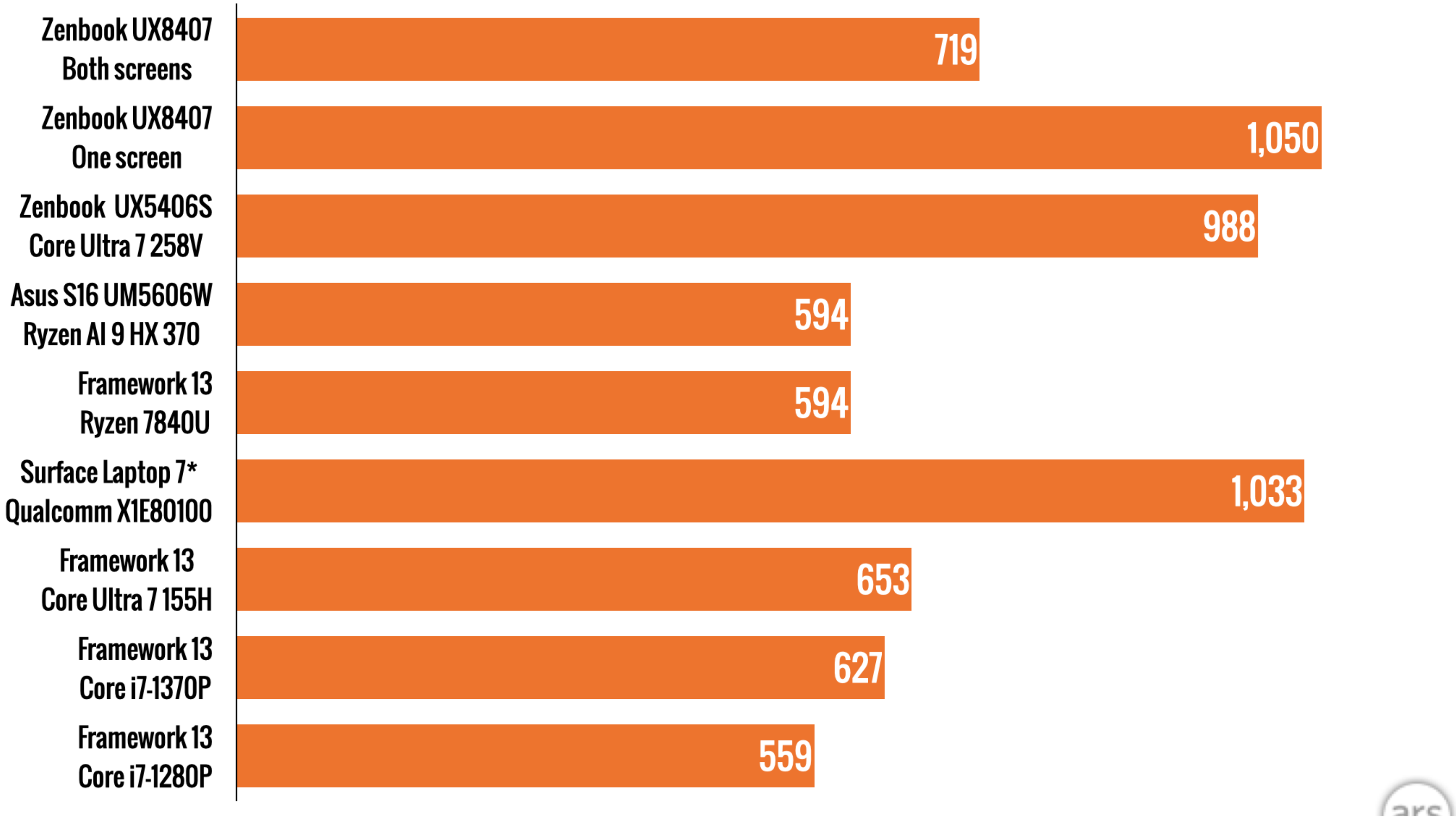
SC and MC performance of the Panther Lake processors using the Geekbench 6 CPU test [227]



Remark

Geekbench 6 is cross-platform benchmark that evaluates how a CPU handles everyday tasks such as email, photo operations, music playback, augmented reality, and machine-learning-style computations.

Battery life comparison using the PCMark Modern Office benchmark [227] (Results in minutes, higher is better) [227]

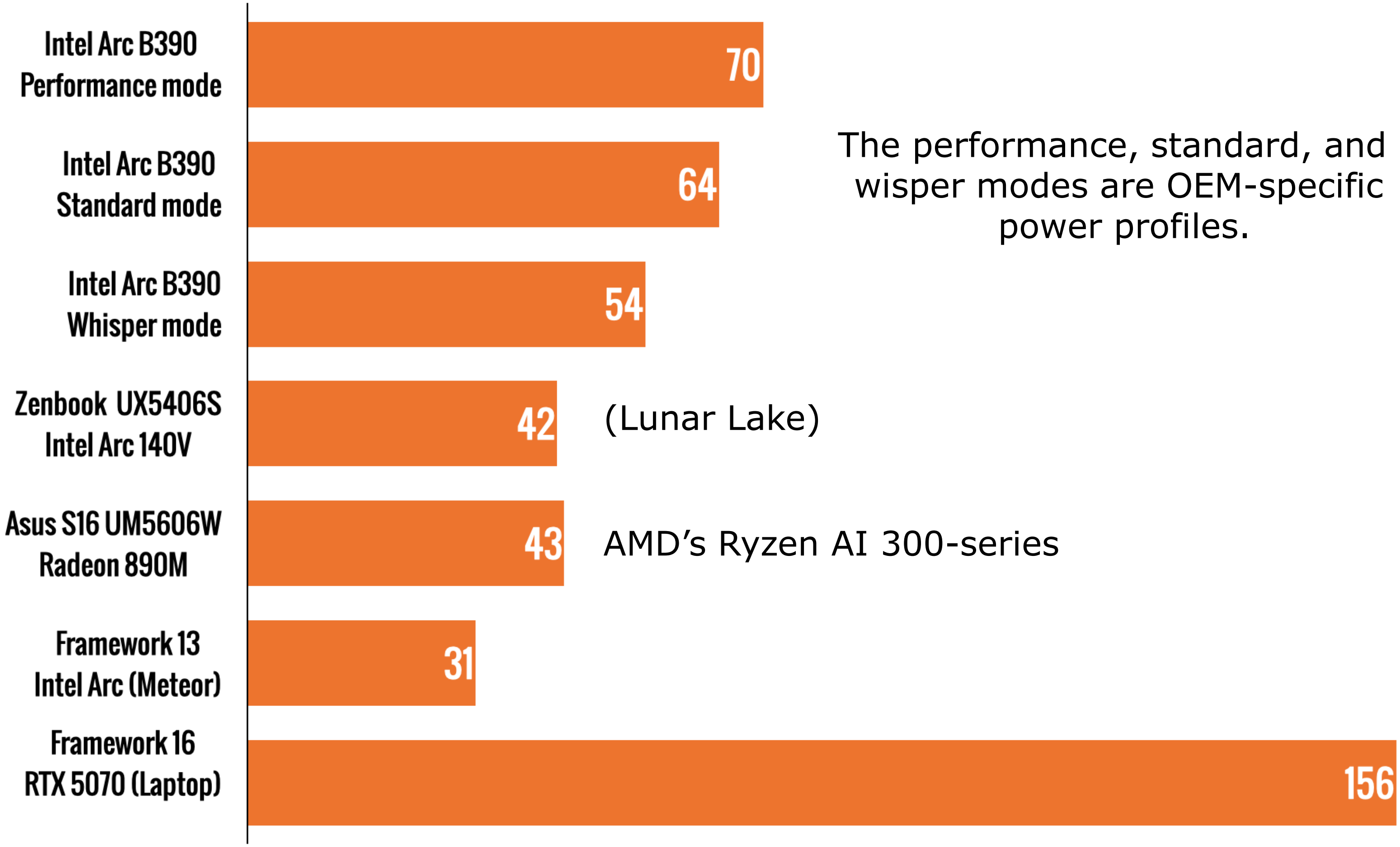


Remark

PCMark measures how well a laptop handles light-to-medium office tasks, such as writing, web browsing, and video conferencing.

It is part of the PCMark 10 benchmark (UL Benchmarks).

Graphics performance comparison while running the Shadow of the tomb raider game (Results in FPS, higher is better) [227]



The performance, standard, and wisper modes are OEM-specific power profiles.

(Lunar Lake)

AMD’s Ryzen AI 300-series

NVIDIA desktop graphics card based on the Blackwell architecture

It is a single-player action-adventure game, introduced in 2018.

In the story, Lara Croft races to stop a Maya apocalypse while exploring jungles, tombs, and underwater environments.

Assessment of the Core Ultra 3. series Panther Lake processors

The previous Figures demonstrate that Intel's Panther Lake processors deliver outstanding performance and power efficiency, surpassing all competitors in both aspects.

As a result, Intel's stock price has seen a significant increase, as illustrated in the following Figure.

How Intel’s stock price responded to market events from Aug. 2025 to Febr. 2026)

The above Figure shows the significant increase in Intel’s stock price following NVIDIA’s stock buy and Intel’s introduction of its Panther Lake family.

Source: Market Watch

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